



The Complete Power Platform Pipelines & ALM Setup Guide

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Introduction

Welcome to The Complete Power Platform Pipelines ALM Setup Guide.

In this guide you will learn how to setup Power Platform Pipelines and the Dataverse Git Integration. I have spent the past 5 years teaching Power Platform ALM best-practices and using them in my own projects. Now I want to share the knowledge I've gained in this set of easy-to-understand, practical instructions.

Power Apps Pipelines already have official documentation by Microsoft. So why did I make my own? A few reasons:

- I wanted a guided-tutorial that anyone could understand which includes clear directions and lots of pictures.
- This tutorial can be continuously improved as new Power Apps Pipelines & ALM features hit "general availability"
- Readers can leave comments on my website describing their own best practices which I can incorporate into future versions

I hope you enjoy my Power Automate Coding Standards For Cloud Flows.



Support The Site ❤️

I don't have ads on my website because I believe in delivering the best learning experience possible. This website is paid for out of my own pocket. If you've found value in the free resources I've created, I'd love to have your support.

Click on the "Buy Me A Cat Treat" button below to support the site.

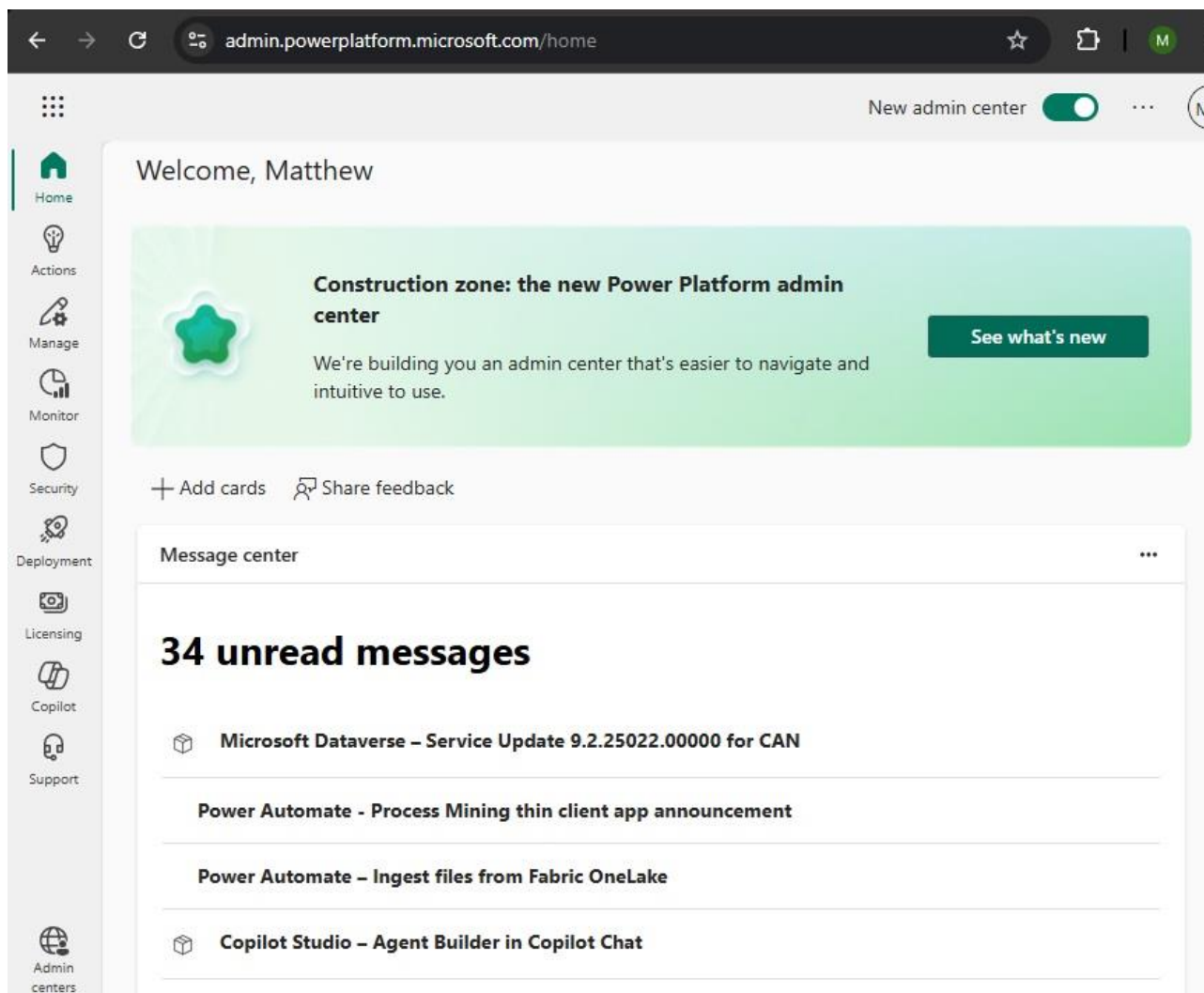
 Buy me a cat treat

Create Power Platform Managed Environments For Dev-Test-Prod

The basic set of Power Platform managed environments required by ALM best practices are a development environment, a test environment and a production environment. A development environment is where makers create the solution. The test environment is where quality assurance testers and end-users test the solution. And the production environment where end-users operate a live version of the solution.

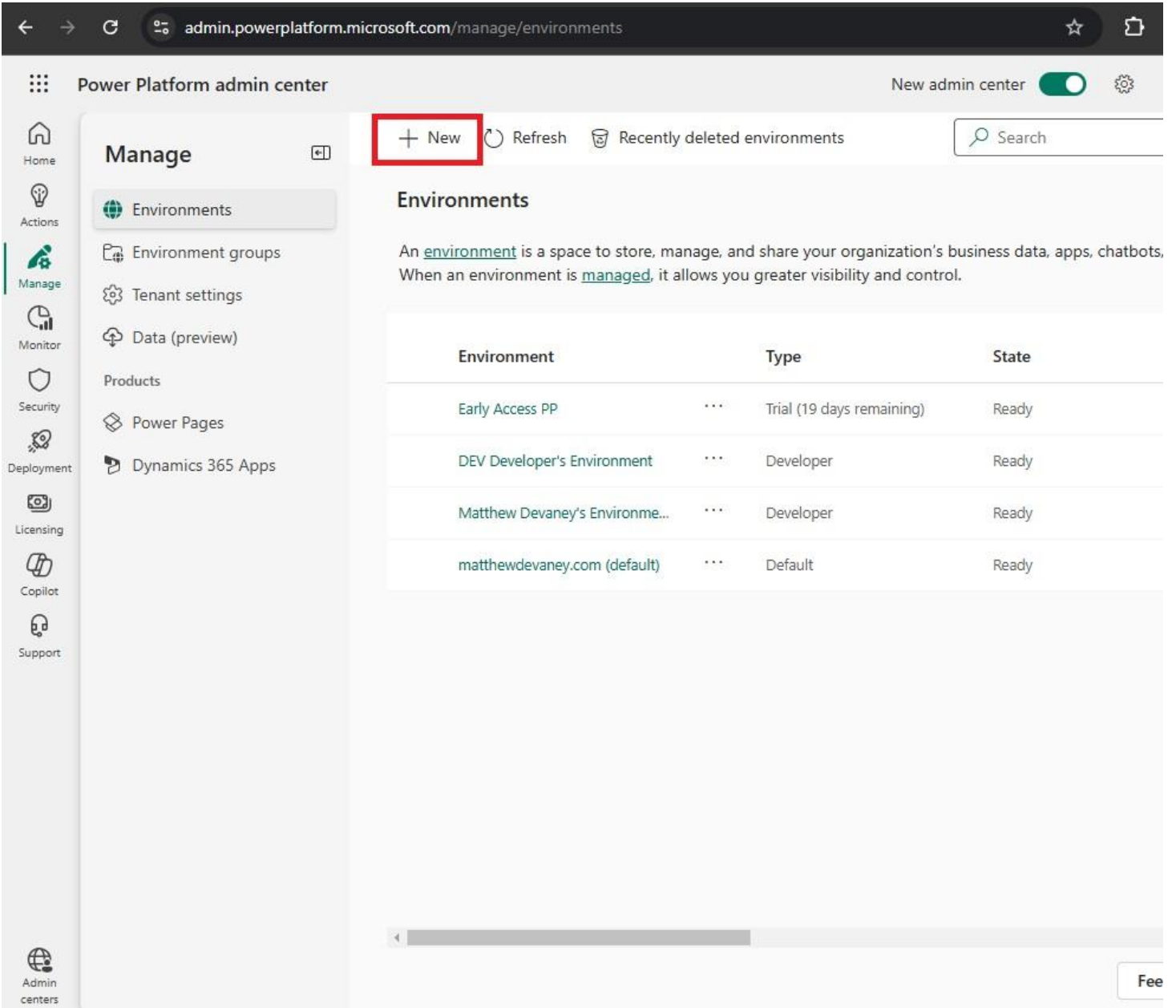
Open The Power Platform Admin Center

Environments are created in the Power Platform Admin Center. Go to admin.powerplatform.com and open the Admin Center homepage.



Create A New Power Platform Environment

To find the Power Platform environments menu, select Manage, and then choose Environments from the submenu. Press the New button to create a new environment.



Configure The Managed Environment For Development

On the New Environment menu select the following setup options:

- Name: choose a name for the environment and use the suffix (DEV). This will let developers know they are in a development environment
- Type: Sandbox for the development environment
- Add a Dataverse data store?: Yes

New environment

i This operation is subject to [capacity constraints](#)

Name *

ALM Tutorial (Dev)

Make this a Managed Environment *i*

☐ No

Group

Choose a group

Region *

United States

A local region can provide quicker data access

Get new features early *i*

☐ No

Type *i* *

Sandbox

Purpose

Describe the environment's purpose

Add a Dataverse data store? *i*

☒ Yes

Next

Cancel

Choose a URL for the Dataverse instance. A common pattern for URLs is *companyname-purpose-environmenttype*. In the example below the environment name is *devaney-almtutorial-dev*. Press the Save button once finished.

←

Add Dataverse

×

This operation is subject to [capacity constraints](#)

Language *

English (United States)

▼

Default language for user interfaces in this environment

Currency *

USD (\$)

▼

Reports will use this currency

Security group *

Restrict environment access to members of a security group or select None to opt for open access across your tenant. [Learn more](#)

None 

(All users across your tenant will have access to the environment)

URL

If you don't enter a domain name, we will pick one for you

devaney-almtutorial-dev

crm.dynamics.com

Enable Dynamics 365 apps?

In addition to Power Apps. [Learn more](#)

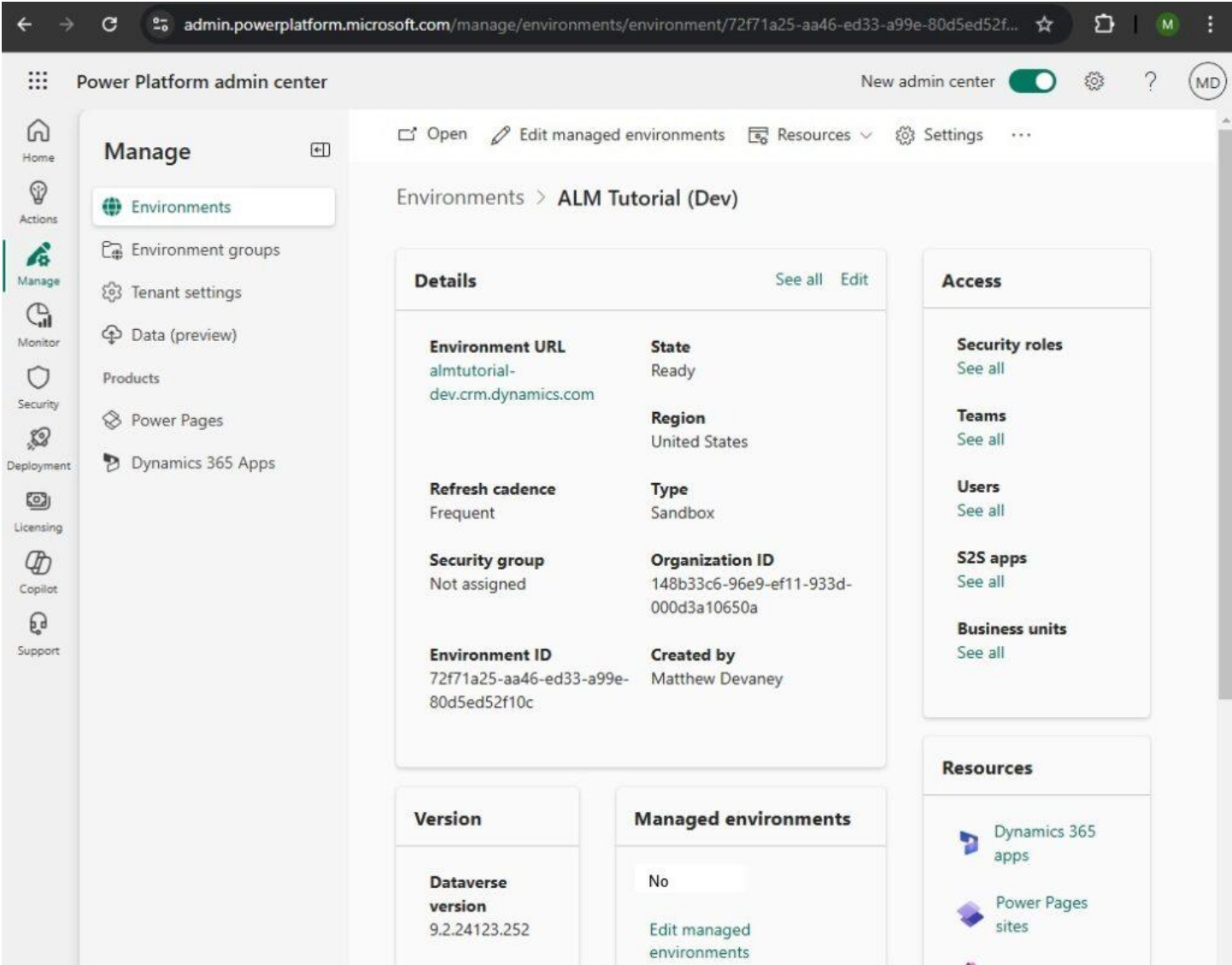
☐

No

Save

Cancel

After waiting a few moments for the environment to be created the browser will redirect to the development environment’s landing page.



Setup The Test And Production Managed Environments

We also require two more environments to be setup for testing and production. Test and production environments are known as target environments. All target environments are required to be managed environments to use Power Platform Pipelines.

Create a new test environment with these settings:

- Name – Your Environment Name (Test)
- Make This A Managed Environment – Yes
- Type – Sandbox

New environment [X]

(i) This operation is subject to [capacity constraints](#)

Name *
ALM Tutorial (Test)

Make this a Managed Environment (i)
☒ Yes

Group
Choose a group [v]

Region *
United States [v]
A local region can provide quicker data access

Get new features early (i)
☐ No

Type (i) *
Sandbox [v]

Purpose
Describe the environment's purpose

Add a Dataverse data store? (i)

Next Cancel

The apply these additional settings and press the Save button:

- Type – Sandbox
- URL – companyname-purpose-test

← Add Dataverse

×

This operation is subject to [capacity constraints](#)

Language *

English (United States) ▾

Default language for user interfaces in this environment

Currency *

USD (\$) ▾

Reports will use this currency

Security group *

Restrict environment access to members of a security group or select None to opt for open access across your tenant. [Learn more](#)

None 

(All users across your tenant will have access to the environment)

URL

If you don't enter a domain name, we will pick one for you

devaney-almtutorial-test

crm.dynamics.com

Enable Dynamics 365 apps?

In addition to Power Apps. [Learn more](#)

Save

Cancel

Repeat the steps to create a new environment for Production with the following settings:

- Name – Your Environment Name (Test)
- Make This A Managed Environment – Yes
- Type – Production

New environment

i This operation is subject to [capacity constraints](#)

Name *

ALM Tutorial (Prod)

Make this a Managed Environment *i*

☒ Yes

Group

Choose a group

Region *

United States

A local region can provide quicker data access

Get new features early *i*

☐ No

Type *i* *

Sandbox

Purpose

Describe the environment's purpose

Add a Dataverse data store? *i*

Next

Cancel

And apply these additional settings, then press Save:

- Type – Production
- URL – companyname-purpose-prod

←

Add Dataverse

×

ⓘ

This operation is subject to [capacity constraints](#)

Language *

English (United States) ▾

Default language for user interfaces in this environment

Currency *

USD (\$) ▾

Reports will use this currency

Security group *

Restrict environment access to members of a security group or select None to opt for open access across your tenant. [Learn more](#)

None 

(All users across your tenant will have access to the environment)

URL

If you don't enter a domain name, we will pick one for you

devaney-almtutorial-prod

crm.dynamics.com

Enable Dynamics 365 apps?

In addition to Power Apps. [Learn more](#)

Save

Cancel

After the setup of all Dev-Test-Prod environments is completed they will appear in the environments menu.

Power Platform admin center

New admin center ☒

Home

Actions

Manage

Monitor

Security

Deployment

Licensing

Copilot

Support

Manage

Environments

Environment groups

Tenant settings

Data (preview)

Products

Power Pages

Dynamics 365 Apps

+ New

Refresh

...

Search

Environments

An [environment](#) is a space to store, manage, and share your organization's business data, ap chatbots, and flows. When an environment is [managed](#), it allows you greater visibility and co

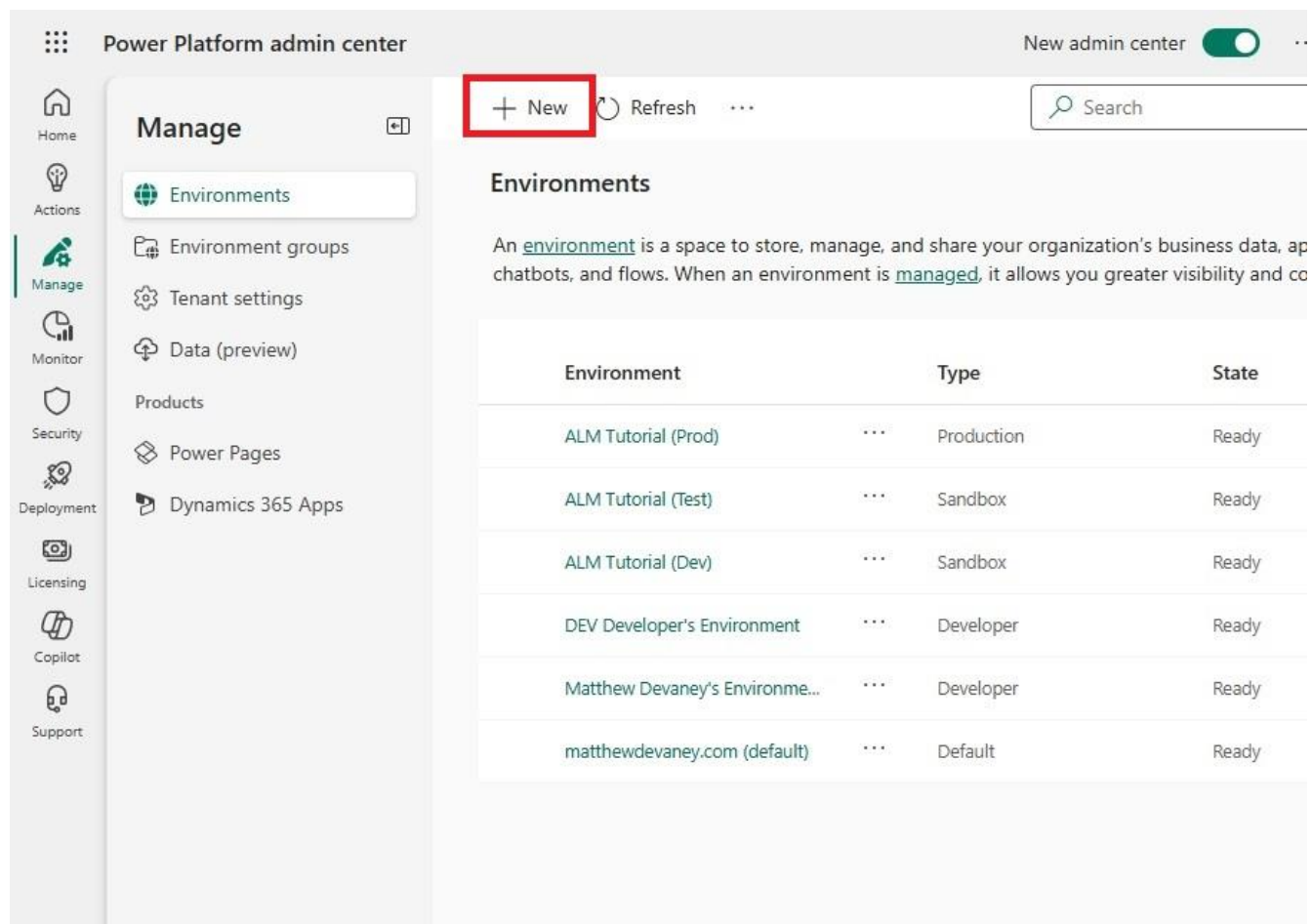
Environment		Type	State
ALM Tutorial (Prod)	...	Production	Ready
ALM Tutorial (Test)	...	Sandbox	Ready
ALM Tutorial (Dev)	...	Sandbox	Ready
DEV Developer's Environment	...	Developer	Ready
Matthew Devaney's Environme...	...	Developer	Ready
matthewdevaney.com (default)	...	Default	Ready

Install Power Platform Pipelines In The Host Environment

The Power Platform Pipelines application allows a user to create and manage pipelines. It must be installed in a host environment. A host environment is simply a production environment where the pipeline definitions are stored. And where the pipelines themselves are executed.

Open The Environments Menu In Power Platform Admin Center

To create a host environment for the Power Platform Pipelines application go to the [Power Platform Admin Center](#). Then open the Environments menu by selecting the option Manage > Environment and press the New button.



The screenshot shows the Power Platform Admin Center interface. The left sidebar contains a 'Manage' section with a list of options: Environments, Environment groups, Tenant settings, Data (preview), Products, Power Pages, and Dynamics 365 Apps. The 'Environments' option is highlighted. The main content area is titled 'Environments' and includes a description: 'An environment is a space to store, manage, and share your organization's business data, apps, chatbots, and flows. When an environment is managed, it allows you greater visibility and control.' Below the description is a table of environments.

Environment	Type	State
ALM Tutorial (Prod)	Production	Ready
ALM Tutorial (Test)	Sandbox	Ready
ALM Tutorial (Dev)	Sandbox	Ready
DEV Developer's Environment	Developer	Ready
Matthew Devaney's Environme...	Developer	Ready
matthewdevaney.com (default)	Default	Ready

Configure The Power Platform Pipelines Host Environment

In the New Environment menu choose the following setup options:

- Name – PP Pipelines Host
- Type – select Production
- Add a Dataverse datastore – Yes

New environment [Close]

i This operation is subject to [capacity constraints](#)

Name *
PP Pipelines Host

Make this a Managed Environment i
☐ No

Group
Choose a group

Region *
United States

A local region can provide quicker data access

Get new features early i
☐ No

Type i *
Production

Purpose
Describe the environment's purpose

Add a Dataverse data store? i
☒ Yes

Next Cancel

Then on the Add Dataverse menu choose a URL. A common URL pattern is *companyname-pipelines-host*. In the example below, the URL is *devaney-pipelines-host*. Press the Save button once setup is finished.

← Add Dataverse

This operation is subject to [capacity constraints](#)

Language *

English (United States) ▾

Default language for user interfaces in this environment

Currency *

USD (\$) ▾

Reports will use this currency

Security group *

Restrict environment access to members of a security group or select None to opt for open access across your tenant. [Learn more](#)

None 

(All users across your tenant will have access to the environment)

URL

If you don't enter a domain name, we will pick one for you

devaney-pipelines-host

crm.dynamics.com

Enable Dynamics 365 apps?

In addition to Power Apps. [Learn more](#)

☐ No

Save

Cancel

After a few moments, the web browser redirects to the landing page for the Power Platform Pipelines host environment.

⋮

Home

Actions

Manage

Monitor

Security

Deployment

Licensing

Copilot

Support

Admin centers

Manage

Environments

Environment groups

Tenant settings

Data (preview)

Products

Power Pages

Dynamics 365 Apps

New admin center

Open

Edit managed environments

Resources

Settings

Environments > PP Pipelines Host

Details

See all

Edit

Environment URL devaney-pipelines-host.crm.dynamics.com	State Ready
Refresh cadence Frequent	Region United States
Security group Not assigned	Type Production
Environment ID 3e40d71a-3daa-ea44-94b8-899c5b543c16	Organization ID c8c2d5de-98e9-ef11-933d-000d3a106b03
	Created by Matthew Devaney

Auditing

Manage

Install The Power Platform Pipelines App

Now that the host environment is setup the Power Platform Pipelines application must be installed. In the Power Platform Admin Center go to Manage > Dynamics 365 Apps. Choose Power Platform Pipelines and select Install.

Power Platform admin center

Home

Actions

Manage

Monitor

Security

Deployment

Licensing

Copilot

Support

Manage

Environments

Environment groups

Tenant settings

Data (preview)

Products

Power Pages

Dynamics 365 Apps

Install

Details

United States

Search

To manage Power Apps and Power Automate resources, select an environment.

Dynamics 365 apps

See which apps are available to install and configure in your org (tenant). [Learn more](#)

Name	Status
Power Pages Stripe v2	enabled
Power Platform Connection References	Enabled
Power Platform Dataflows	Enabled
Power Platform Environment Settings App	Enabled
Power Platform Pipelines	Enabled
Property Casualty Insurance Data Model	Not Configured
Retail banking core sample data	Not Configured
Retail banking data model	Not Configured
SAP Setup Assistant	Enabled
ServiceTeam ITAM	Enabled

Choose the Power Platform Pipelines host environment and press the Install button.

Install Power Platform Pipelines

Name

Power Platform Pipelines

Description

Publisher

Microsoft Dynamics 365

Select an environment *

PP Pipelines Host

Don't see your environment?

Package(s)

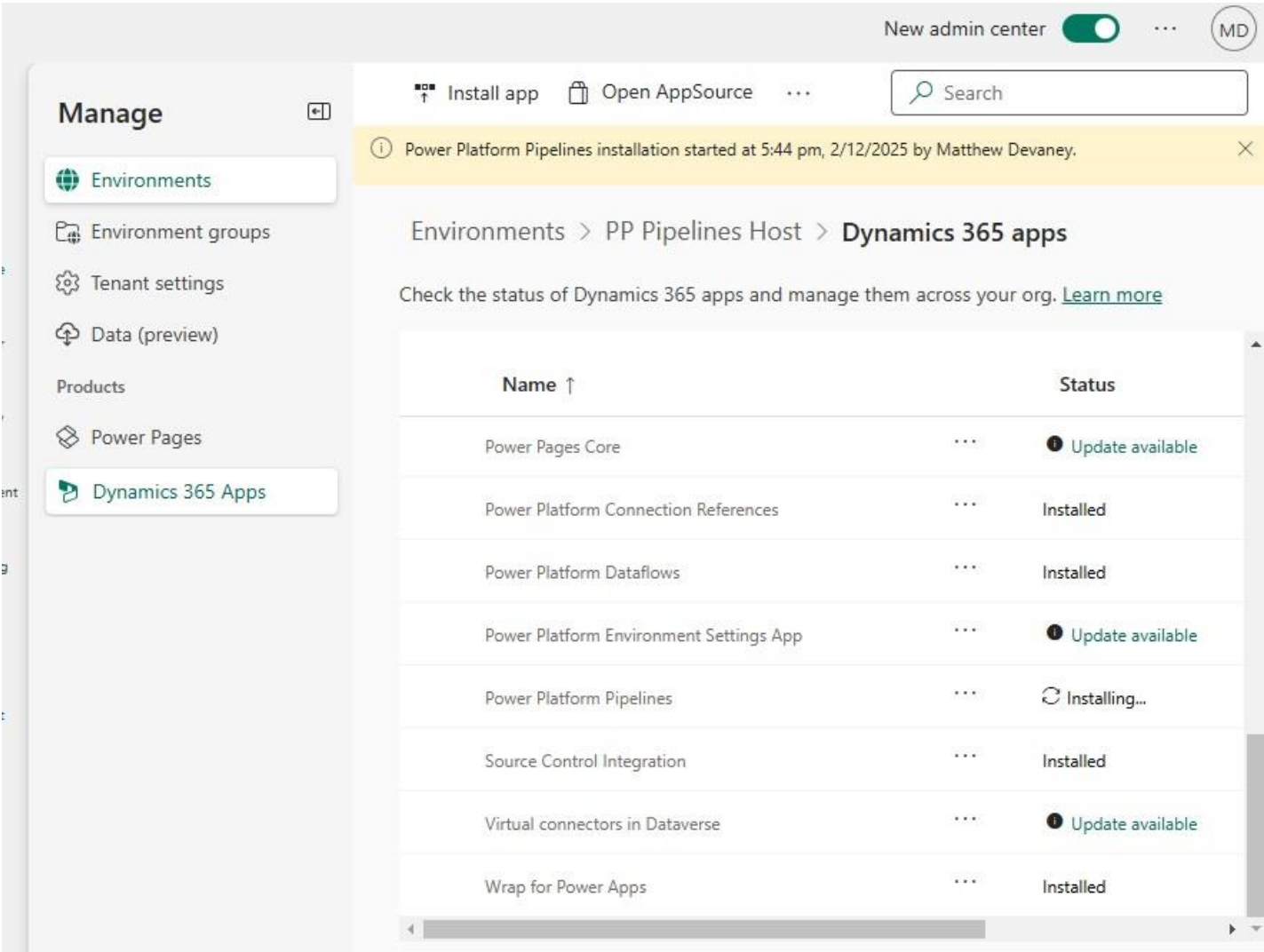
Name	Version	
Microsoft App Deployment Anchor Solution	9.1.24123.24121...	Terms of service

☒ I agree to the terms of service

Install

Cancel

The app installation process begins.



Verify The Model-Driven App Installation In The Host Environment

After a few minutes the Power Platform Pipelines installation will complete.

Home

Actions

Manage

Monitor

Security

Deployment

Licensing

Copilot

Support

Admin centers

Manage

Environments

Environment groups

Tenant settings

Data (preview)

Products

Power Pages

Dynamics 365 Apps

New admin center

United States

Search

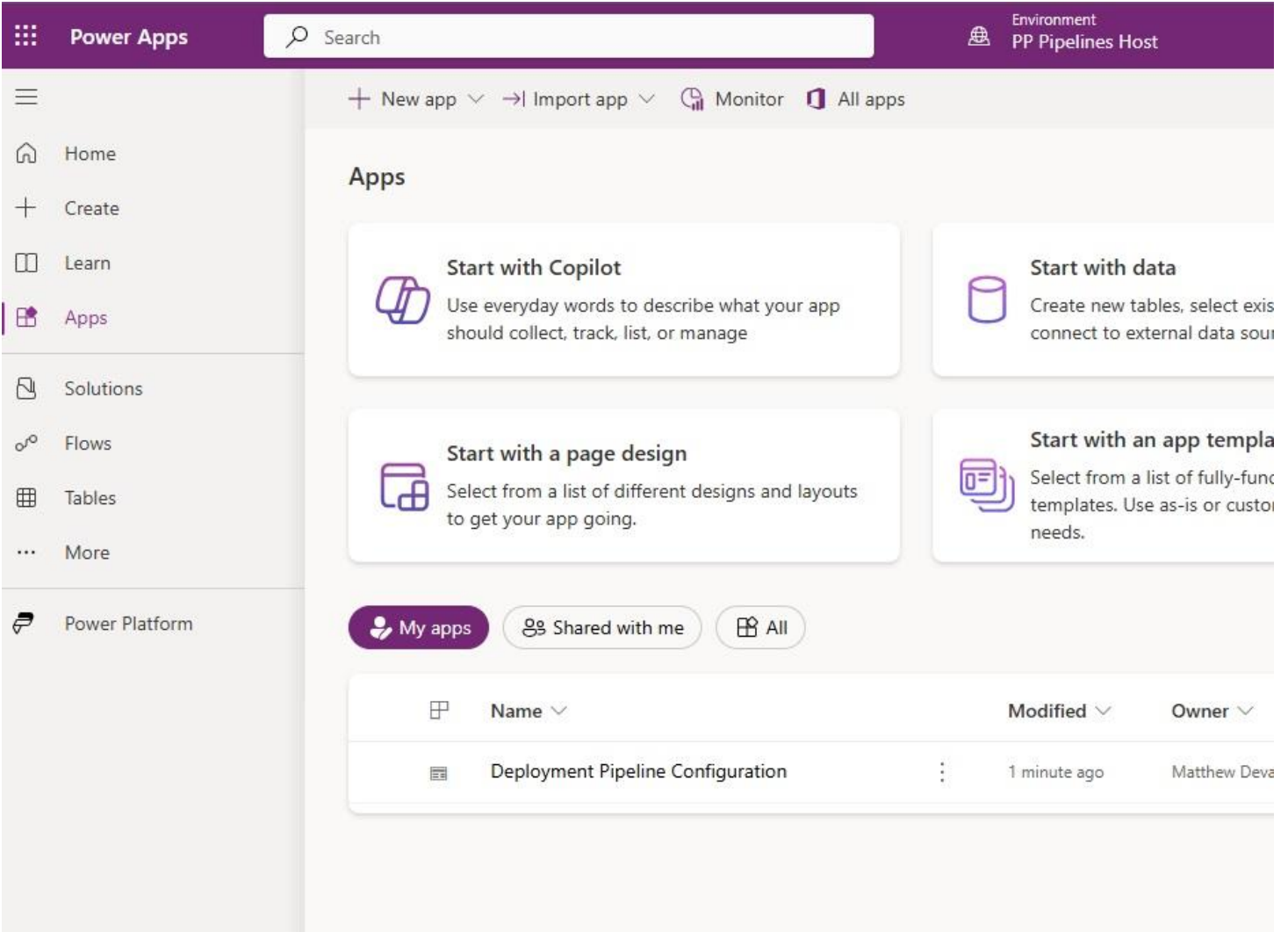
Dynamics 365 apps

See which apps are available to install and configure in your org (tenant). [Learn more](#)

Name ↑	Status ⓘ
Power Platform Connection References	Enabled
Power Platform Dataflows	Enabled
Power Platform Environment Settings App	Enabled
Power Platform Pipelines	Enabled
Property Casualty Insurance Data Model	Not Configured
Retail banking core sample data	Not Configured
Retail banking data model	Not Configured
SAP Setup Assistant	Enabled
ServiceTeam ITAM	Enabled

Feedback

Go to the Power Platform Pipelines Host environment and look for a model-driven app named Deployment Pipelines Configuration. This is the application used to create and manage pipelines.

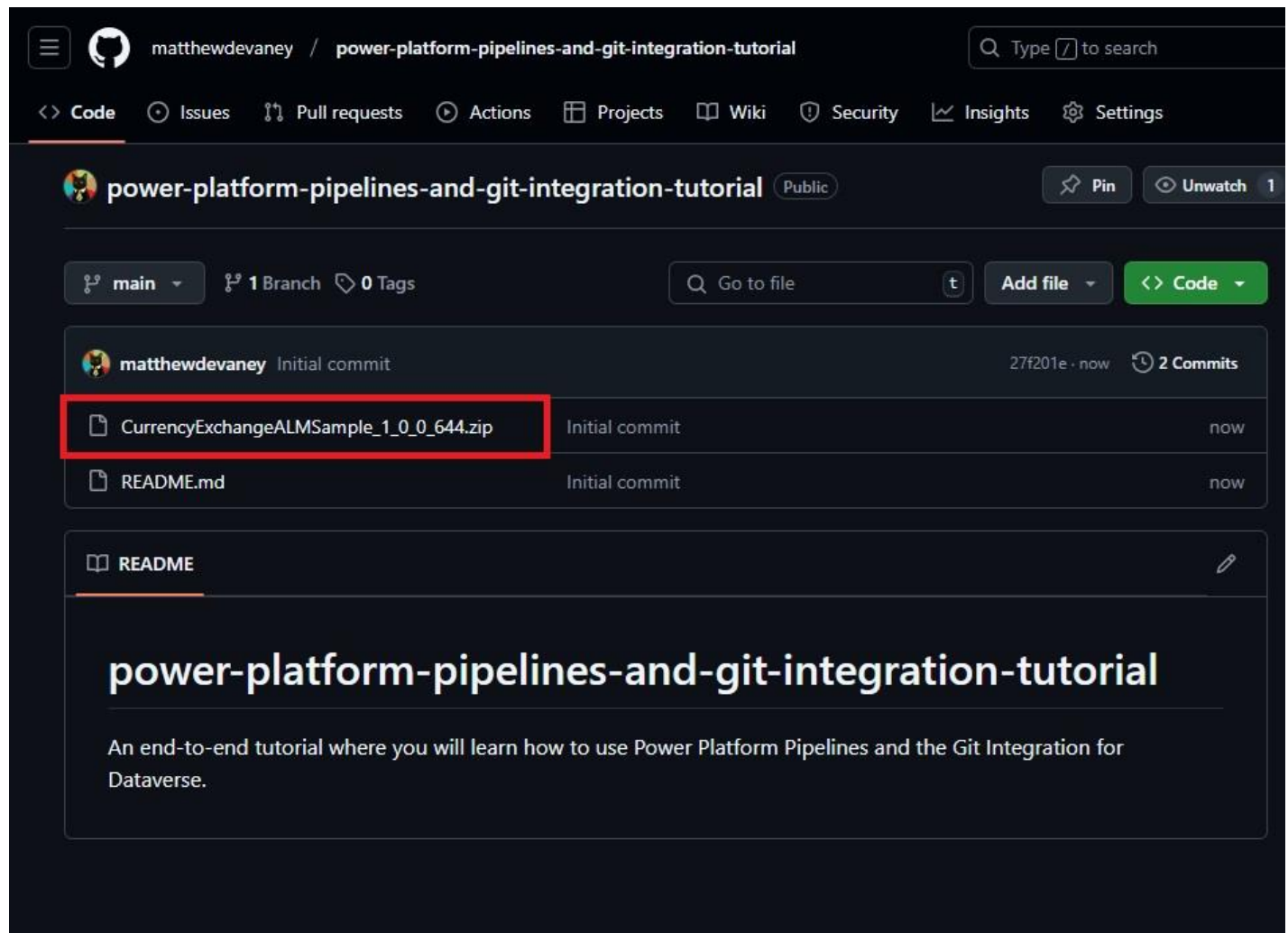


Import The Sample Solution Into The Dev Environment

This Power Platform Pipelines tutorial uses a sample solution with a canvas app, a model driven app, a flow, connections references, environment variables and tables. Rather than build them from scratch, we will import them into the development environment. That way we can keep focused on learning Power Platform application lifecycle management.

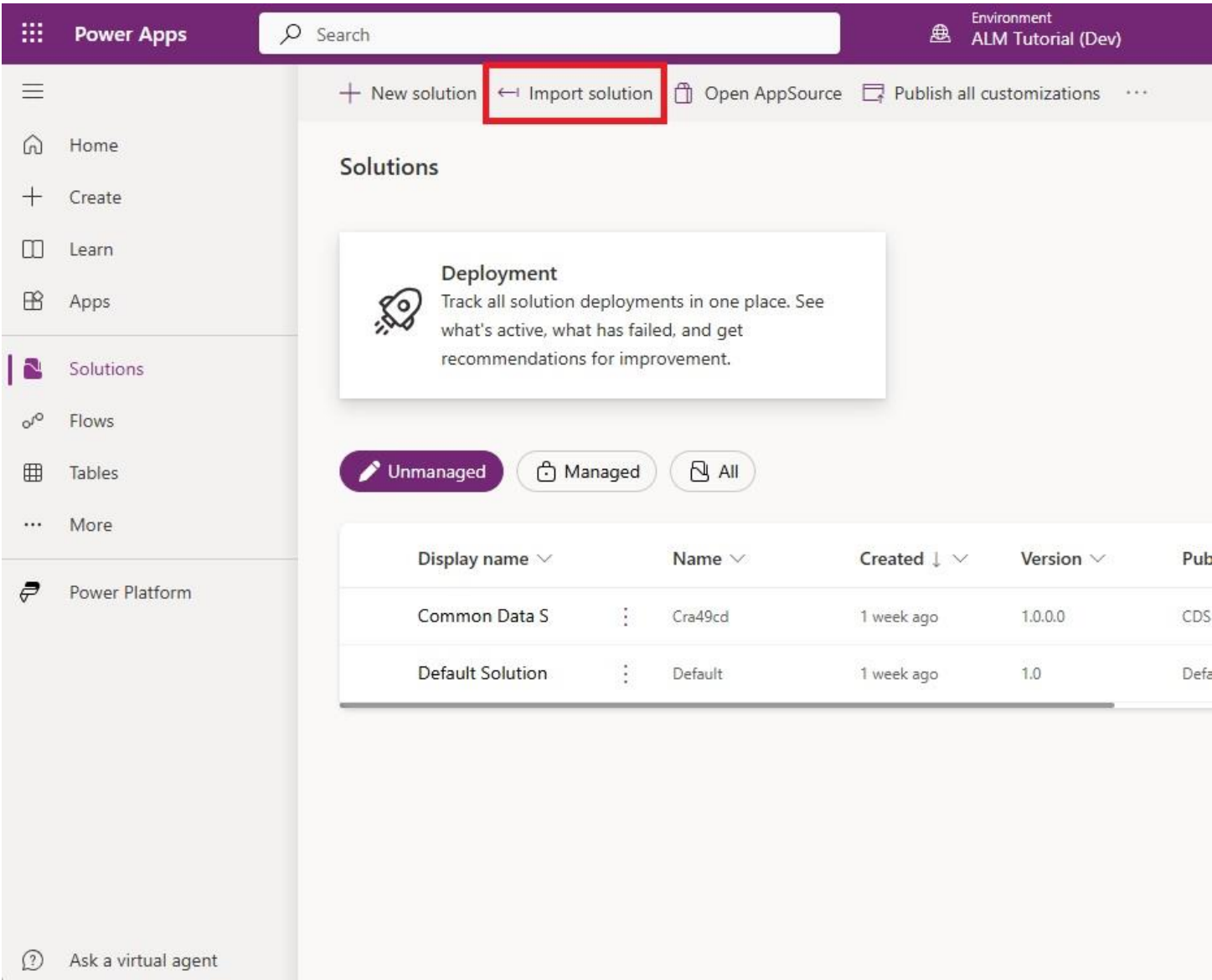
Download The Currency Exchange Sample Solution

The Currency Exchange sample solution is available for free in this Github repository. Download the file to your local machine.

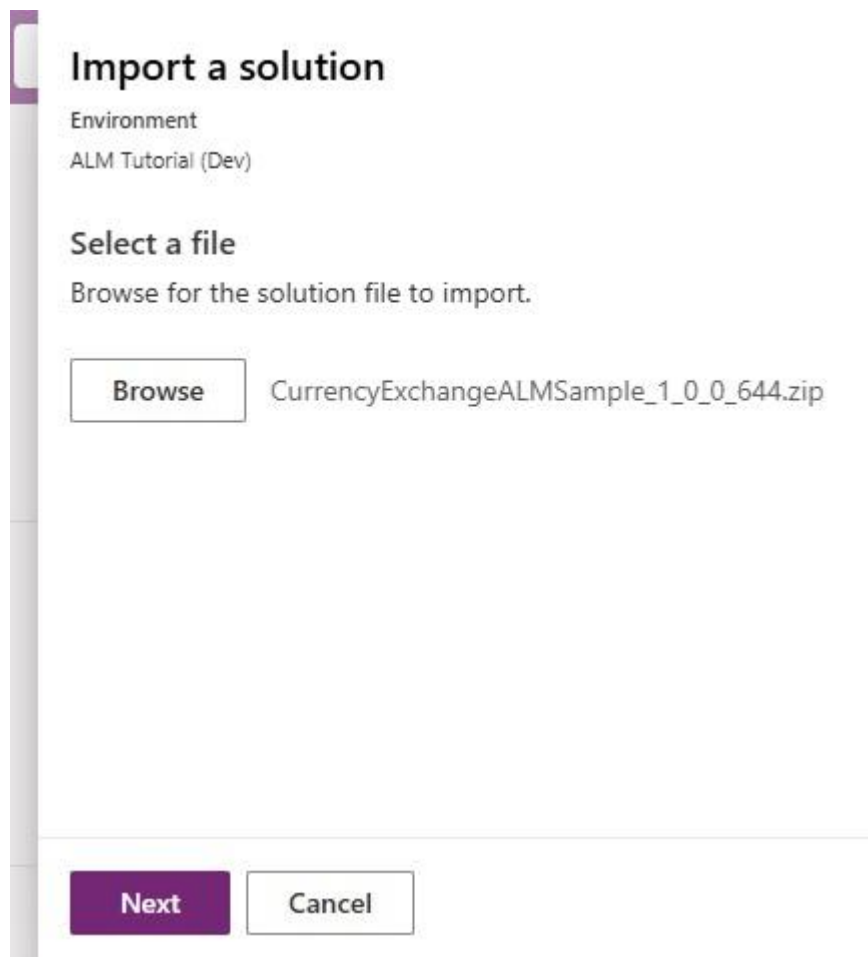


Import The Solution Into the Development Environment

Open the Development environment in Power Apps and go to the Solutions menu. Press the Import solution button.



Select the Currency Exchange sample solution from your local machine and press Next.



The dialog box has a purple header bar with a white square icon on the left. The main content area is white. At the bottom, there is a horizontal line separating the main content from the footer. The footer contains two buttons: a purple 'Next' button and a white 'Cancel' button with a grey border.

Import a solution

Environment
ALM Tutorial (Dev)

Select a file

Browse for the solution file to import.

CurrencyExchangeALMSample_1_0_0_644.zip

Review the import details and press the Next button again.

ps

← Import a solution

Environment
ALM Tutorial (Dev)

Details

Name
CurrencyExchangeALMSample

Type
Unmanaged

Publisher
Matthew Devaney

Version
1.0.0.644

Patch
No

Advanced settings ▾

Next

Cancel

Configure The Connection References

The sample solution contains two connection references used in a Power Automate flow. One for Dataverse and another for Outlook. Wait for the green checkmark to appear beside each connection reference. If it does not appear, press the three dots to troubleshoot the issue. Then press the Next button.

 **Import a solution** 

Environment
ALM Tutorial (Dev)

Connections

Re-establish connections to activate your solution.

2 updates needed

Sign in

These services use your credentials to sign into apps and create connections. A green check means you're ready to go.



Microsoft Dataverse CurrencyExchangeRates-b00e5
md_sharedcommondataserviceforapps_b00e5



Office 365 Outlook
md_Office365Outlook
[Permissions](#)

Office 365 Outlook

 md@matthewdevaney.com

 + Add new connection

Next

Cancel

Import The Sample Solution Into The Dev Environment

29

Set The Environment Variable Value

The sample solution also contains an environment variable for Email Notifications. The email address in this environment variable will receive email. Set this value to your personal email then press Import.

← Import a solution



Environment

ALM Tutorial (Dev)

Environment Variables

Enter information for each field, so your app works properly. You can edit your environment variables later.

1 updates needed

Email Notifications Target

 Reset

Import

Cancel

The sample solution will begin importing.

Power Apps

Home

Create

Learn

Apps

Solutions

Flows

Tables

More

Power Platform

New solution

Import solution

Open AppSource

Publish all customizations

Set preferences

Currently importing solution "Currency Exchange ALM Sample".

Solutions

Deployment

Track all solution deployments in one place. See what's active, what has failed, and get recommendations for improvement.

Unmanaged

Managed

All

Display name		Name	Created
Common Data Services Def...	Preferred solution	Cra49cd	1 week ago
Default Solution		Default	1 week ago

After a few minutes the import process will complete.

Power Apps

Home

Create

Learn

Apps

Solutions

Flows

Tables

More

Power Platform

New solution

Import solution

Open AppSource

Publish all customizations

Solution "Currency Exchange ALM Sample" imported successfully.

Solutions

Deployment

Track all solution deployments in one place. See what's active, what has failed, and get recommendations for improvement.

Unmanaged

Managed

All

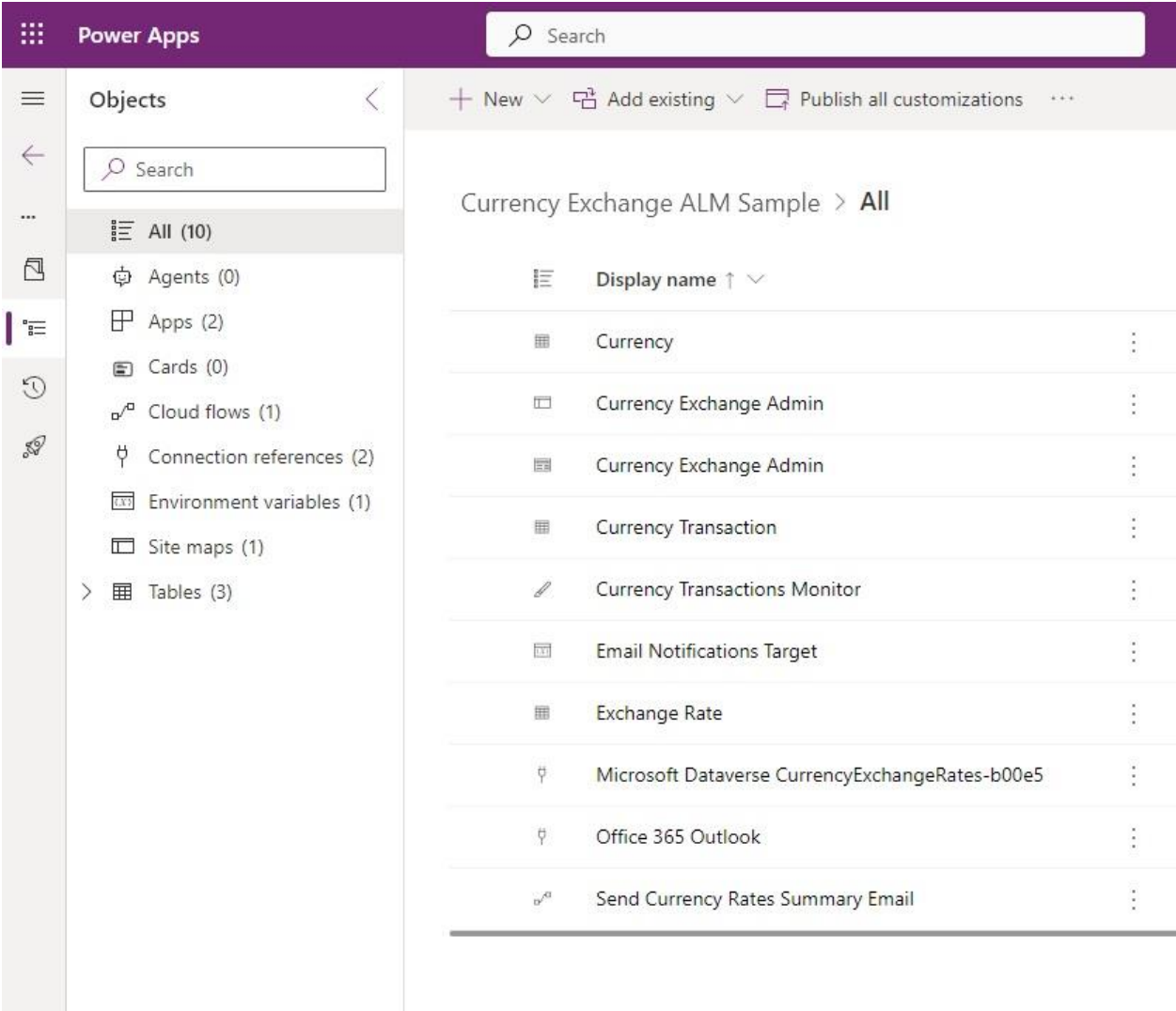
Display name		Name
Currency Exchange ALM Sample		CurrencyExchang...
Common Data Services Def...	Preferred solution	Cra49cd
Default Solution		Default

Import The Sample Solution Into The Dev Environment

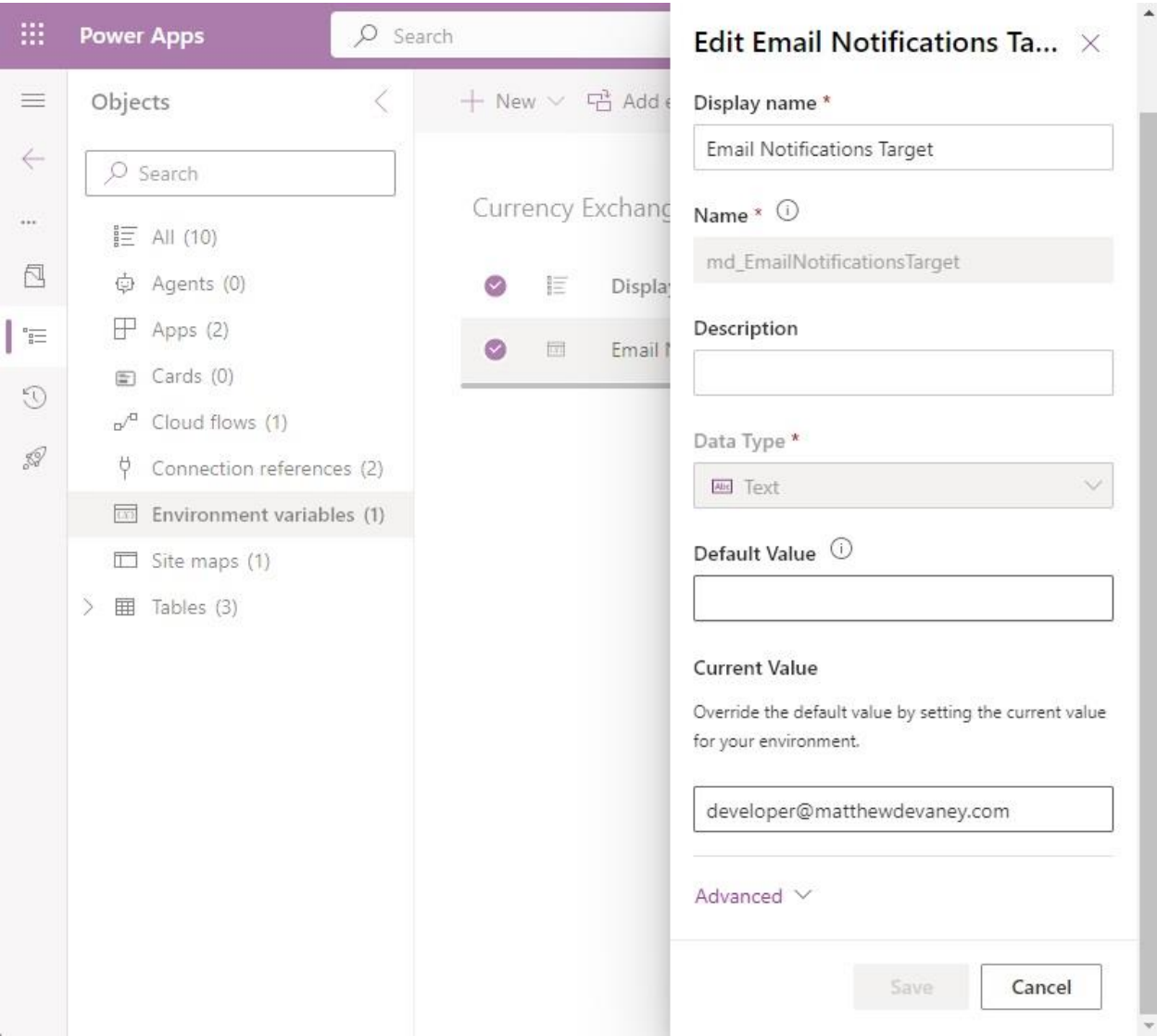
32

Review The Sample Solution In The Development Environment

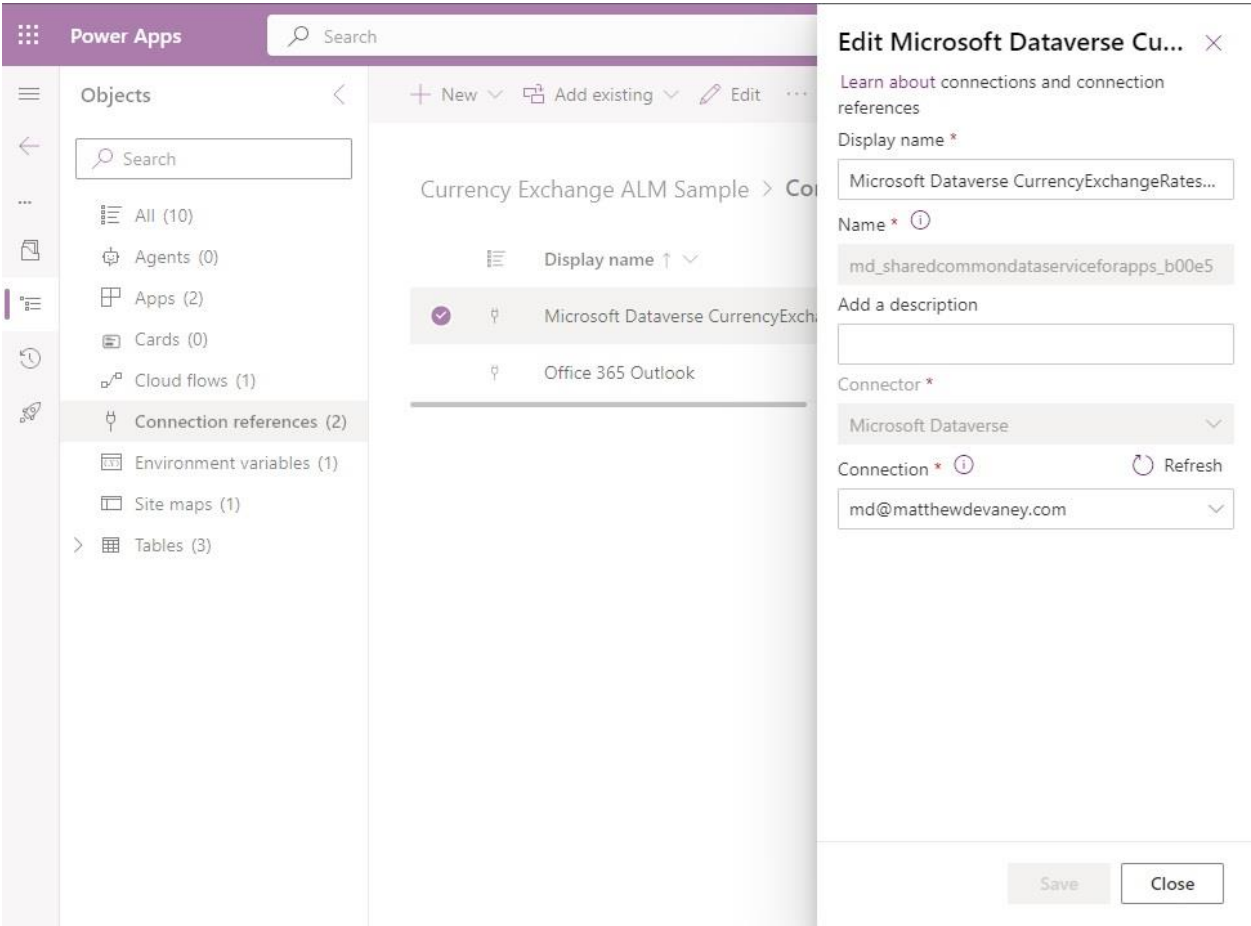
Open up the sample solution in the development environment and get to know its components.



Note that the environment variable is set to the value we defined during the import process.



And the connection references for Dataverse and Outlook have been configured with a connection.



To see the connections used in the connection references go to the Connections menu.

Power Apps

Home

Create

Learn

Apps

Solutions

Flows

Tables

Connections

More

Power Platform

Search

New connection

Connections in ALM Tutorial (Dev)

Canvas

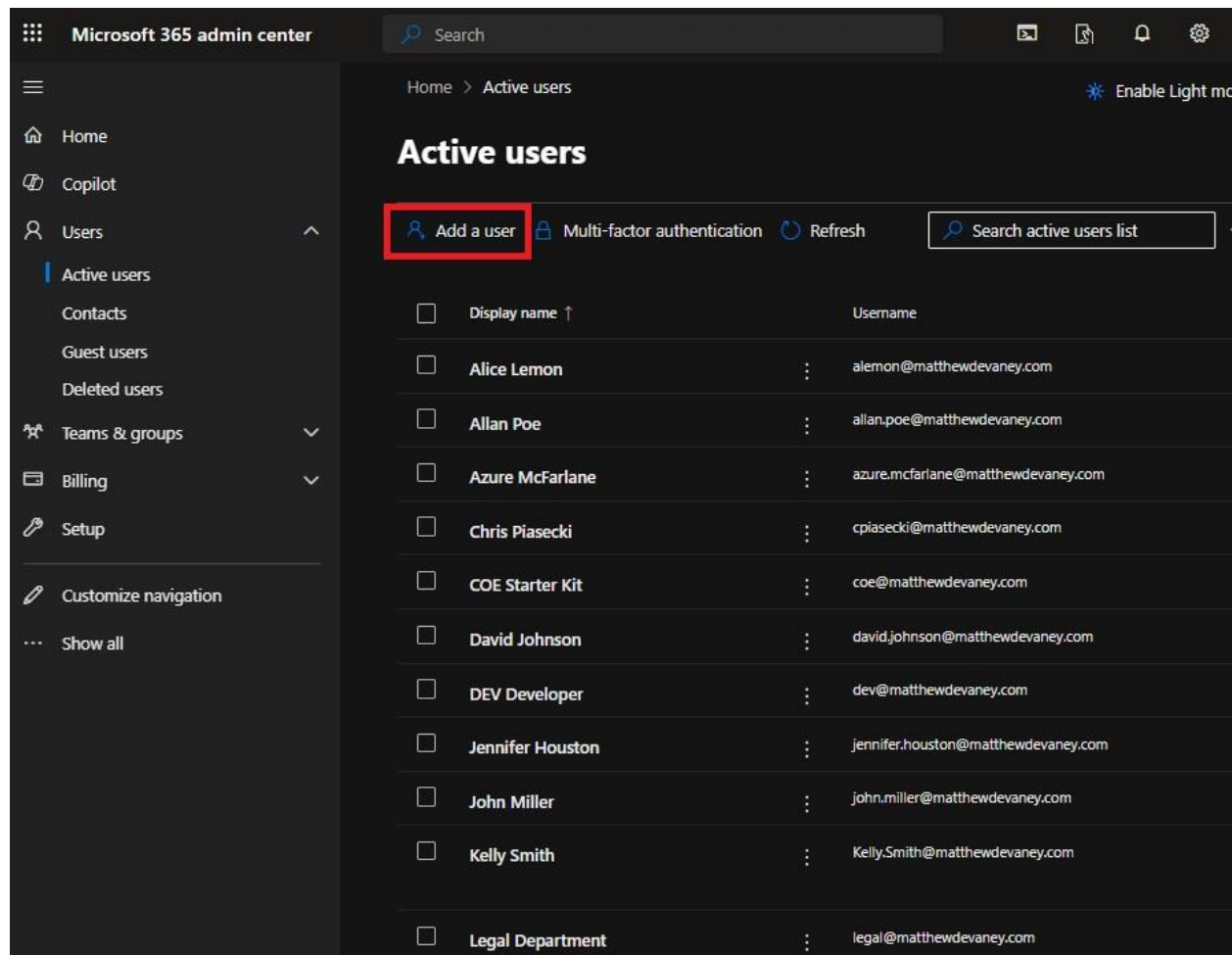
	Name		Modified
	md@matthewdevaney.com Office 365 Outlook	...	34 sec ago
	md@matthewdevaney.com Microsoft Dataverse	...	43 sec ago

Setup A Power Platform Service Account

A Power Platform Service Account is a Microsoft 365 account used by the system to perform background operations. Service Accounts often have an elevated security role like System Administrator. By using a Service Account, Power Platform Pipelines can deploy solutions as the service account instead of the user who started the deployment. This avoids giving the user system administrator privileges to the target environment

Add A New User In The Microsoft 365 Admin Center

To create a Power Platform Service Account go to the [Microsoft 365 Admin Center](#). Then browse to Users > Active Users and press the Add A User button.



Set the first name and last name to Service Account. And choose a username such as powerplatformserviceaccount@domain.com. Then press the Next button.

Microsoft 365 admin center

Search

Menu

Basics

Product licenses

Optional settings

Finish

Add a user

Set up the basics

To get started, fill out some basic information about who you're adding as a user.

First name

Last name

Service

Account

Display name *

Power Platform Service Account

Username *

Domains

powerplatformserviceaccount

@ matthewdevaney.com

☐ Automatically create a password

Passwords must be between 8 and 256 characters and use a combination of at least three of the following: uppercase letters, lowercase letters, numbers, and symbols.

Password *

.....

Strong

☐ Require this user to change their password when they first sign in

Next

Cancel

Assign Power Apps And Power Automate Licenses

The Power Platform Service Account must have premium licenses assigned to it. Apply a Power Apps Premium license and a Power Automate license. Then press Next.

The screenshot shows the 'Add a user' process in the Microsoft 365 admin center. The left sidebar contains a progress indicator with four steps: 'Basics' (completed), 'Product licenses' (active), 'Optional settings', and 'Finish'. The main content area is titled 'Assign product licenses' and includes the instruction 'Assign the licenses you'd like this user to have.' Below this is a 'Select location' dropdown menu set to 'United States'. A section titled 'Licenses (2)' shows a list of available licenses. The 'Assign user a product license' radio button is selected. Two licenses are checked: 'Power Apps Premium' (2 of 5 licenses available) and 'Power Automate Premium' (2 of 5 licenses available). Other unselected licenses include 'Microsoft Dynamics AX7 User Trial' (9999 of 10000 licenses available), 'Microsoft Fabric (Free)' (Unlimited licenses available), 'Microsoft Power Apps Plan 2 Trial' (9996 of 10000 licenses available), and 'Microsoft Power Apps for Developer' (9994 of 10000 licenses available). At the bottom are 'Back' and 'Next' buttons.

Microsoft 365 admin center

Search

Add a user

- Basics
- Product licenses**
- Optional settings
- Finish

Assign product licenses

Assign the licenses you'd like this user to have.

Select location *

United States

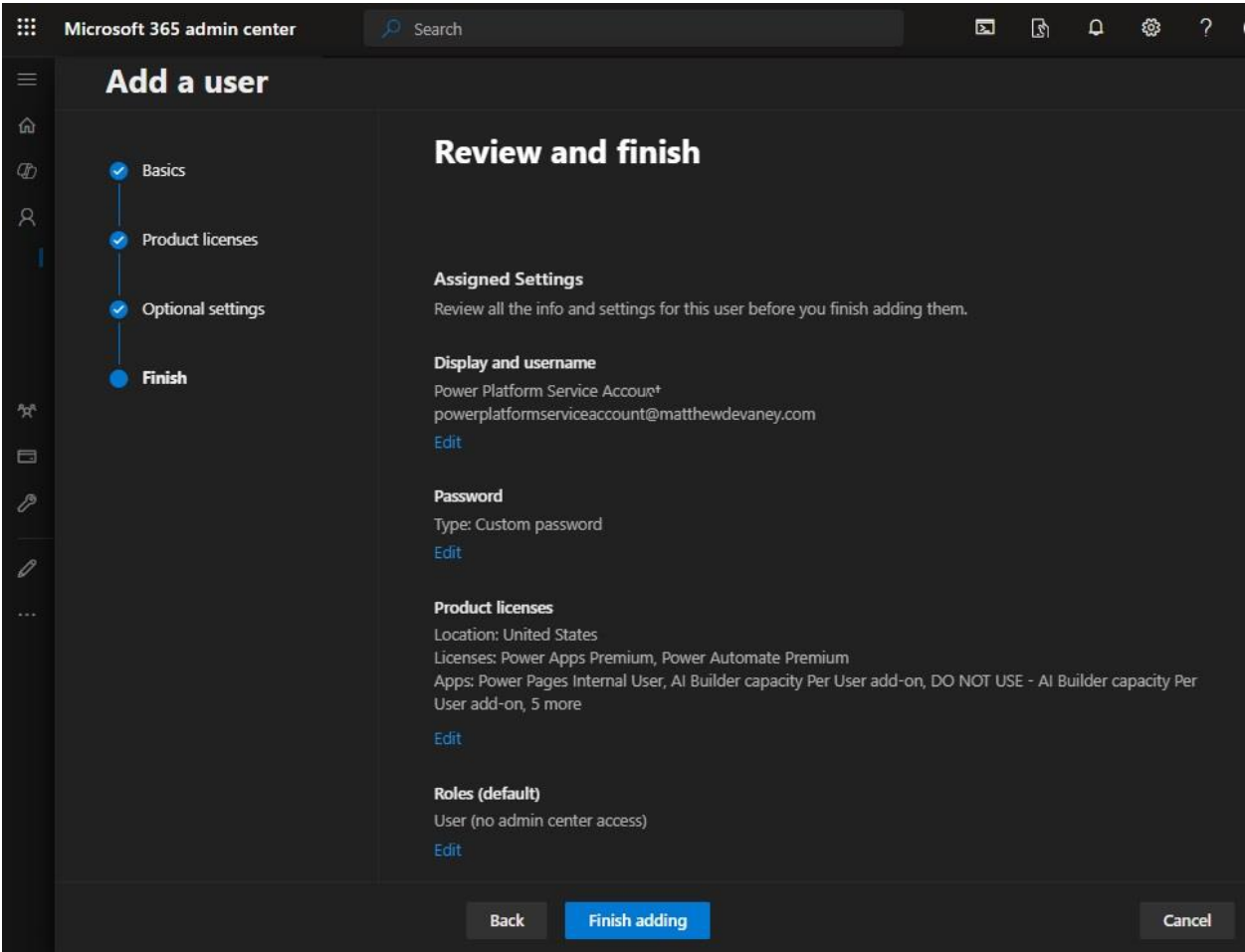
Licenses (2) ^

☒ Assign user a product license

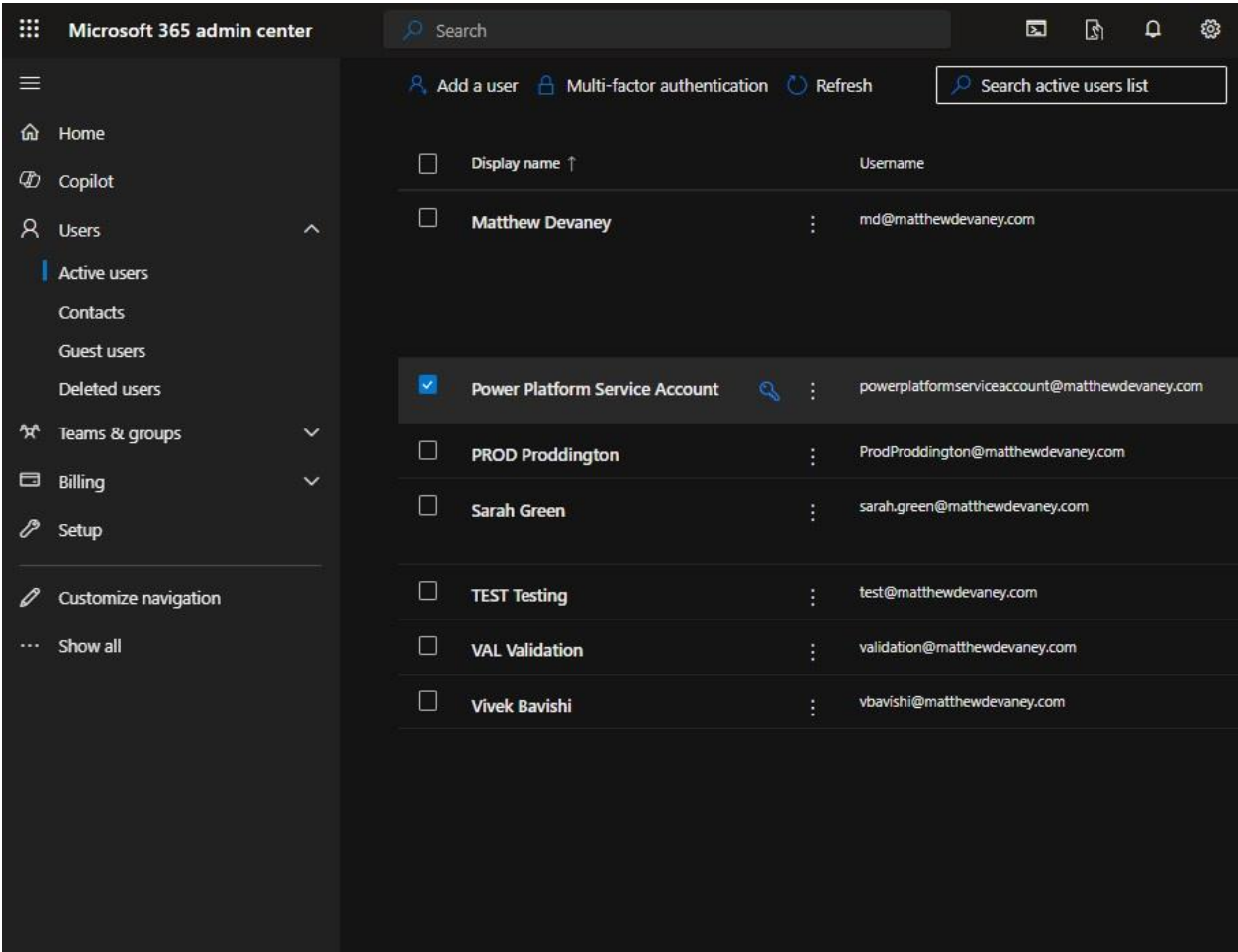
- ☒ **Power Apps Premium**
2 of 5 licenses available
- ☒ **Power Automate Premium**
2 of 5 licenses available
- ☐ **Microsoft Dynamics AX7 User Trial**
9999 of 10000 licenses available
- ☐ **Microsoft Fabric (Free)**
Unlimited licenses available
- ☐ **Microsoft Power Apps Plan 2 Trial**
9996 of 10000 licenses available
- ☐ **Microsoft Power Apps for Developer**
9994 of 10000 licenses available

Back Next

On the final screen, review the setup details. Then press the Finish button.

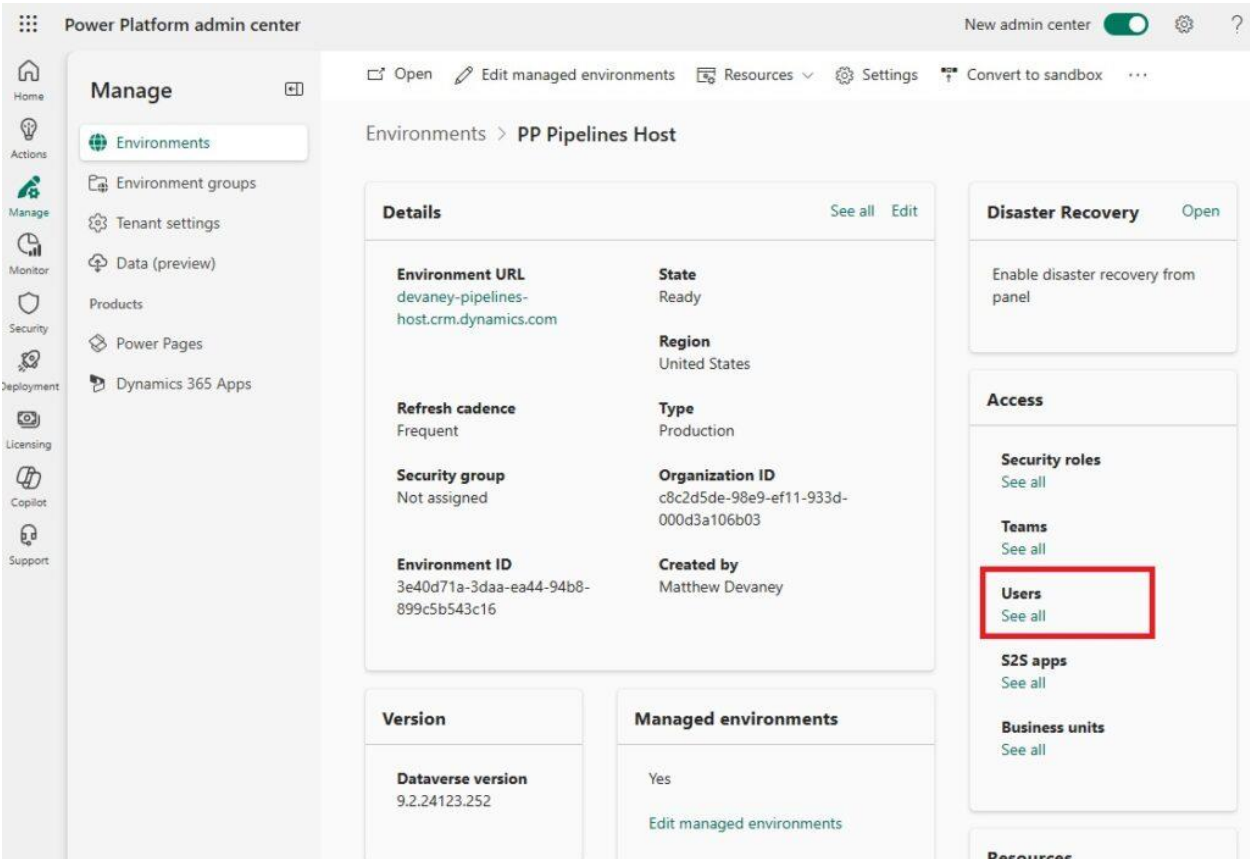


The Power Platform Service Account now appears in the list of Users.



Grant The System Administrator Role

The Power Platform Service Account must be given a System Administrator role in all environments: Development, Test, Production and the Pipelines Host environment. For each environment, go-to the landing page and open the Users menu.



Add the Power Platform Service Account as a new user.

Power Platform admin center

New admin center

Home

Actions

Manage

Monitor

Security

Deployment

Licensing

Copilot

Support

Manage

Environments

Environment groups

Tenant settings

Data (preview)

Products

Power Pages

Dynamics 365 Apps

Refresh

Add user

Manage users in Dynamics 365

Filter

Search

Environments > PP Pipelines Host > Settings > Users

Manage users so they can access data within their environment. This list includes users with disa statuses. [Learn more](#)

Looking for application users? Click here to go to the [app users list](#)

To validate user permissions for specific app(s), go to [app access checker](#).

Name		Username
Matthew Devaney	:	md@matthewdevaney.com
Service Account	:	powerplatformserviceaccount@matthewdevaney.com
Support User	:	crmoln@microsoft.com

Assign the System Administrator role to the Service Account.

Power Platform admin center

Manage

Environments

Environment groups

Tenant settings

Data (preview)

Products

Power Pages

Dynamics 365 Apps

Run diagnostics

Manage

Environments > PP Pipeline

Manage users so they can access statuses. [Learn more](#)

Looking for application users? Click

To validate user permissions for spe

Name
Matthew Devaney
Service Account
Support User

SA

Service Account

User status: Enabled

Run diagnostics

Refresh user

Summary

User Name

powerplatformserviceaccount@matthewdevaney.com

Direct Assigned Roles

System Administrator

Manage roles

Teams

devaney-pipelines-host

Manage teams

Contact Information

First Name

Service

Last Name

Account

Primary email

powerplatformserviceaccount@matthewdevaney.com

Phone number

Mobile phone

Organization Information

Business Unit: devaney-pipelines-host

Change business unit

Manager

None provided

Change Manager

Reassign records

Position

None provided

Change position

Administration

Read-Write

Client Access License (CAL) Information

Setup A Power Platform Service Account

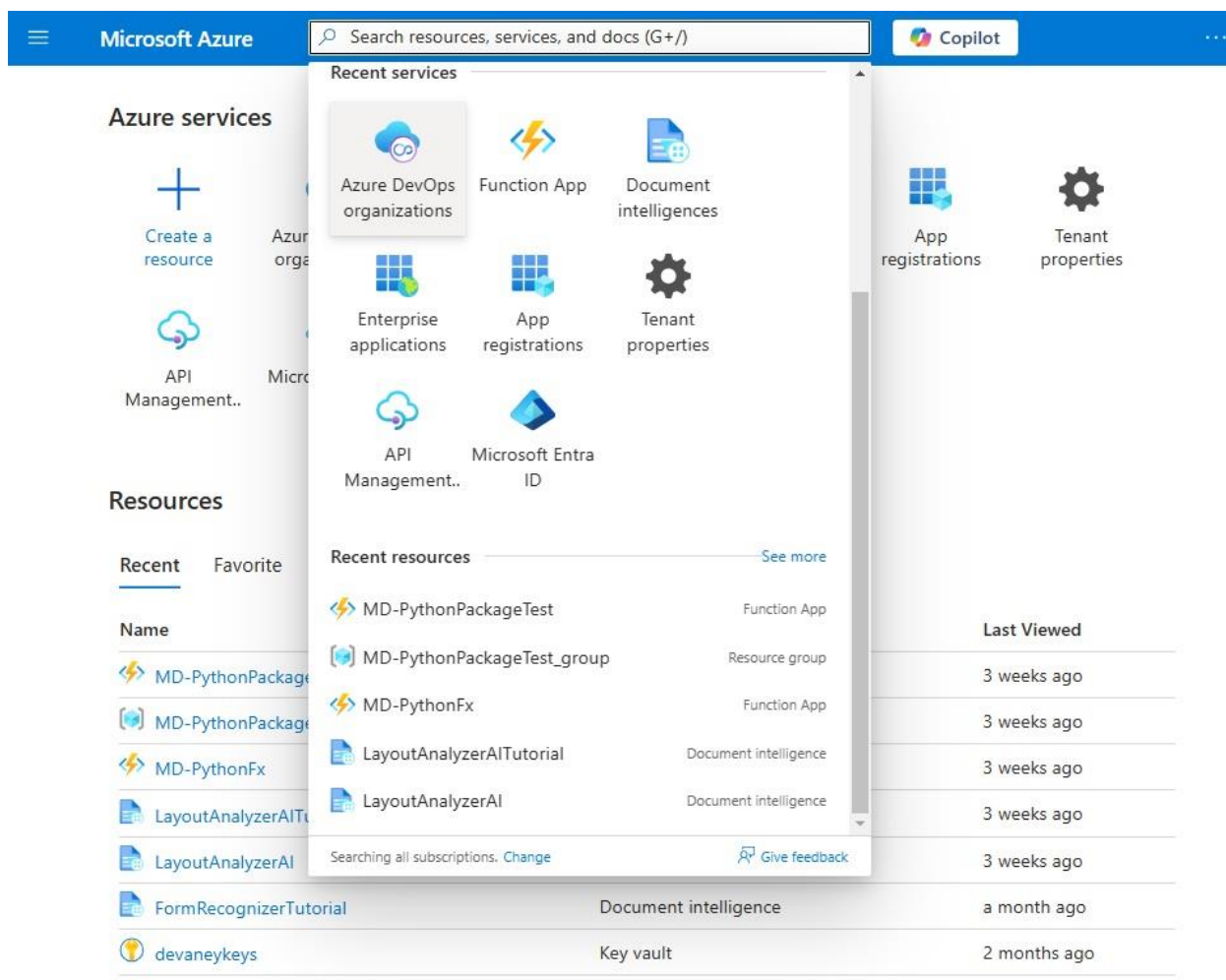
44

Start A New Azure DevOps Project

An Azure DevOps project must be setup to take advantage of Power Platform Git Integration. Then we can add an Azure Repo to the DevOps project to store code in Git. An Azure DevOps license is free for the first 5 users in an organization and only takes a few minutes to setup.

Create A New Azure DevOps Organization

Azure DevOps can be accessed through [the Azure Portal](#). Open the Azure Portal and search for Azure DevOps organizations.



An Azure DevOps organization is needed to create Azure DevOps projects. Press the My Azure Dev Ops Organizations page.

Microsoft Azure

Search resources, services, and docs (G+/)

Copilot

Home >

Azure DevOps

Azure DevOps

Plan smarter, collaborate better, and ship faster with a set of modern dev services

My Azure DevOps Organizations

Get started using Azure DevOps

Billing management for Azure DevOps

Give feedback

Tell us about your experience with the Azure DevOps page



Press the Create new organization button or select an existing organization.

Microsoft

Matthew DevaneySign out



Matthew Devaney
Edit profile
md@matthewdevaney.com

matthewdevaney.com

Canada
md@matthewdevaney.com

Visual Studio Dev Essentials
Get everything you need to build

Azure DevOps Organizations

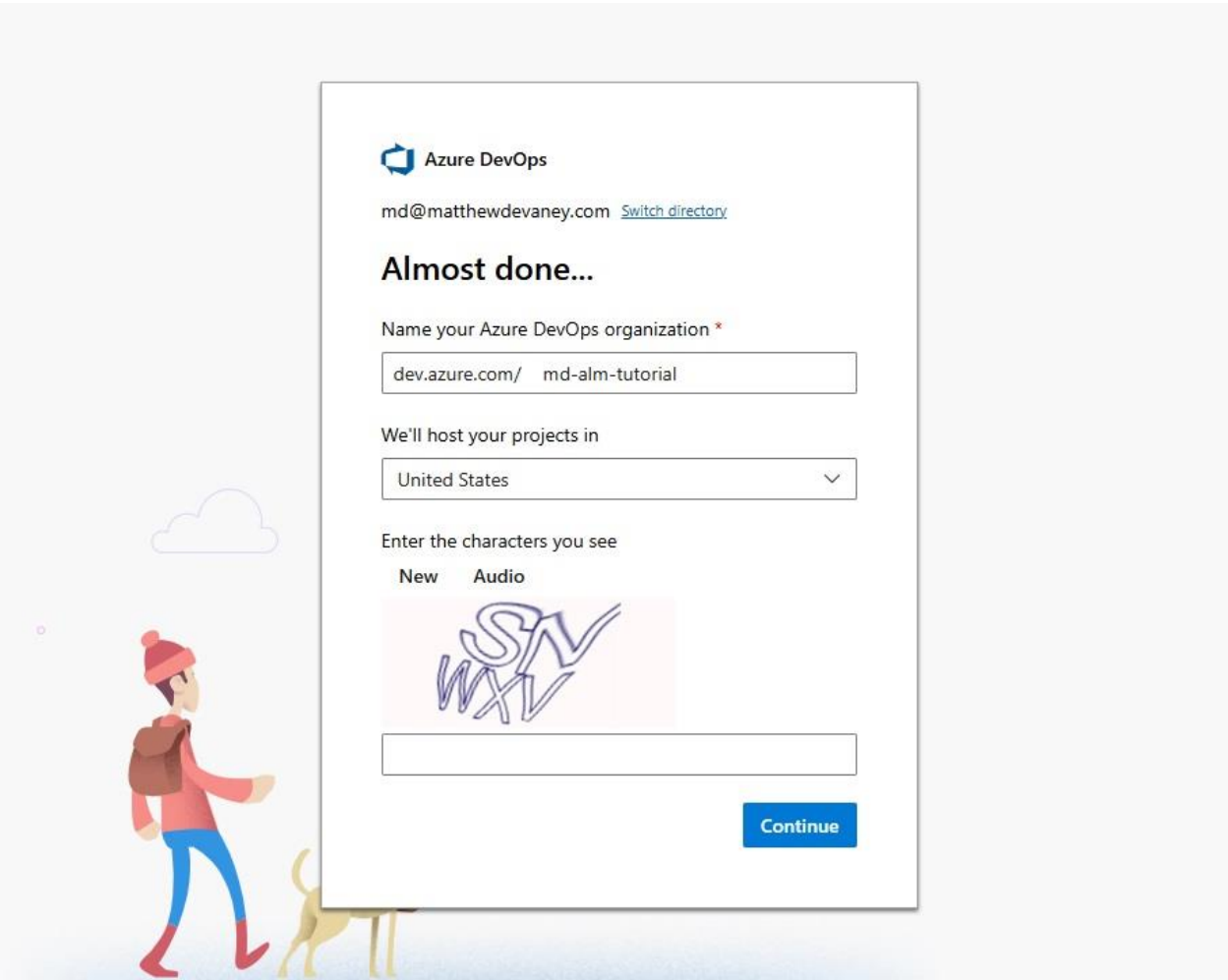
Create new organization

dev.azure.com/md0453 (Owner)

Projects	Actions
 Power Apps Co-Authoring	Open in Visual Studio
 Device Management	
 ALM Demo	
 alm-sandbox	
 PortlandALMDemo	

New project

If starting a new organization, select a URL and a host country. Then press Continue.



Setup An Azure DevOps Project

On the Azure DevOps home page, select your organization once again, and fill-in the create a project menu. Choose the name ALM Tutorial Project. Set visibility to Private and version control to Git. Then press the Create project button.

Azure DevOps

Search

md-alm-tutorial

md0453

[New organization](#)

Create a project to get started

Project name *

ALM Tutorial Project

Description

Visibility

☐ Public
Anyone on the internet can view the project. Certain features like TFVC are not supported.

☒ Private
Only people you give access to will be able to view this project.

Public projects are disabled for your organization. You can turn on public visibility with [organization policies](#).

^ Advanced

Version control ⓘ

Git

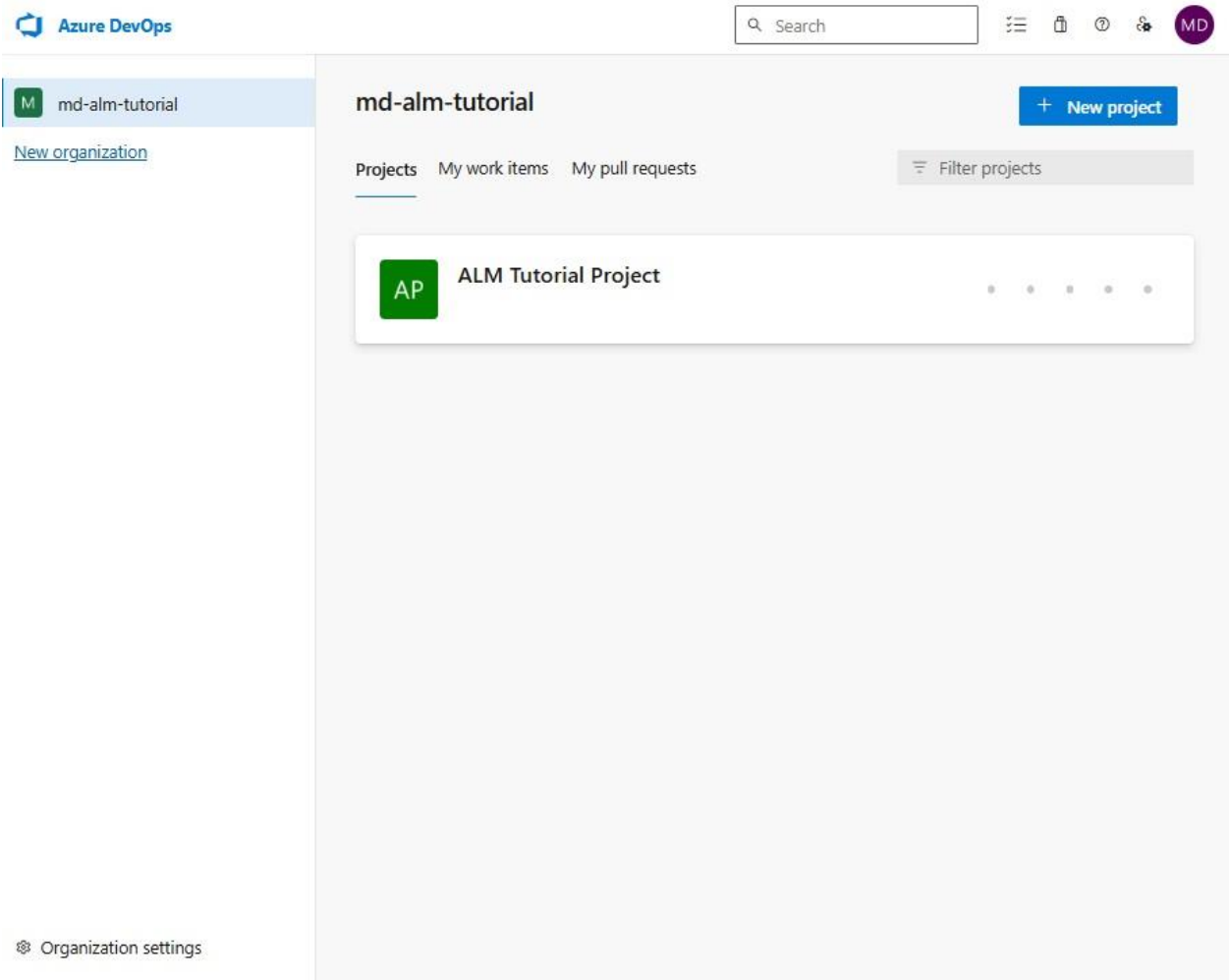
Work item process ⓘ

Scrum

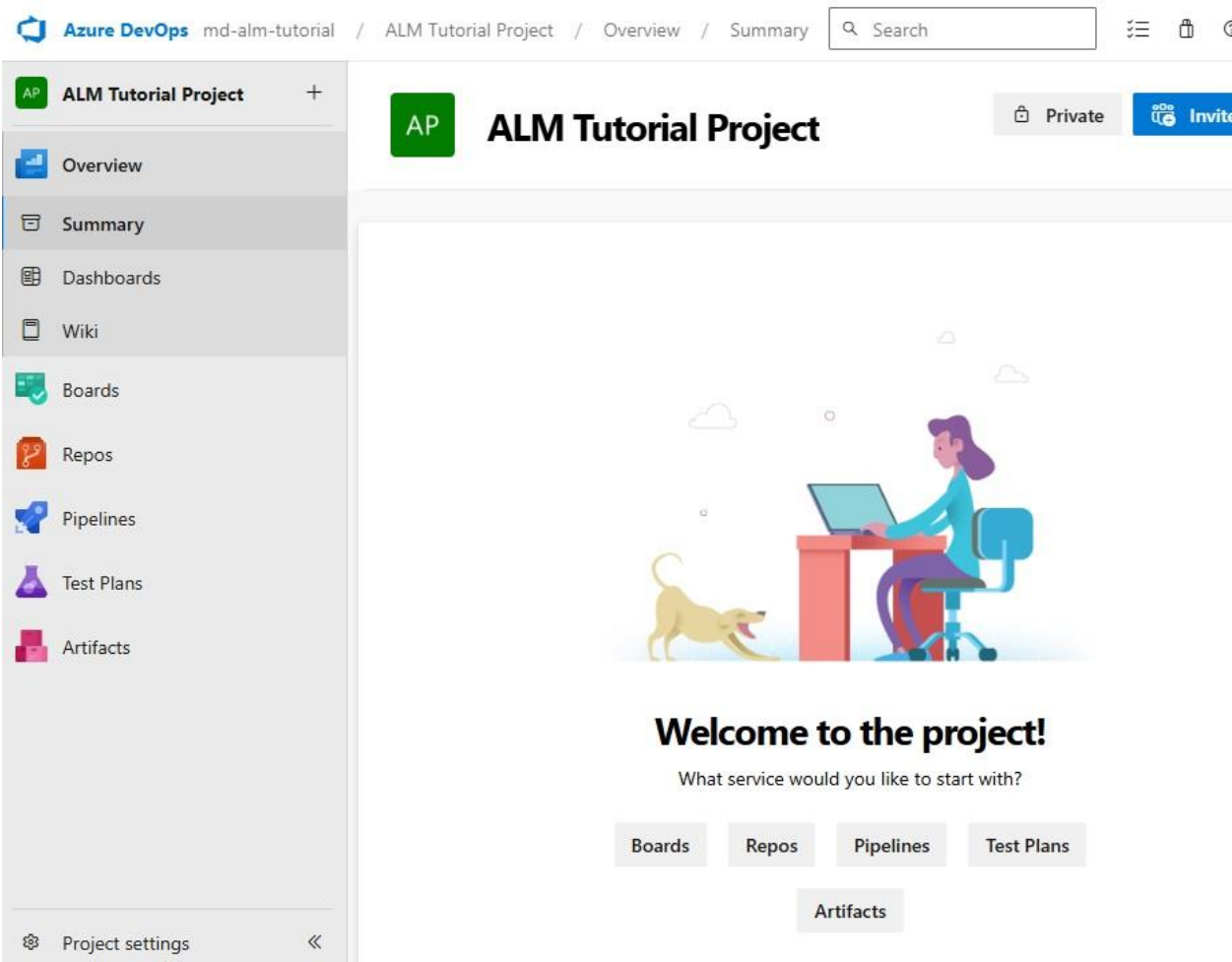
+ Create project

Organization settings

The new project will appear inside the selected organization. Open the project.



The project landing screen appears with a welcome message.



Configure Development And Target Environments

Power Platform Pipelines must be connected to a development environment and one or more target environments. A development environment is where makers develop solutions. Target environments are where solutions get deployed to. For example, the test and production environments. Before creating the pipeline we will add environments to the pipeline configuration app.

Open The Deployment Pipeline Configuration App

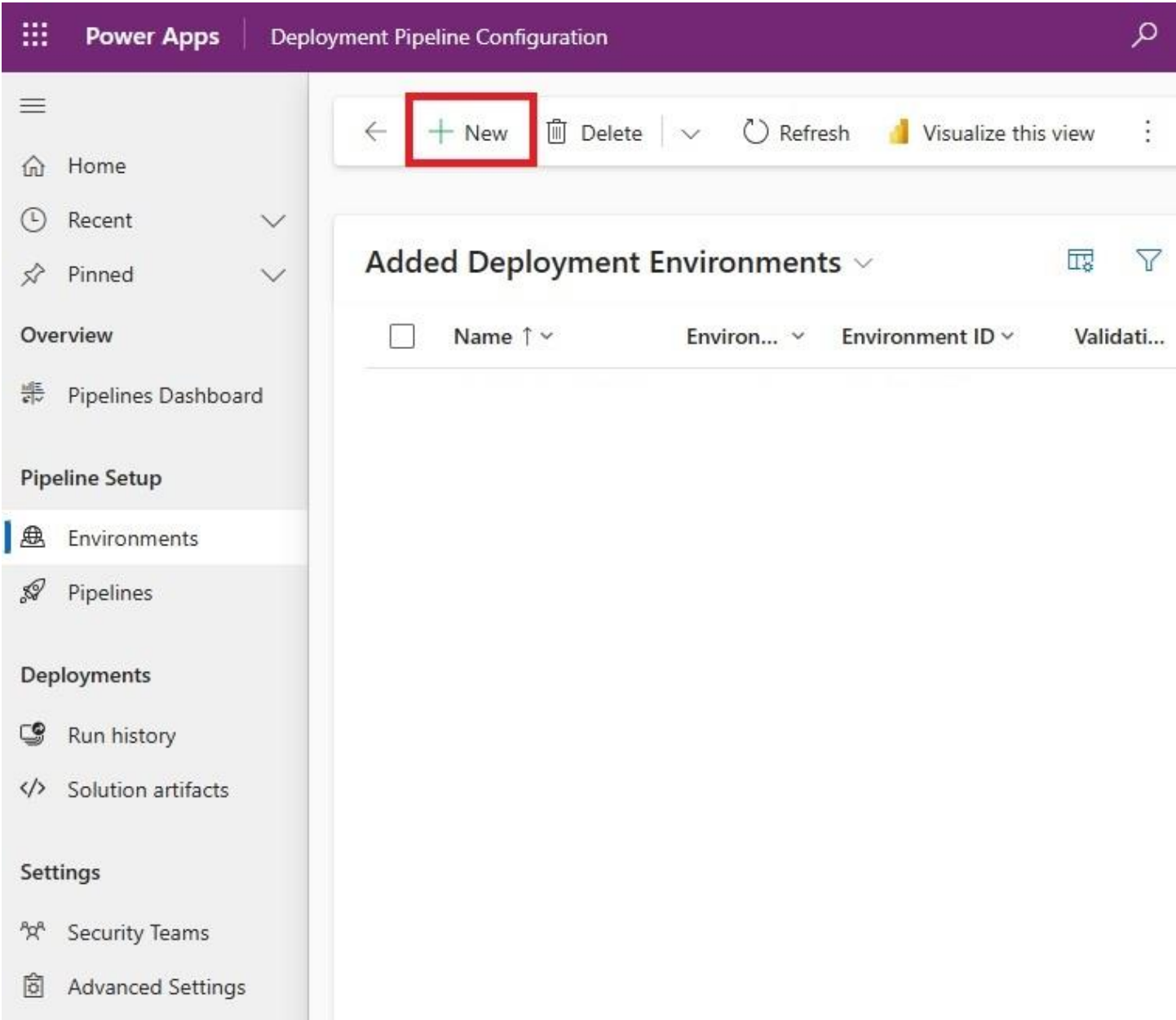
Go to the Power Platform Pipelines host environment and open the Deployment Pipelines Configuration app.

The screenshot shows the Power Apps interface. The top navigation bar includes 'Power Apps', a search bar, and 'Environment PP Pipelines Host'. The left sidebar contains navigation options: Home, Create, Learn, Apps (selected), Solutions, Flows, Tables, More, and Power Platform. The main area is titled 'Apps' and features four templates: 'Start with Copilot', 'Start with data', 'Start with a page design', and 'Start with an app template'. Below these templates are filters for 'My apps', 'Shared with me', and 'All'. A table lists the apps, with 'Deployment Pipeline Configuration' highlighted.

Name	Modified	Owner
Deployment Pipeline Configuration	1 minute ago	Matthew Deva

Add The Development Environment To The Pipelines App

First, we will add a development environment. Go to the Environments page and press the New button.



Fill-in the environment details then press the save button. The validation status will switch to pending.

- Name – ALM Tutorial (Dev)
- Owner – set to your own user account
- Environment Type – Development Environment
- Environment ID – the environment ID from the environment’s developer resources menu.

Power Apps

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Advanced Settings

←

↗

Save

Save & Close

+ New

Force Link

Flow

New Deployment Environment - Unsaved

General

Name

*

ALM Tutorial (Dev)

Owner

*

MD

Matthew Devaney (Offline) X

Environment Type

*

Development Environment

Environment ID

*

72f71a25-aa46-ed33-a99e-80d5ed52f10c

Validation Status

🔒

Pending

Error Message

🔒

Description

After a few moments pass press the refresh button. The validation status will be updated to success.

Power Apps

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Save Save & Close New Force Link Deactiva

ALM Tutorial (Dev) - Saved

Deployment Environment

General Related

Name

ALM Tutorial (Dev)

Owner

Matthew Devaney (Offline)

Environment Type

Development Environment

Environment ID

72f71a25-aa46-ed33-a99e-80d5ed52f10c

Validation Status

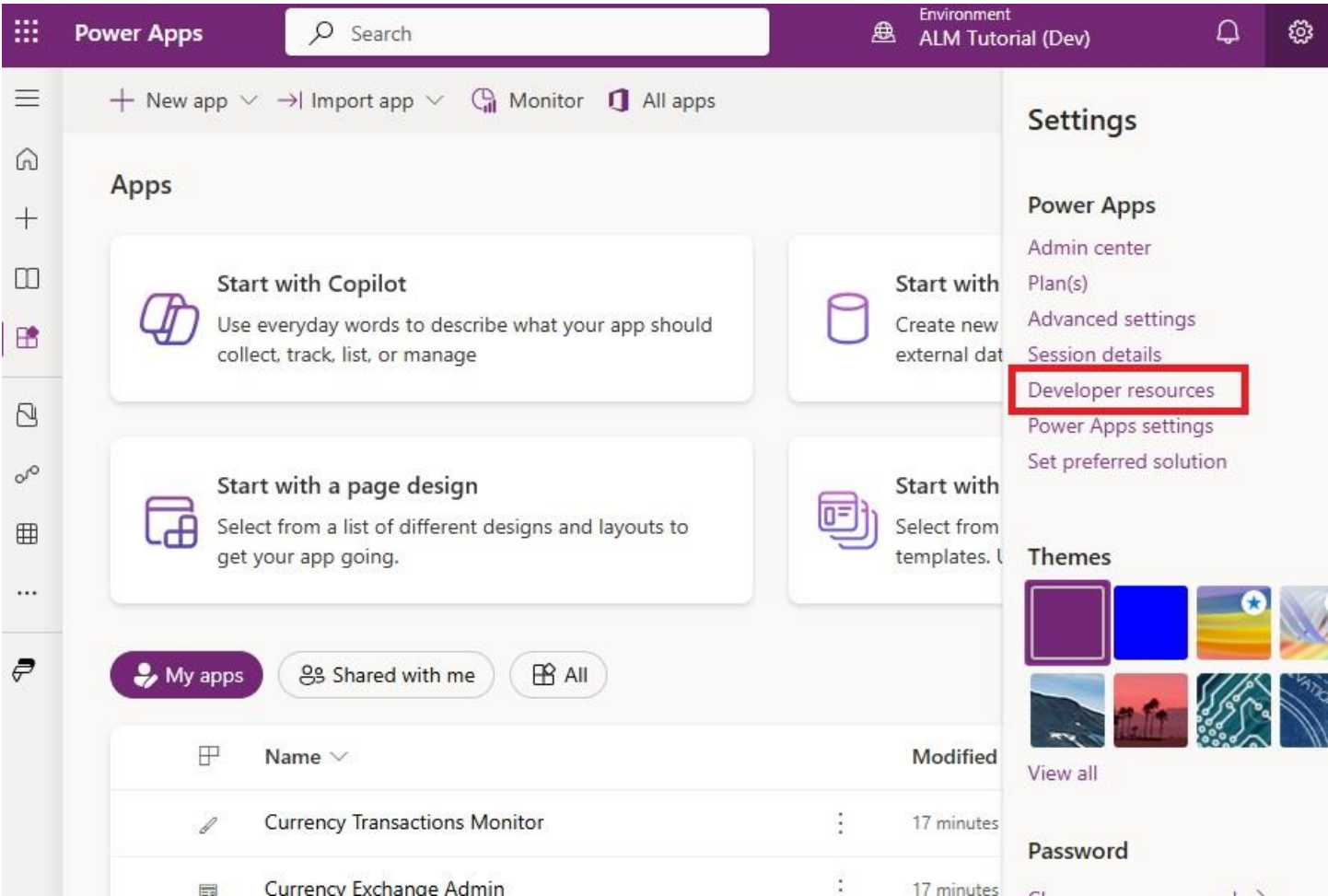
Success

Error Message

Description

Get The Environment ID From Developer Resources

The environment id is necessary to setup the developer and target environments in Power Platform Pipelines. Go to any environment, press the settings button and select developer resources.



The unique identifier of the environment is shown as Environment ID.

The screenshot shows the Dynamics 365 Developer portal interface. At the top, a purple header bar contains the text 'Environment ALM Tutorial (Dev)' and several icons: a globe, a bell, a gear, a question mark, and a circular profile icon labeled 'MD'. On the left, a sidebar menu is partially visible with items like 'Monitor', 'be what you', 'designs and', 'All', 'lonitor', and 'iin'. The main content area is titled 'Developer resources' with a close button (X) in the top right corner. It lists several environment details: 'Environment unique name' (unq148b33c696e9ef11933d000d3a106), 'Environment ID' (72f71a25-aa46-ed33-a99e-80d5ed52f10c, highlighted with a red box), 'Organization ID' (148b33c6-96e9-ef11-933d-000d3a10650a), 'Web API endpoint' (https://devaney-almtutorial-dev.api.crm.dynamics.com/api/data/v9.2) with a 'Learn more' link, 'Discovery endpoint' (https://globaldisco.crm.dynamics.com/api/discovery/v2.0/Instances) with a 'Learn more' link, and a 'Documentation' section with links to 'Developer center', 'Official NuGet feed', 'PowerShell Gallery feed', and 'Samples'.

Environment
ALM Tutorial (Dev)

Monitor

Developer resources

Environment unique name
unq148b33c696e9ef11933d000d3a106

Environment ID
72f71a25-aa46-ed33-a99e-80d5ed52f10c

Organization ID
148b33c6-96e9-ef11-933d-000d3a10650a

Web API endpoint
https://devaney-almtutorial-dev.api.crm.dynamics.com/api/data/v9.2
[Learn more](#)

Discovery endpoint
https://globaldisco.crm.dynamics.com/api/discovery/v2.0/Instances
[Learn more](#)

Documentation
[Developer center](#)
[Official NuGet feed](#)
[PowerShell Gallery feed](#)
[Samples](#)

Setup The Test Environment As A Target Environment

Continue on with adding the Test Environment as a target environment. Target environments are environments where managed solutions are deployed to.

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New

Delete

Refresh

Visualize this view

Added Deployment Environments

☐	Name ↑	Environ...	Environment ID	Validati...
☐	ALM Tutorial (Dev)	Develo...	72f71a25-aa46-e...	Success

Add the environment details then press the save button. Wait for the environment to pass validation.

- Name – ALM Tutorial (Test)
- Owner – set to your own user account
- Environment Type – Target Environment
- Environment ID – the environment ID from the environment’s developer resources menu.

The screenshot displays the 'Power Apps' interface for 'Deployment Pipeline Configuration'. The left sidebar contains navigation options: Home, Recent, Pinned, Overview, Pipelines Dashboard, Pipeline Setup, Environments (selected), Pipelines, Deployments, Run history, Solution artifacts, Settings, Security Teams, and Advanced Settings. The main area shows the configuration for 'ALM Tutorial (Test) - Saved' under the 'General' tab. The configuration fields are as follows:

Field	Value
Name	ALM Tutorial (Test)
Owner	Matthew Devaney (Offline)
Environment Type	Target Environment
Environment ID	55087040-c33e-ef70-a0d5-704279d55d00
Validation Status	Success
Error Message	
Description	---

Configure The Prod Environment As A Target Environment

Add a new environment one more time. Supply the environment details then press the save button. Wait for the environment to pass validation.

- Name – ALM Tutorial (Prod)
- Owner – set to your own user account
- Environment Type – Target Environment
- Environment ID – the environment ID from the environment’s developer resources menu.

The screenshot displays the 'Power Apps' interface for 'Deployment Pipeline Configuration'. The left sidebar contains navigation options: Home, Recent, Pinned, Overview, Pipelines Dashboard, Pipeline Setup, Environments (selected), Pipelines, Deployments, Run history, Solution artifacts, Settings, Security Teams, and Advanced Settings. The main panel shows the configuration for 'ALM Tutorial (Prod) - Saved' under the 'General' tab. The configuration details are as follows:

Field	Value
Name	ALM Tutorial (Prod)
Owner	Matthew Devaney (Offline)
Environment Type	Target Environment
Environment ID	45cfb346-bddb-efa5-afb4-71f8a4c94c00
Validation Status	Success
Error Message	
Description	---

All environments have now been added to the Deployment Pipeline Configuration app. The Developer environment is included as development environment. And the Test & Production environments are target environments.

Power Apps

Deployment Pipeline Configuration

+

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New

Delete

Refresh

Visualize this view

Added Deployment Environments*

Filter

☐	Name ↑	Environment Type	Environment ID
☐	ALM Tutorial (Dev)	Development Environme...	72f71a25-aa46-ed33-a99e-80d
☐	ALM Tutorial (Prod)	Target Environment	45cfb346-bddb-efa5-afb4-71f8
☐	ALM Tutorial (Test)	Target Environment	55087040-c33e-ef70-a0d5-704

Configure Development And Target Environments

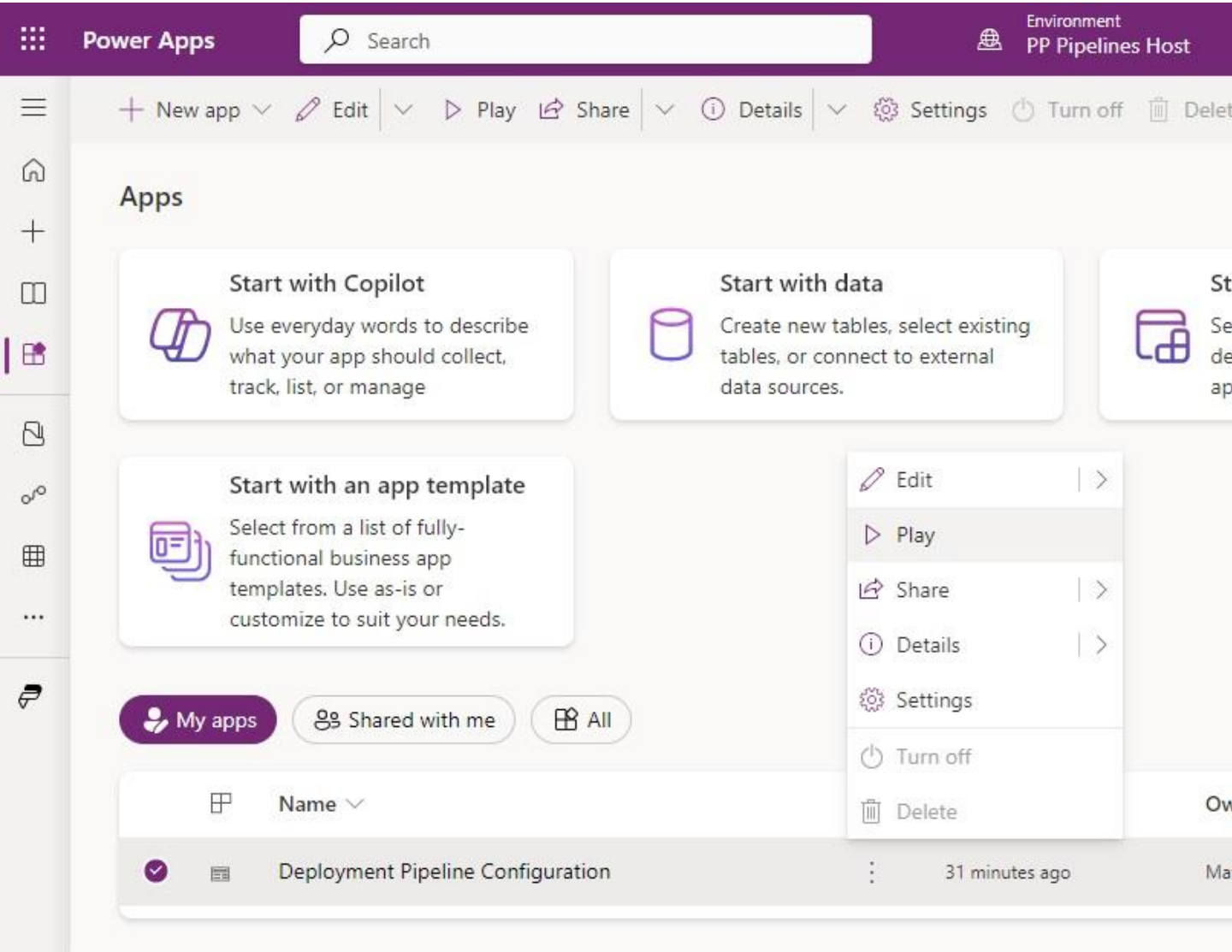
61

Create A Power Platform Pipeline

Power Platform Pipelines move solutions from a development environment to a target environment such as the test environment or the production environment. They ensure deployments to the correct environment are performed in a consistent manner. Configuration of components such as connection references and environment variables can be controlled through deployment settings. And a copy of the solution is stored for easy redeployment if needed.

Create A New Power Platform Pipeline

Go to the pipelines host environment and open the deployment pipeline configuration model-driven app.



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+ New

🗑 Delete

⌵

🔄 Refresh

📊 Visualize this view

⋮

Share

Active Deployment Pipelines

🏠

🔍

🔍 Filter by keyword

Name ↑

Description

Modifie...

Owner

C

We didn't find anything to show here

Rows: 0

Name the pipeline ALM Tutorial Pipeline. Allow redeployments of older versions. Then save the pipeline.

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📁

Save

Save & Close

+ New

Flow

New Deployment Pipeline - Unsaved

GeneralDeployment stagesRun history

Pipeline Details

Name

* ALM Tutorial Pipeline

AI deployment notes

* Enabled

Description

Allow this pipeline to generate deployment notes using AI. [Learn more](#)

Owner

*

MDMatthew Devar

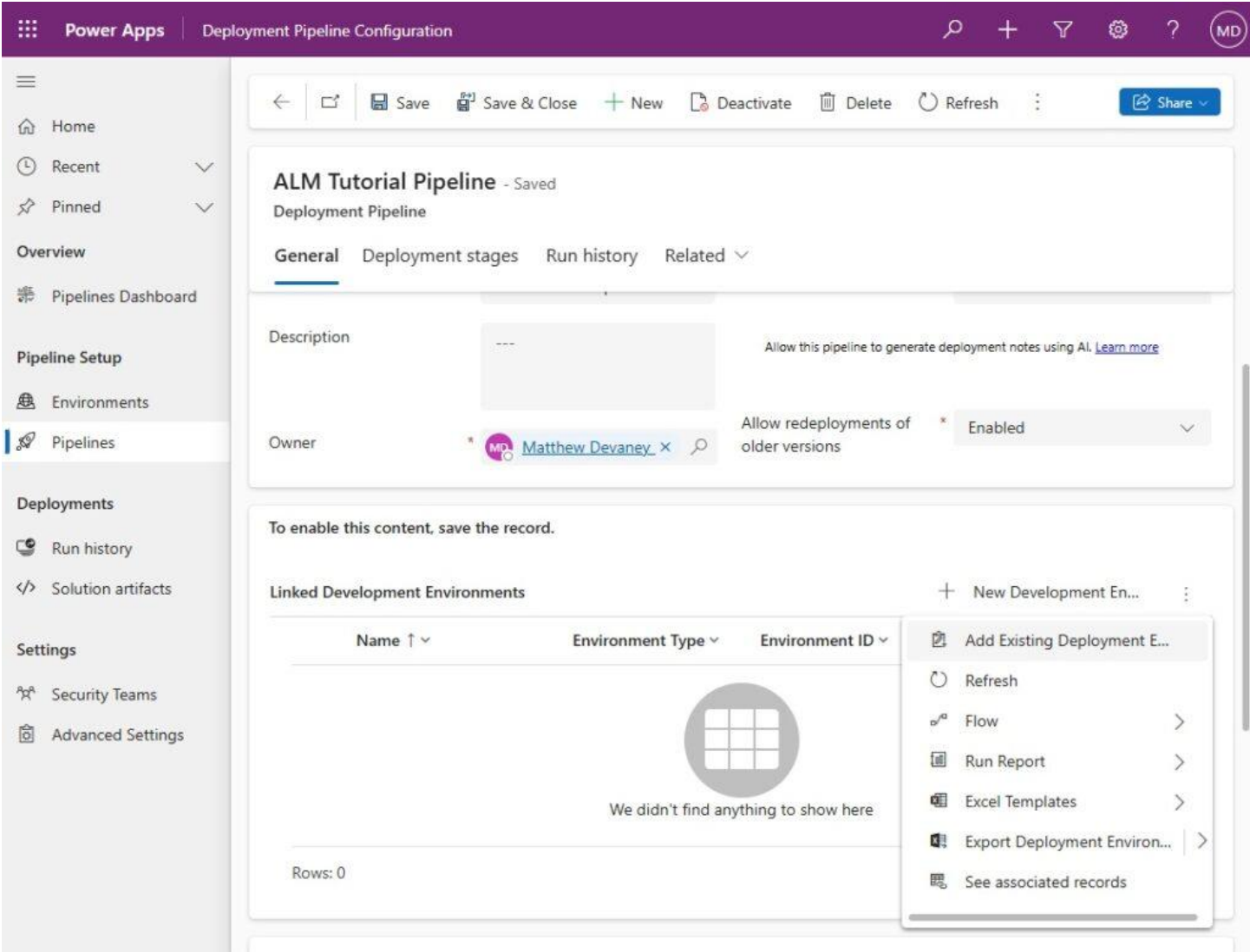
Allow redeployments of older versions

* Enabled

To enable this content, save the record.

Link A Development Environment To The Power Platform Pipeline

We want the pipeline to export solutions form the development environment and import them into the test and production environments. Add an existing development environment to the pipeline.



Select the ALM Tutorial (Dev) development environment to the pipeline. We created this record in a previous module of the tutorial.

Lookup Records

Select record

ALM Tutorial (Dev) ×

Add more records

Recent records

All records

ALM Tutorial (Prod)

ALM Tutorial (Test)

+ New Record

Add

Cancel

The ALM Tutorial (Dev) environment now appears in the linked development environments subgrid.

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SaveSave & CloseNewDeactivateDeleteShare

ALM Tutorial Pipeline - Saved

Deployment Pipeline

GeneralDeployment stagesRun historyRelated

Description

Allow this pipeline to generate deployment notes using AI. [Learn more](#)

Owner

Matthew Devane

Allow redeployments of older versions

Enabled

To enable this content, save the record.

Linked Development Environments

+New Development En...

Name	Environment Type	Environment ID	Validation Status
☐ ALM Tutorial (Dev)	Development En...	72f71a25-aa46-ed33-a...	Success

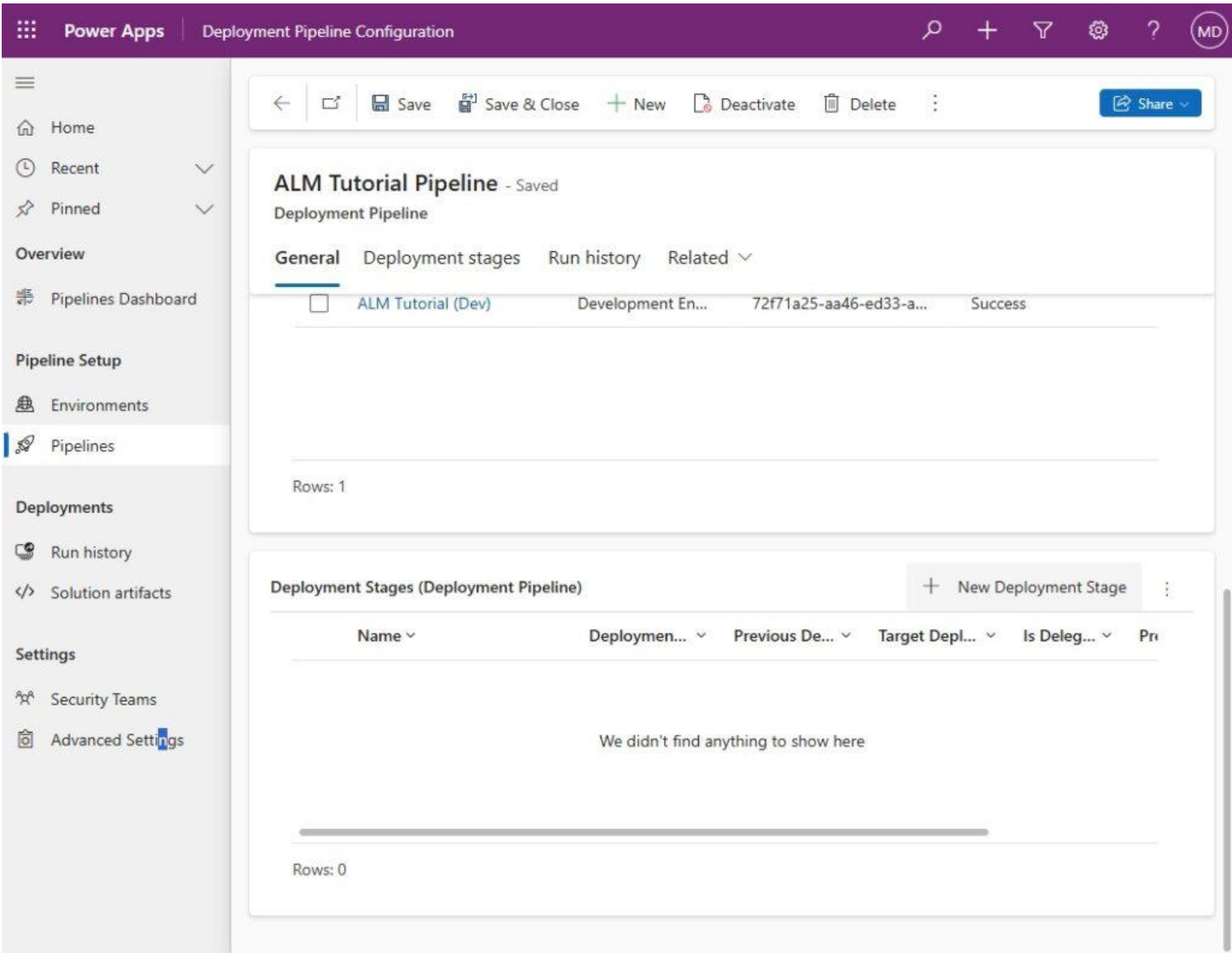
Rows: 1

Create A Power Platform Pipeline

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Add A Test Environment Deployment Stage To The Pipeline

The test environment is the first target environment we will add to the Power Platform pipeline. Look for the deployment stages subgrid and press the new deployment stage button.



Setup the test environment deployment stage as follows. Then save and close the stage.

- Name – ALM Tutorial Deployment Stage (Test)
- Owner – your Microsoft 365 account
- Deployment Pipeline – ALM Tutorial Pipeline
- Target Deployment Environment ID – ALM Tutorial (Test)

The screenshot shows the 'Power Apps' interface with the 'Deployment Pipeline Configuration' tab selected. The left sidebar contains navigation options: Home, Recent, Pinned, Overview, Pipelines Dashboard, Pipeline Setup (Environments, Pipelines), Deployments (Run history, Solution artifacts), and Settings (Security Teams, Advanced Settings). The main area displays the 'New Deployment Stage - Unsaved' form. The form has a 'General' tab and several fields: Name (ALM Tutorial Pipeline), Owner (Matthew Devaney (Offline)), Description (empty), Name (ALM Tutorial Deployment Stage (Test)), Owner (Matthew Devaney (Offline)), Description (empty), Deployment Pipeline (ALM Tutorial Pipeline), Previous Deployment Stage (empty), Target Deployment Environment ID (ALM Tutorial (Test)), Pre-Export Step Required (checkbox), Pre-Deployment Step Required (checkbox), and Is Delegated Deployment (checkbox).

Power Apps | Deployment Pipeline Configuration

← ↻ Save Save & Close + New Flow

New Deployment Stage - Unsaved

General

Name * ALM Tutorial Pipeline

Owner * Matthew Devaney (Offline)

Description

Name * ALM Tutorial Deployment Stage (Test)

Owner * Matthew Devaney (Offline)

Description

Deployment Pipeline * ALM Tutorial Pipeline

Previous Deployment Stage

Target Deployment Environment ID * ALM Tutorial (Test)

Pre-Export Step Required ☐

Pre-Deployment Step Required ☐

Is Delegated Deployment ☐

The test environment deployment stage appears in the subgrid.

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ALM Tutorial Pipeline - Saved

Deployment Pipeline

General

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Related

Linked Development Environments

	Name	Environment	Environment ID	Validation Status
☐	ALM Tutorial (Dev)	Developmen...	72f71a25-aa46-ed3...	Success

Rows: 1

Deployment Stages (Deployment Pipeline)

+ New Deployment Stage

	Name	Deployment Pipeline	Previous De...	Ta
☐	ALM Tutorial Deployment Stage (Test)	ALM Tutorial Pipeline		AI

Rows: 1

Configure A Production Environment Deployment Stage

The final deployment stage of the pipeline is the production environment. Add another new deployment stage to the pipeline with the following details. Then save and close the stage.

- Name – ALM Tutorial Deployment Stage (Prod)
- Owner – your Microsoft 365 account
- Deployment Pipeline – ALM Tutorial Pipeline
- Previous Deployment Stage – ALM Tutorial (Test)
- Target Deployment Environment ID – ALM Tutorial (Prod)

The screenshot shows the 'Power Apps | Deployment Pipeline Configuration' interface. The left sidebar contains navigation links: Home, Recent, Pinned, Overview, Pipelines Dashboard, Pipeline Setup (Environments, Pipelines), Deployments (Run history, Solution artifacts), and Settings (Security Teams, Advanced Settings). The main area is titled 'New Deployment Stage - Unsaved' and has a 'General' tab. The form contains the following fields:

- Name:** * ALM Tutorial Pipeline
- Owner:** * Matthew Devaney (Offline)
- Description:**
- Name:** * ALM Tutorial Deployment Stage (Prod)
- Owner:** * Matthew Devaney (Offline)
- Description:** ---
- Deployment Pipeline:** * ALM Tutorial Pipeline
- Previous Deployment Stage:** ALM Tutorial Deployment Stage (Test)
- Target Deployment Environment ID:** * ALM Tutorial (Prod)
- Pre-Export Step Required:** ☐
- Pre-Deployment Step Required:** ☐
- Is Delegated Deployment:** ☐

The production environment appears in the deployment stages subgrid.

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Save & Close

New

Deactivate

Delete

Share

ALM Tutorial Pipeline

- Saved

Deployment Pipeline

General

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Related

Linked Development Environments

Name

Environment

Environment ID

Validation Status

ALM Tutorial (Dev)

Developmen...

72f71a25-aa46-ed3...

Success

Rows: 1

Deployment Stages (Deployment Pipeline)

New Deployment Stage

Name

Deployment Pipeline

Previous De...

Ta

ALM Tutorial Deployment Stage (Test)

ALM Tutorial Pipeline

AL

ALM Tutorial Deployment Stage (Prod)

ALM Tutorial Pipeline

ALM Tutorial ...

AL

Rows: 2

Create A Power Platform Pipeline

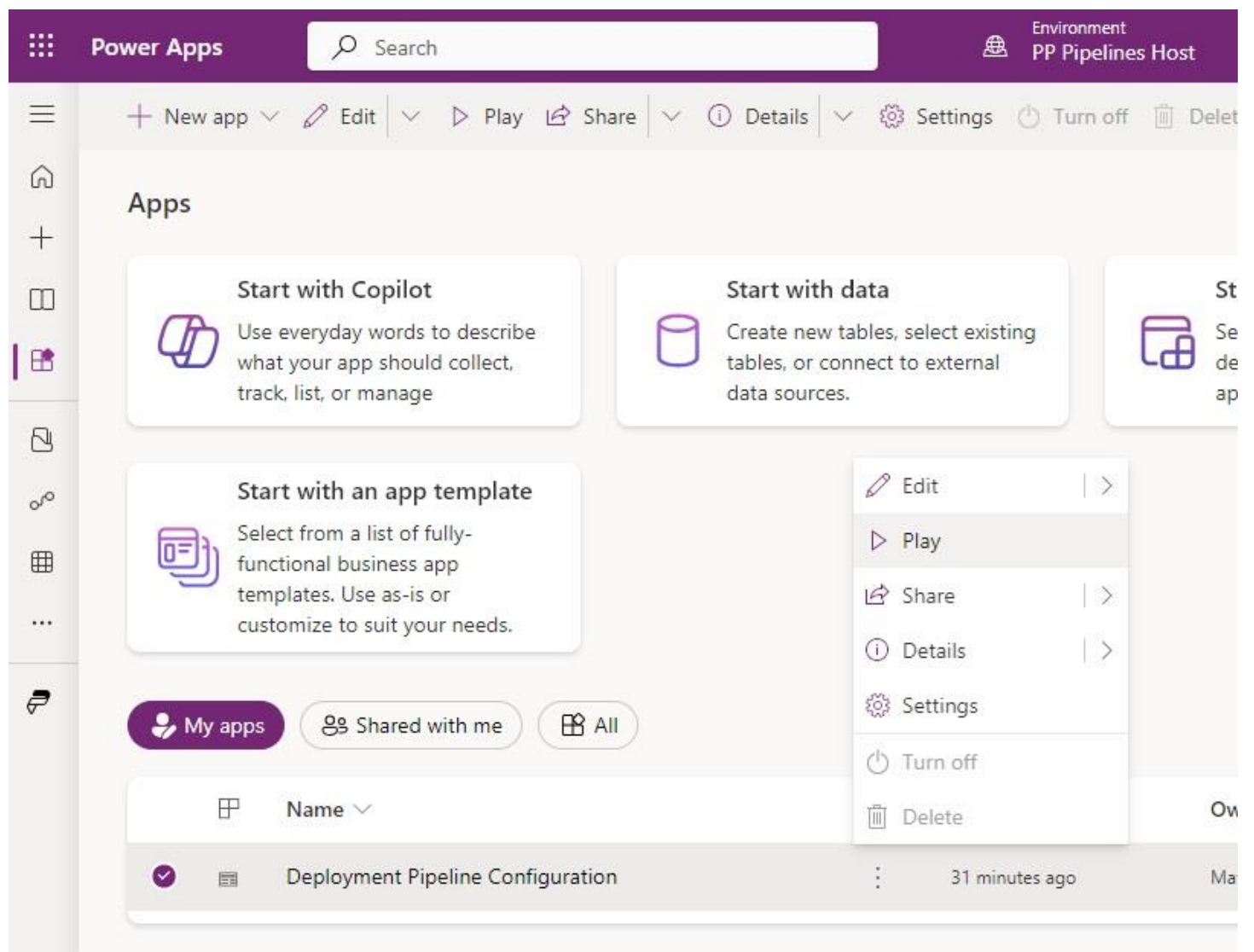
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Assign Power Platform Pipelines Security Roles

A Power Platform Pipelines security role is required to create or use Power Platform pipelines if you are not the system administrator. We can assign a user to the deployment pipeline administrators role or the deployment pipeline users role indirectly through security teams. Security teams are found in the pipelines model-driven app.

Open The Pipelines Model-Driven App

Go to the pipelines host environment and open the deployment pipeline configuration model-driven app.



Assign A User To The Deployment Pipeline Administrator Role

The deployment pipeline administrator role gives users access the to deployment pipeline configuration app and the ability to create & edit pipelines in the host environment. To assign a new pipelines administrator, open the security teams page and select deployment pipeline administrators.

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Show Chart

Delete

Email a Link

Flow

Share

Pipeline Security Teams*

Filter by keyword

Team Name	Description
☒ Deployment Pipeline Administrators	Deployment Pipeline Administrators have full access to all pipelines.
☐ Deployment Pipeline Makers	Deployment Pipeline Makers can create and consume a personal pipeline.
☐ Deployment Pipeline Users	Deployment Pipeline Users may trigger a pipeline that is shared with them

Rows: 3 Selected: 1

Press the add existing user button in the Team members subgrid.

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Save Save & Close New Connect Delete Share

Deployment Pipeline Administrators - Saved

<Deployment Pipeline Administrators> Default Queue

General Related

Team Name *

Deployment Pipeline Administrators

Business Unit *

devaney-pipelines-host

Administrator *

SYSTEM

Team Type *

Owner

Object Id for a group

Description

Deployment Pipeline Administrators have full access to all pipelines.

Team members

Full Name

We didn't find any records.

Rows: 0

Add Existing User

Refresh

Flow

Run Report

Excel Templates

Export Users

See all records

Assign Power Platform Pipelines Security Roles

75

Choose the user you want to become a Power Platform pipelines administrator. Then press the Add button.

Lookup Records

×

Select record

MD

Matthew Devaney (Offline)

×

Add more records

🔍

← All

Users

No records found.

🔍

Advanced lookup

Add

Cancel

The user appears in the team members subgrid.

Power Apps

Deployment Pipeline Configuration

MD

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<

Save

Save & Close

+ New

Connect

Share

Deployment Pipeline Administrators

- Saved

Team

Team

<Deployment Pipeline Administrators>

Default Queue

General

Related

Team Name *

Deployment Pipeline Administrators

Business Unit *

devaney-pipelines-host

Administrator *

SYSTEM

Team Type *

Owner

Object Id for a group

Team members

Select all

MD

Matthew Devaney

devaney-pipelines-host

Add Users To The Deployment Pipeline Users Role

The deployment pipeline user role gives the user an ability to run a pipeline that is shared with them. To add a new pipelines user, go to the security teams page and select deployment pipeline users

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←

Delete

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Share

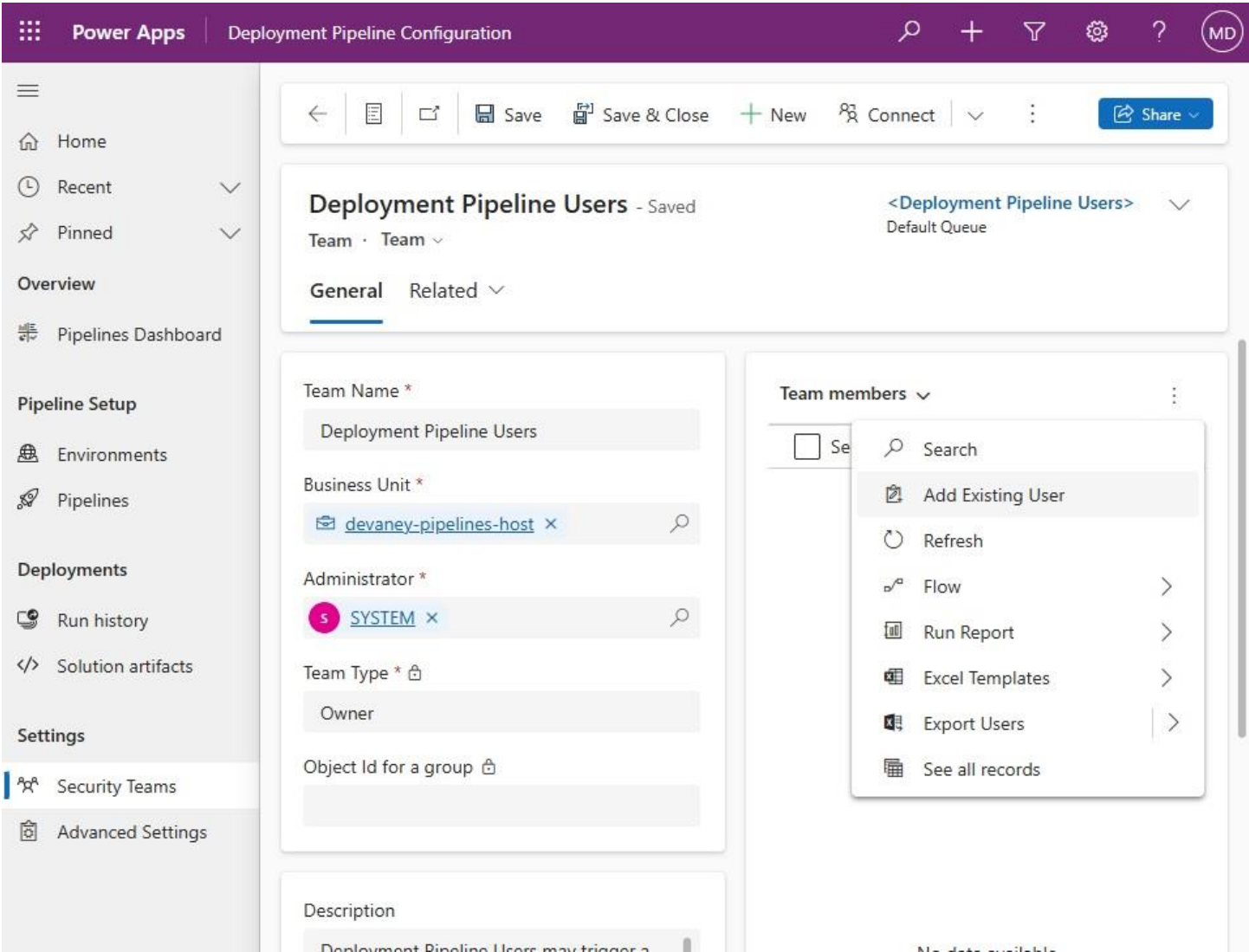
Pipeline Security Teams*

Filter by keyword

☒	Team Name	Description
☐	Deployment Pipeline Admin...	Deployment Pipeline Administrators have full access to all pipeline
☐	Deployment Pipeline Makers	Deployment Pipeline Makers can create and consume a personal p
☒	Deployment Pipeline Users	Deployment Pipeline Users may trigger a pipeline that is shared w

Rows: 3 Selected: 1

Look for the team members subgrid and press the add existing user button.



Choose one or more users to become Power Platform pipeline users. The press the add button.

Lookup Records

×

Select record

 David Johnson (Offline) ×

 Sarah Green ×

>

Add more records 

← All

Users

No records found.

 Advanced lookup

Add

Cancel

The Power Platform pipelines users now show up in the team members subgrid.

Power Apps

Deployment Pipeline Configuration

MD

Home

Recent

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Overview

Pipelines Dashboard

Pipeline Setup

Environments

Pipelines

Deployments

Run history

Solution artifacts

Settings

Security Teams

Advanced Settings

Deployment Pipeline Users - Saved

<Deployment Pipeline Users>

Team · Team

Default Queue

General

Related

Team Name

Deployment Pipeline Users

Business Unit

devaney-pipelines-host

Administrator

SYSTEM

Team Type

Owner

Object Id for a group

Description

Team members

Select all

DJ

David Johnson

devaney-pipelines-host

SG

Sarah Green

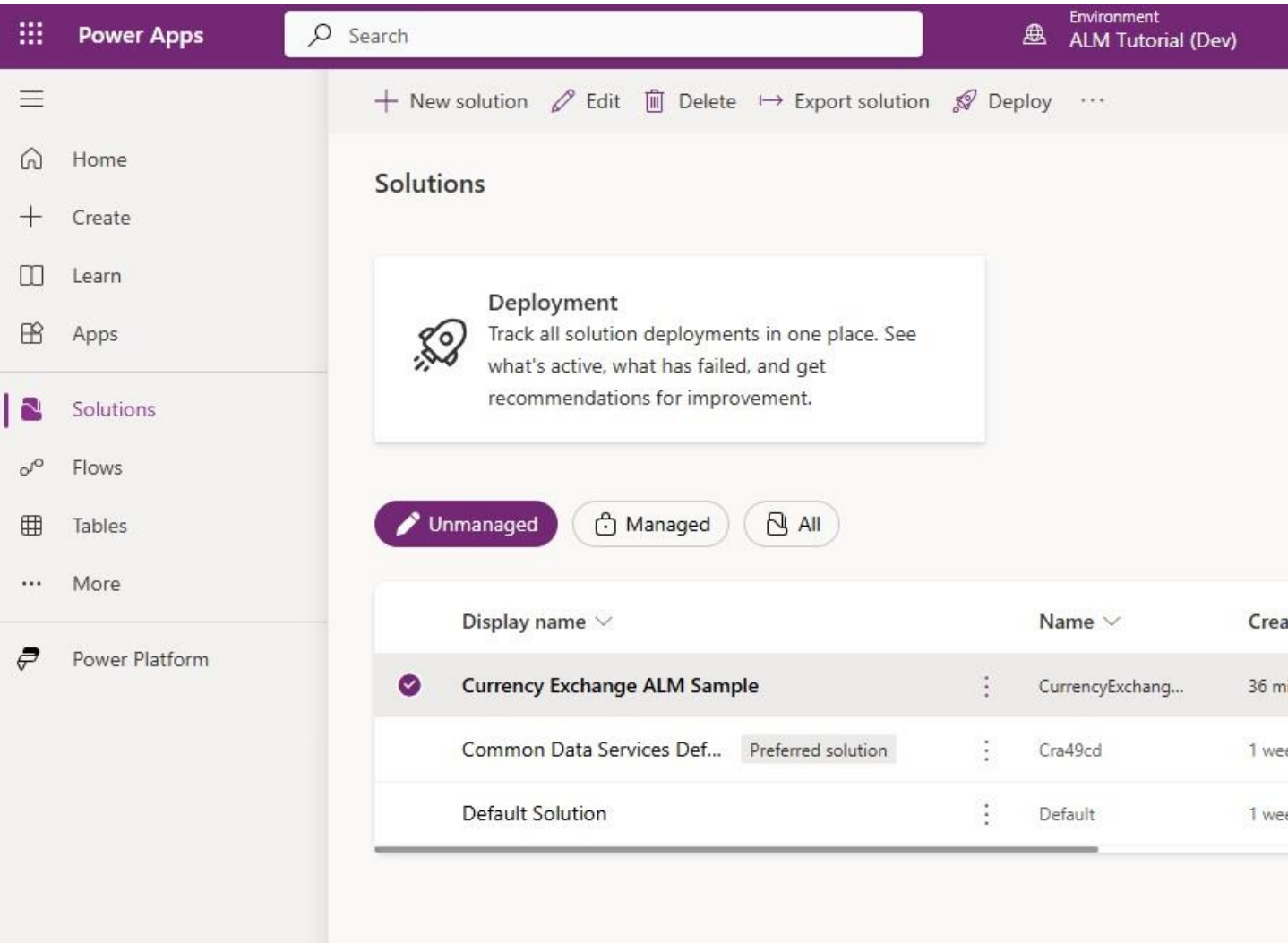
devaney-pipelines-host

Deploy A Solution To The Test Environment

After creating a Power Platform Pipeline, we can use it to import a solution from the development environment into the test environment, before pushing it to production. The test environment allows us to perform quality tests on the solution and validate the deployment while not blocking development. It is recommended to check solutions into git-based source control before moving them into test.

Deploy The Solution To The Test Environment With Pipelines

Go to the development environment and open the solution you want to move to the test environment.



Select the pipelines menu item from the left navigation.

Power Apps

Search

Environment
ALM Tutorial (Dev)

MD

Objects

Search

All (10)

Agents (0)

Apps (2)

Cards (0)

Cloud flows (1)

Connection references (2)

Environment variables (1)

Site maps (1)

Tables (3)

+ New

Add existing

Publish all customizations

Search

Currency Exchange ALM Sample > All

Display name ↑	Name	Type	Managed	Cus
Currency	md_currency	Table	No	Yes
Currency Exchange Ad...	md_CurrencyExc...	Site Map	No	Yes
Currency Exchange Ad...	md_CurrencyExc...	Model-Driven App	No	Yes
Currency Transaction	md_currencytran...	Table	No	Yes
Currency Transactions ...	md_currencytran...	Canvas App	No	Yes
Email Notifications Tar...	md_EmailNotific...	Environment Vari...	No	No
Exchange Rate	md_exchangerate	Table	No	Yes
Microsoft Dataverse C...	md_sharedcomm...	Connection Refer...	No	Yes
Office 365 Outlook	md_Office365Ou...	Connection Refer...	No	Yes
Send Currency Rates S...	Send Currency R...	Cloud Flow	No	Yes

On the Pipelines page, select the ALM Tutorial Pipeline. Then press the Deploy here button.

Power Apps

Search

Environment
ALM Tutorial (Dev)

Create pipelineAdd stageDelete pipelineRefreshManage pipelines

Pipelines

As your app advances, you'll start sharing it to be used in your organization. As it gets widely adopted, it's important to practice healthy application lifecycle management by building securely in an isolated location so the people using your app don't run into issues while you work on it. Once everything is ready, you'll move it to a more permanent place and share there. [Learn more](#)

Target Environments in pipelines must be enabled as Managed Environments. [Learn more](#)

Pipeline

ALM Tutorial Pipeline

Create new pipeline

DetailsRun history

Development

Securely test and verify in isolation. After you've got everything working properly, it's time to deploy. [Learn more](#)

Solution version
1.0.0.644

ALM Tutorial Deployment Stage (Test)

Go to this environment

ALM Tutorial (Test)

Deploy here

Choose to deploy the solution now, then press the Next button.

Deploying Solution

Destination

Summary

Select deployment target

Select the deployment pipeline you'd like to use. Each deployment pipeline has different stages, after you choose a pipeline - select the stage.

Pipeline

ALM Tutorial Pipeline

+ Create new pipeline

ALM Tutorial Deployment Stage (Test)

ALM Tutorial Deployment Stage (Prod)

Deployment schedule

Now

Later

An AI-generated description for this solution will be created when you click Next. [Learn more](#)

Back

Next

Close

Deploy A Solution To The Test Environment

85

Configure Connection References And Environment Variables

The sample solution contains two connection references used in a Power Automate flow. One for Dataverse and another for Outlook. Wait for the green checkmark to appear beside each connection reference. If it does not appear, press the three dots to troubleshoot the issue. Then press the Next button.

Deploying Solution

close

Destination

Connections

Environment variables

Summary

Connections

Re-establish connections to activate your solution.

2 updates needed

Sign in

These services use your credentials to sign into apps and create connections.
A green check means you're ready to go.

Microsoft Dataverse CurrencyExchange...

md_sharedcommondataserviceforapps_b...

✓ ...

Office 365 Outlook

md_Office365Outlook

Permissions

✓ ...

Back

Next

Close

Deploy A Solution To The Test Environment

86

The sample solution also contains an environment variable for Email Notifications. The email address in this environment variable will receive email. Set this value to the tester’s email address then press Next.

Deploying Solution

✓ Destination

✓ Connections

● Environment variables

○ Summary

Environment Variables

Enter information for each field, so your app works properly. You can edit your environment variables later.

1 updates needed

Email Notifications Target

test@matthewdevaney.com

Reset

Back

Next

Close

Write Deployment Notes To Explain Solution Updates

Each deployment will update a the solution in the target environment. To make it easier to track changes it is important to write good deployment notes. Explain the difference between each deployment as opposed to giving an overview of the entire solution. Do not use AI generated deployment notes.

Press the deploy button once finished.

Deploying Solution

✓ Destination

✓ Connections

✓ Environment variables

✓ Summary

Pipeline	Stage
ALM Tutorial Pipeline	ALM Tutorial Deployment Stage (Test)
Display name	Name
Currency Exchange ALM Sample	CurrencyExchangeALMSample
Publisher	Prefix
Matthew Devaney	md

Version *

1.0.0.645

Deployment notes

The Currency Exchange ALM Sample solution is designed to manage and monitor currency exchange transactions. It includes various components that facilitate administration, tracking, and communication of currency-related data. The solution provides a structured approach to handle currency information and automate notifications.

Key Components:

1. Site Map - Currency Exchange Admin: Organizes navigation within the solution, allowing users to easily access tables like Currency, Currency Transaction, and Exchange Rate.
2. Classic Canvas App - Currency Transactions Monitor: A user interface

X Clear

👍👎

ⓘ

AI-generated content can have mistakes. Make sure it's accurate and appropriate before using it. [Read terms](#)

Back

Deploy

Close

Wait For The Test Stage Deployment To Complete

Power Platform Pipelines will begin to the deploy the solution to the test environment. This operation takes a few minutes to complete.

Pipeline

ALM Tutorial Pipeline

+ Create new pipeline

Details

Run history

Development

Securely test and verify in isolation. After you've got everything working properly, it's time to deploy. [Learn more](#)

Solution version
1.0.0.644

ALM Tutorial Deployment Stage (Test)

ALM Tutorial (Test)

Go to this environment

Currently deploying version 1.0.0.645

Deploy here

ALM 1

ALM

When the deployment successfully completes the last installed version and the deployment date appear in the test stage.

Power Apps

Search

Environment
ALM Tutorial (Dev)

Create pipeline

Add stage

Delete pipeline

Refresh

Manage pipelines

Pipelines

As your app advances, you'll start sharing it to be used in your organization. As it gets widely adopted, it's important to practice healthy application lifecycle management by building securely in an isolated location so the people using your app don't run into issues while you work on it. Once everything is ready, you'll move it to a more permanent place and share there. [Learn more](#)

Target Environments in pipelines must be enabled as Managed Environments. [Learn more](#)

Pipeline

ALM Tutorial Pipeline

Create new pipeline

Details

Run history

Development

Securely test and verify in isolation. After you've got everything working properly, it's time to deploy. [Learn more](#)

Solution version
1.0.0.645

ALM Tutorial Deployment Stage (Test)

ALM Tutorial (Test)

Last successfully installed version
1.0.0.645

Last deployed
Feb 12, 2025 6:47 PM (an hour ago)

Deploy here

Review The Deployment To The Test Environment

Go to the test environment and open the solutions menu to check on the deployment of the sample solution. Look under the managed solutions tab.

Power Apps

Search

Environment
ALM Tutorial (Test)

Home

Create

Learn

Apps

Solutions

Flows

Tables

More

Power Platform

New solution

Import solution

Open AppSource

Solutions

Deployment

Track all solution deployments in one place. See what's active, what has failed, and get recommendations for improvement.

Unmanaged

Managed

All

Display name	Name	Created
Currency Exchange ALM Sample	CurrencyExchang...	54 minutes a...
Power Apps Checker Base	msdyn_PowerAp...	1 week ago
Power Apps Checker	msdyn_PowerAp...	1 week ago
Contextual Help Base	msdyn_Contextu...	1 week ago
Contextual Help	msdyn_Contextu...	1 week ago

It is recommended to verify the deployment by opening applications and automations to ensure they are working in the new environment.

Power Apps

Search

Objects

Search

All (9)

Agents (0)

Apps (1)

Cards (0)

Cloud flows (1)

Connection references (2)

Environment variables (1)

Site maps (1)

Tables (3)

+ New

Add existing

...

Search

You cannot directly edit the objects within a managed solution. If the managed properties for solution objects are to allow customization, you can edit them from another unmanaged solution.

Currency Exchange ALM Sample > All

Display name ↑	Name
Currency	md_currency
Currency Exchange Admin	md_CurrencyExchangeAdm
Currency Transaction	md_currencytransaction
Currency Transactions Monitor	md_currencytransactionsr
Email Notifications Target	md_EmailNotificationsTar
Exchange Rate	md_exchangerate
Microsoft Dataverse CurrencyExchangeRates-...	md_sharedcommondatas
Office 365 Outlook	md_Office365Outlook
Send Currency Rates Summary Email	Send Currency Rates Sum

View The Pipeline Run History

To look at the run history of the Power Platform Pipeline, go to the host environment and open the Deployment Pipeline Configuration app. Then go to the run history page and select a run.

Power Apps

Deployment Pipeline Configuration

Home

Recent

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Overview

Pipelines Dashboard

Pipeline Setup

Environments

Pipelines

Deployments

Run history

Solution artifacts

Settings

Security Teams

Advanced Settings

Show As

Show Chart

Edit

Activate

Deactivate

Run history*

Deployment Pipeline (...)

Name (Deployment Stage)

Start Time

ALM Tutorial Pipeline

ALM Tutorial Deployment St...

2/13/2025 12:4...

The run details page looks like this.

Power Apps

Deployment Pipeline Configuration

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ALM Tutorial Deployment Stage

(Test)_CurrencyExchangeALMSample_ddda7af8-006b-4483-8c4f-6209e4ce7adf - Saved

Deployment Stage Run

General

Operation details

Deployment settings

Deployment notes

Related

Stage Run Status

Succeeded

Development Deployment Environment ID

ALM Tutorial (Dev)

Deployment Stage

ALM Tutorial Depl...

Target Deployment Environment ID

ALM Tutorial (Test)

Operation

Deploy

Artifact ID

CurrencyExc...

Start Time

2/13/2025

12:42 AM

Artifact Name

CurrencyExchangeALM...

End Time

2/13/2025

12:47 AM

Solution Artifact Version

1.0.0.645

Solution Artifact Current Version

1.0.0.644

Scheduled Time

Is Redeployment

* ☐

Pre-Export Step Required

* ☐

Pre-Deployment Step Required

* ☐

Is Delegated Deployment

☐

Get The Managed & Unmanaged Solution ZIP File

During manual exports we download a solution zip file to our machine then import it into the test environment. This gives us a valuable copy of the solution zip file which we can use in redeployments. The Power Platform Pipeline also stores the solution zip files as solution artifacts. Go to the solution artifacts page and open an artifact.

Power Apps

Deployment Pipeline Configuration

🔍

+

⌵

☰

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📊

Pipelines Dashboard

Pipeline Setup

🌐

Environments

🚀

Pipelines

Deployments

📋

Run history

</>

Solution artifacts

Settings

🛡️

Security Teams

⚙️

Advanced Settings

⬅️

✎ Edit

📄 Activate

🚫 Deactivate

🗑 Delete

⌵

👤 Assign

⋮

Active Deployment Artifacts*

🏠

⌵

🔍 Filter by

☒	Name ↑	Artifact Version ↓	Generated On ↓
☒	CurrencyExchangeALMSample	1.0.0.645	2/13/2025 12:42 AM

The managed and unmanaged solution zip files are found on the solution artifact details page.

Power Apps

Deployment Pipeline Configuration

MD

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Save

Save & Close

New

Deactivate

Delete

Share

CurrencyExchangeALMSample - Saved

Deployment Artifact

General

Deployment information

Related

Name

CurrencyExchangeA...

Owner

MD

Matthew D...

Artifact Version

1.0.0.645

Managed Artifact File

CurrencyExchangeALMSample_1.0.0.645_managed.zip

Delete

Unmanaged Artifact File

CurrencyExchangeALMSample_1.0.0.645.zip

Delete

Generated On

2/13/2025

12:42 AM

Deploy A Solution To The Test Environment

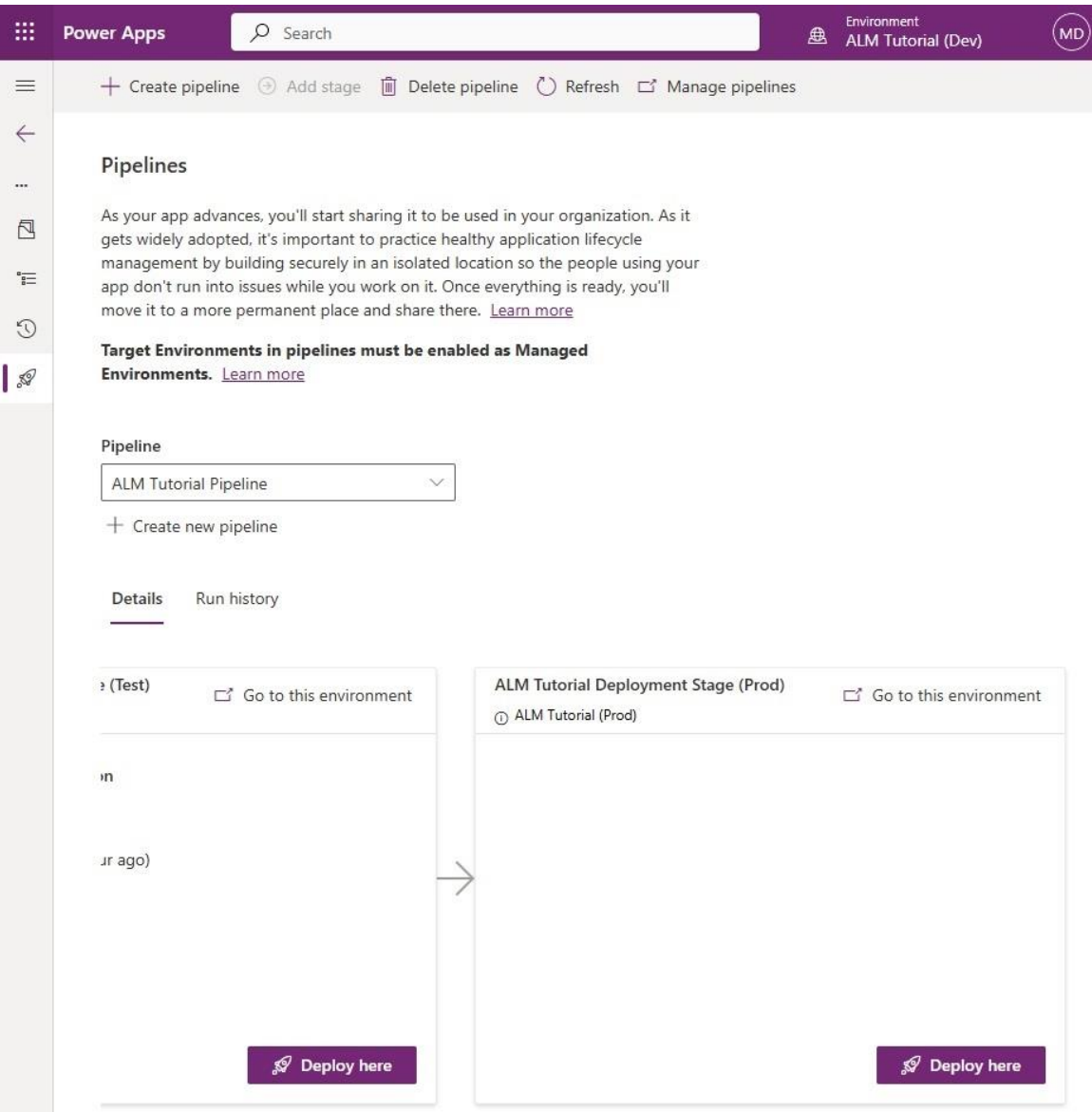
96

Deploy A Solution To The Production Environment

Once testing is completed we can use Power Platform Pipelines to move the managed solution to the production environment and go-live. Using a pipeline ensures the deployment to production is performed in a consistent manner. And it forces all changes to pass through the test environment first.

Deploy The Solution To The Production Environment With Pipelines

Go to the development environment and open the solution you want to move to the production environment.



Schedule the deployment now and press the Next button.

Deploying Solution



Select deployment target

Select the deployment pipeline you'd like to use. Each deployment pipeline has different stages, after you choose a pipeline - select the stage.

Pipeline

ALM Tutorial Pipeline 

+ Create new pipeline

☐ ALM Tutorial Deployment Stage (Test)

☒ ALM Tutorial Deployment Stage (Prod)

Deployment schedule

☒ Now

☐ Later

 An AI-generated description for this solution will be created when you click Next. [Learn more](#)

Back

Next

Close

Set Connection References And Environment Variables

The sample solution contains two connection references used in a Power Automate flow. One for Dataverse and another for Outlook. Wait for the green checkmark to appear beside each connection reference. If it does not appear, press the three dots to troubleshoot the issue. Then press the Next button.

Deploying Solution

✓ Destination

● Connections

○ Environment variables

○ Summary

Connections

Re-establish connections to activate your solution.

2 updates needed

Sign in

These services use your credentials to sign into apps and create connections. A green check means you're ready to go.



Microsoft Dataverse CurrencyExchang...
md_sharedcommondataserviceforapps...

✓

...



Office 365 Outlook
md_Office365Outlook
[Permissions](#)

✓

...

Back

Next

Close

Deploy A Solution To The Production Environment

99

The sample solution also contains an environment variable for Email Notifications. The email address in this environment variable will receive email. Set this value to the production email address then press Next.

Deploying Solution

✓ Destination

✓ Connections

● Environment variables

○ Summary

Environment Variables

Enter information for each field, so your app works properly. You can edit your environment variables later.

1 updates needed

Email Notifications Target

prod@matthewdevaney.com

Reset

Back

Next

Close

Give Deployment Notes For Release To Production

Similar to the deployment to the test environment, we want to write an update explaining the changes to production. Highlight the differences between the solution we are about to deploy and the current version in production. This helps us to track changes when troubleshooting issues with a past deployment. Do not use AI generated deployment notes.

Press the deploy button once finished.

Deploying Solution

✓ Destination

✓ Connections

✓ Environment variables

✓ Summary

Pipeline	Stage
ALM Tutorial Pipeline	ALM Tutorial Deployment Stage (Prod)

Display name	Name
Currency Exchange ALM Sample	CurrencyExchangeALMSample

Publisher	Prefix
Matthew Devaney	md

Version
1.0.0.645

Deployment notes

1. Classic Canvas App - Currency Transactions Monitor: A user interface for monitoring and managing currency transactions.

2. Entity - Currency: Stores information about different currencies.

3. Entity - Currency Transaction: Records details of currency transactions.

4. Entity - Exchange Rate: Maintains exchange rate data between different currencies.

5. Environment Variable Definition - Email Notifications Target: Defines the target email address for sending notifications.

6. Modern Flow - Send Currency Rates Summary Email: Automates the process of sending summary emails about currency rates.

✕ Clear

👍🗨

ⓘ

AI-generated content can have mistakes. Make sure it's accurate and appropriate before using it. [Read terms](#)

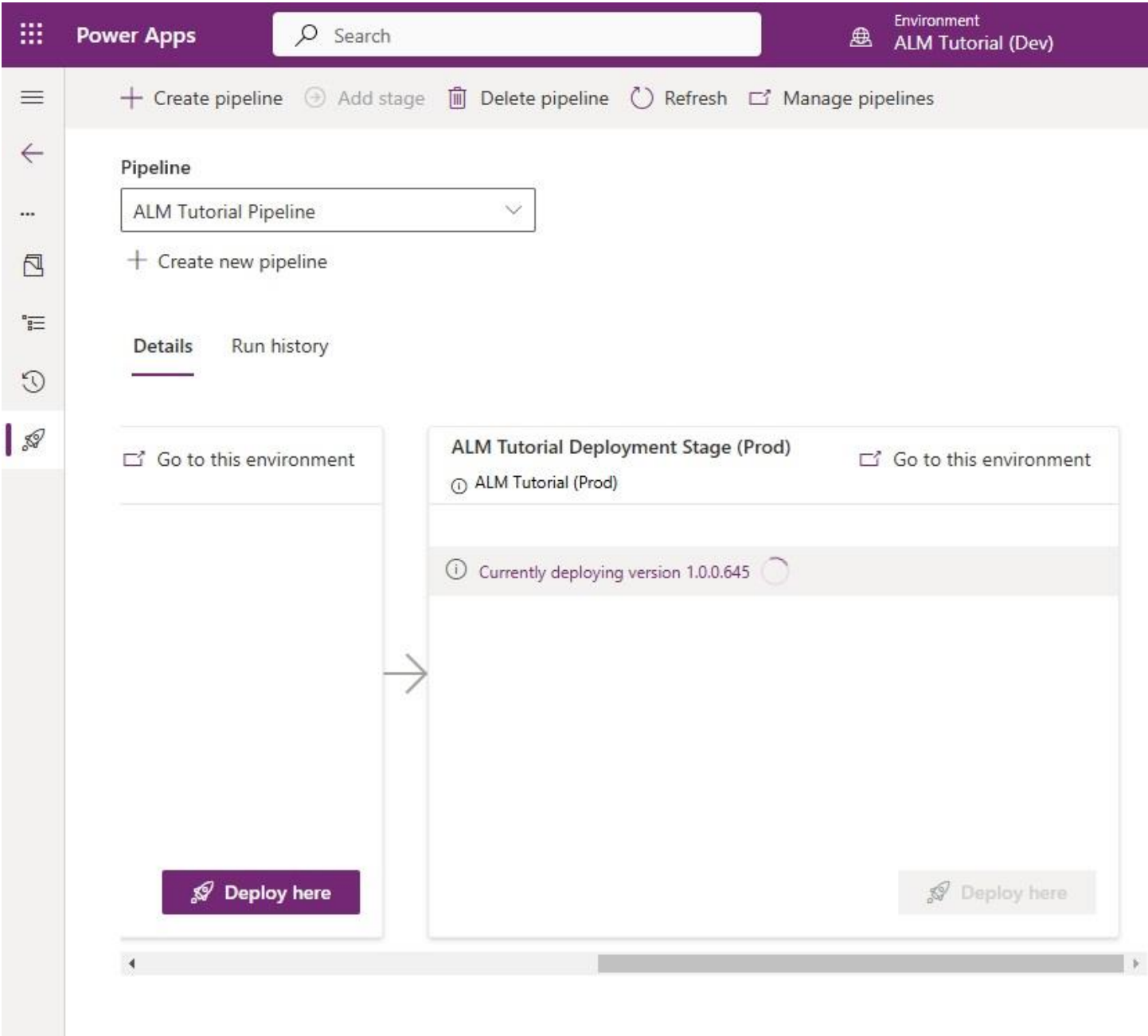
Back

Deploy

Close

Wait For The Production Stage Deployment To Complete

Power Platform Pipelines needs a few minutes to import the solution into the Production environment.



Once the solution import is completed the last successfully installed version and last deployed date will appear in the production stage card.

Validate The Deployment To Production

After a deployment it is important to make sure the solution is working in the target environment. Go to the production environment, browse to the solutions menu, select the Managed tab and open the sample solution.

Power Apps

Search

Environment
ALM Tutorial (Prod)

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Apps

Solutions

Flows

Tables

More

Power Platform

New solution Import solution Open AppSource

Solutions

Deployment

Track all solution deployments in one place. See what's active, what has failed, and get recommendations for improvement.

Unmanaged

Managed

All

Display name	Name	Created	Version
Currency Exchange ALM Sample	CurrencyExchang...	5 minutes ago	1.0.0
Power Apps Checker Base	msdyn_PowerAp...	1 week ago	2.0.0
Power Apps Checker	msdyn_PowerAp...	1 week ago	2.0.0
Contextual Help Base	msdyn_Contextu...	1 week ago	1.0.0
Contextual Help	msdyn_Contextu...	1 week ago	1.0.0

Go through each of the solution components and give them a brief test.

Power Apps

Search

Environment
ALM Tutorial (Prod)

Objects

Search

All (9)

Agents (0)

Apps (1)

Cards (0)

Cloud flows (1)

Connection references (2)

Environment variables (1)

Site maps (1)

Tables (3)

New

Add existing

Publish all customizations

You cannot directly edit the objects within a managed solution. If the managed properties for solution objects are set to can edit them from another unmanaged solution.

Currency Exchange ALM Sample > All

	Display name ↑ ↓	Name ↓	Type ↓	Managed ↓
	Currency	md_currency	Table	Yes
	Currency Exchange Ad...	md_CurrencyExc...	Site Map	Yes
	Currency Transaction	md_currencytran...	Table	Yes
	Currency Transactions ...	md_currencytran...	Canvas App	Yes
	Email Notifications Tar...	md_EmailNotific...	Environment Vari...	Yes
	Exchange Rate	md_exchangerate	Table	Yes
	Microsoft Dataverse C...	md_sharedcomm...	Connection Refe...	Yes
	Office 365 Outlook	md_Office365Ou...	Connection Refe...	Yes
	Send Currency Rates S...	Send Currency R...	Cloud Flow	Yes

View The Pipeline Run History

To look at the run history of the Power Platform Pipeline, go to the host environment and open the Deployment Pipeline Configuration app. Then go to the run history page and select the latest run.

Power Apps

Deployment Pipeline Configuration

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Pipeline Setup

Environments

Pipelines

Deployments

Run history

Solution artifacts

Settings

Security Teams

Advanced Settings

Run history*

Show As

Show Chart

Delete

Refresh

Visualize this view

	Deployment Pipeline (...)	Name (Deployment Stage)	Start Time ↓	En
	ALM Tutorial Pipeline	ALM Tutorial Deployment Stage (Prod)	2/13/2025 1:48 ...	2,
	ALM Tutorial Pipeline	ALM Tutorial Deployment Stage (Test)	2/13/2025 12:4...	2,

1 - 2 of 2

Deploy A Solution To The Production Environment

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The run details page looks like this.

Power Apps

Deployment Pipeline Configuration

MD

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Pipelines Dashboard

Pipeline Setup

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Run history

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Settings

Security Teams

Advanced Settings

Save

Save & Close

Deactivate

Cancel Deployment

Delete

Share

ALM Tutorial Deployment Stage

(Prod)_CurrencyExchangeALMSample_1.0.0.645_034e75c0-9712-46dc-98f8-5b64686d1d6f - Saved

Deployment Stage Run

Matthew Devaney

Owner

General

Operation details

Deployment settings

Deployment notes

Related

Stage Run Status

Succeeded

Deployment Stage

ALM Tutorial Deployment ...

Operation

Deploy

Start Time

2/13/2025

1:48 AM

End Time

2/13/2025

1:51 AM

Scheduled Time

Pre-Export Step Required

Pre-Deployment Step Required

Is Delegated Deployment

Development Deployment Environment ID

ALM Tutorial (Dev)

Target Deployment Environment ID

ALM Tutorial (Prod)

Artifact ID

CurrencyExchangeAL...

Artifact Name

CurrencyExchangeALMSample

Solution Artifact Version

1.0.0.645

Solution Artifact Current Version

1.0.0.645

Is Redeployment

Deploy A Solution To The Production Environment

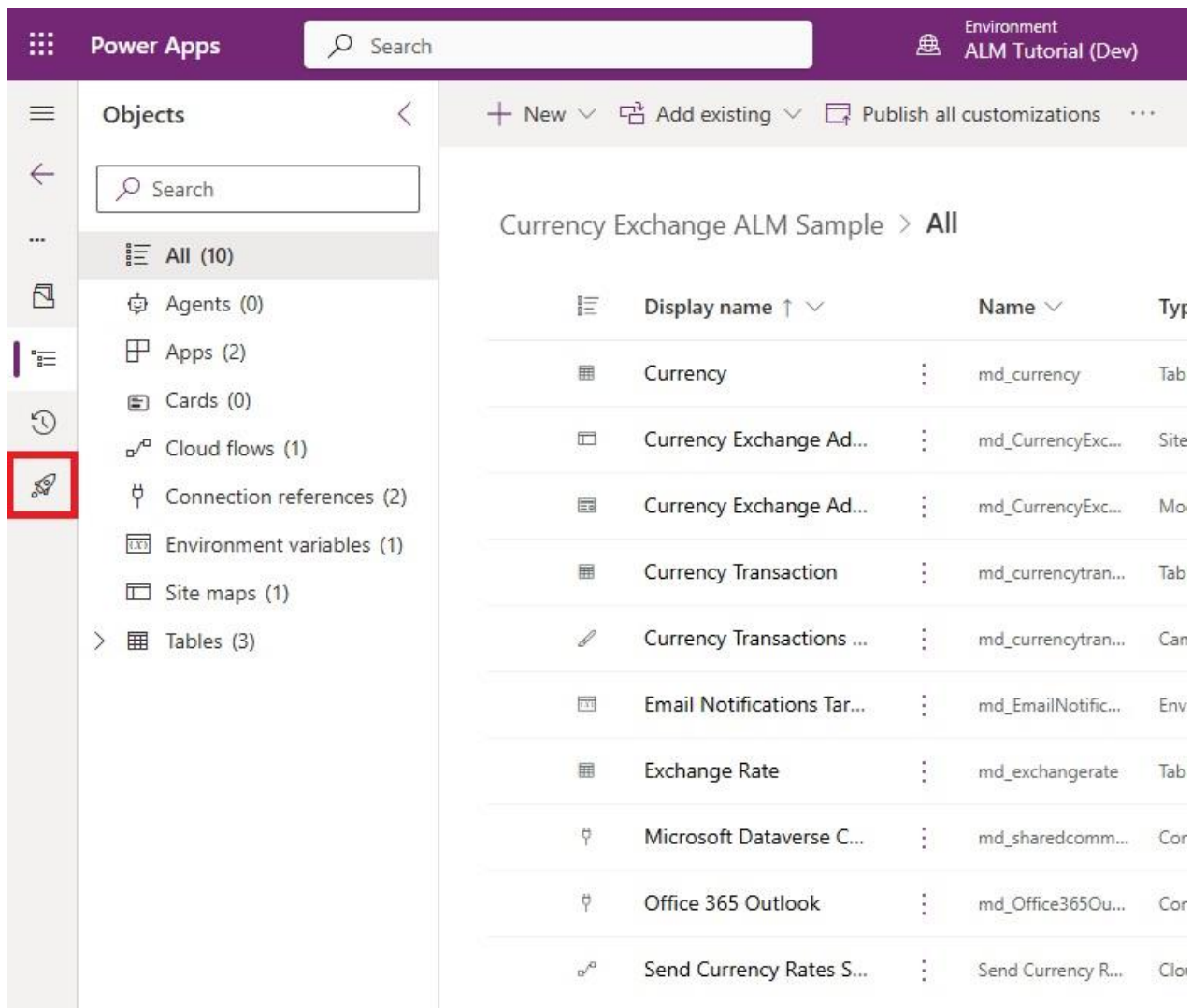
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Redeploy A Past Solution Version To A Target Environment

Do you need to roll-back a solution version? This is possible using the Power Platform Pipelines redeploy feature. It allows you to pick a solution and downgrade to an earlier version. Be careful though, if you are removing Dataverse tables or columns then the data in stored them will be lost.

Open The Power Platform Pipelines Run History

In the development environment, go to the solution you want to rollback and open the Pipelines menu.



The screenshot shows the Power Apps interface. The top bar includes the 'Power Apps' logo, a search bar, and the environment name 'Environment ALM Tutorial (Dev)'. The left sidebar contains a navigation menu with icons for Objects, Apps, Cards, Cloud flows, Connection references, Environment variables, Site maps, and Tables. The 'Pipelines' icon (a rocket) is highlighted with a red box. The main area displays a list of objects for the 'Currency Exchange ALM Sample' solution, categorized under 'All'.

Display name ↑ ↓	Name ↓	Type
Currency	md_currency	Tab
Currency Exchange Ad...	md_CurrencyExc...	Site
Currency Exchange Ad...	md_CurrencyExc...	Mo
Currency Transaction	md_currencytran...	Tab
Currency Transactions ...	md_currencytran...	Can
Email Notifications Tar...	md_EmailNotific...	Env
Exchange Rate	md_exchangerate	Tab
Microsoft Dataverse C...	md_sharedcomm...	Cor
Office 365 Outlook	md_Office365Ou...	Cor
Send Currency Rates S...	Send Currency R...	Clo

Select the run history tab.

Power Apps

Search

Environment
ALM Tutorial (Dev)

+

Create pipeline

→

Add stage

🗑

Delete pipeline

↺

Refresh

🔗

Manage pipelines

Pipelines

As your app advances, you'll start sharing it to be used in your organization. As it gets widely adopted, it's important to practice healthy application lifecycle management by building securely in an isolated location so the people using your app don't run into issues while you work on it. Once everything is ready, you'll move it to a more permanent place and share there. [Learn more](#)

Target Environments in pipelines must be enabled as Managed Environments. [Learn more](#)

Pipeline

ALM Tutorial Pipeline

+

Create new pipeline

Details

Run history

Development

Securely test and verify in isolation. After you've got everything working properly, it's time to deploy. [Learn more](#)

ALM Tutorial Deployment Stage (Test)

🔗 Go to this en

🕒 ALM Tutorial (Test)

Last successfully installed version

1.0.0.645

Redeploy A Solution With Power Platform Pipelines

The run history shows all past deployments to a target environment. To redeploy a past solution version, select the three dots beside it and press choose the redeploy option.

Power Apps

Search

Environ
ALM T

+ Create pipeline

+ Add stage

Delete pipeline

Refresh

Manage pipelines

Pipelines

As your app advances, you'll start sharing it to be used in your organization. As it gets widely adopted, it's important to practice healthy application lifecycle management by building securely in an isolated location so the people using your app don't run into issues while you work on it. Once everything is ready, you'll move it to a more permanent place and share there. [Learn more](#)

Target Environments in pipelines must be enabled as Managed Environments. [Learn more](#)

Pipeline

ALM Tutorial Pipeline

+ Create new pipeline

Details

Run history

Start time ↓

End time

Start time ↓

End time

Start time ↓

End time

Feb 12, 2025 7:48 PM (2 days ago)

Feb 12, 2025 7:51 PM (2 days ago)

Feb 12, 2025 6:42 PM (2 days ago)

Feb 12, 2025 6:42 PM (2 days ago)

Cancel deployment

Redeploy

Information

Details

Stage

ALM Tutorial Deployment Stage (Prod)

Source

ALM Tutorial (Dev)

Target

ALM Tutorial (Prod)

Start time

Feb 12, 2025 7:48 PM (2 days ago)

End time

Feb 12, 2025 7:51 PM (2 days ago)

Solution

CurrencyExchangeALMSample

Version

1.0.0.645

Deployed by

Matthew Devaney

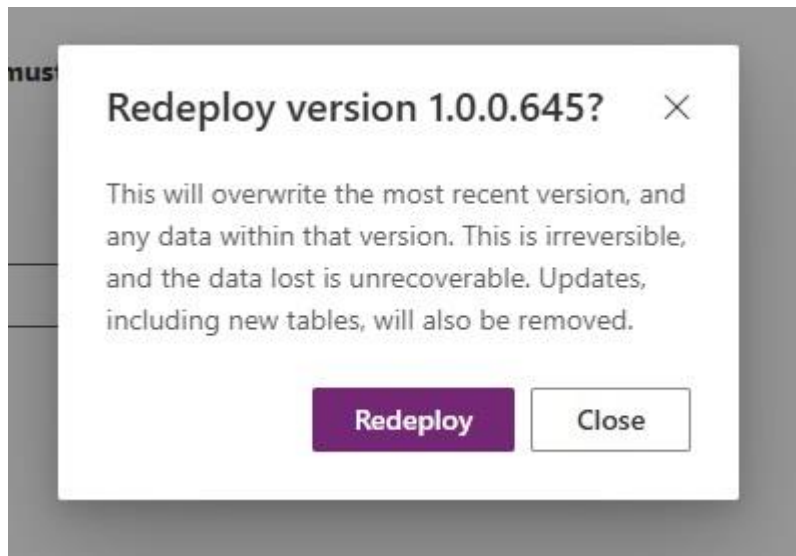
Deployment notes

The Currency Exchange ALM Sample solution is designed to manage and monitor currency exchange

Operation

Deploy

A redeployment will overwrite the current solution in the target environment. If that solution has a different set of Dataverse tables and columns it could result in data loss. Before you redeploy, take a moment to determine the differences between solutions and understand what will be deleted.



Perform A Delegated Deployment Using A Service Account

A delegated deployment in Power Platform Pipelines is performed by the service account rather than the user who pressed the deploy button. This is useful because we can grant the service account elevated permissions to the target environment while the user is prevented from making any direct changes inside of the environment. It forces them to use pipelines for deployment, which is a good thing. And if the service account performs the initial deployment of a solution the apps and flows within it will be owned by the service account in the new environment.

Create A Power Automate Flow To Approve Delegated Deployments

Power Platform Pipelines delegated deployments require an approval. Currently, the only way to perform this approval is by using a Power Automate flow.

Go to the Pipelines Host environment and create a new solution named *Pipelines Extensions*.

The screenshot shows the 'New solution' dialog in the Power Apps interface. The background shows the 'Solutions' page with a 'Deployment' card and a table of solutions. The dialog box is open on the right side of the screen.

New solution

Display name *
MD Pipelines Extensions

Name *
MDPipelinesExtensions

Publisher *
Default Publisher for devaney-pipeline

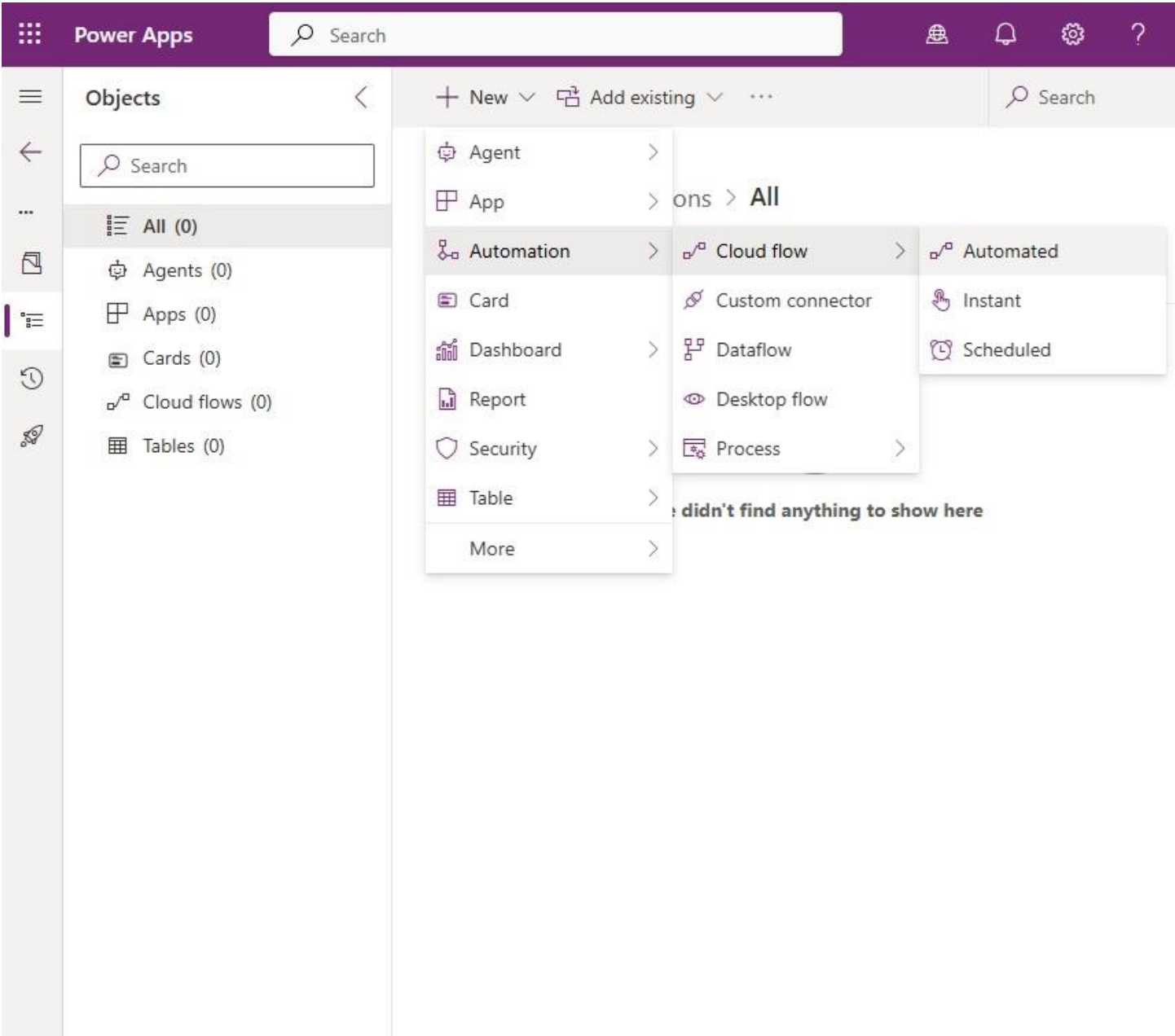
+ New publisher

Version *
1.0.0.0

More options

Create Cancel

Then create a new Power Automate flow with an automated trigger inside of the Pipelines Extensions solution.



Name the solution Approve Pipelines Deployment and select the when an action is performed Dataverse trigger.

Build an automated cloud flow

Flow name: Approve Pipelines Deployment

Choose your flow's trigger * ⓘ

Search or select a trigger from the list below to create a flow. (Required)

Dataverse

- ☒ When an action is performed (Microsoft Dataverse) ⓘ
- ☐ When a row is added, modified or del... (Microsoft Dataverse) ⓘ

Free yourself from repetitive work just by connecting the apps you already use—automate alerts, reports, and other tasks.

Examples:

- Automatically collect and store data in business solutions
- Generate reports via custom queries on your SQL database

Skip Create Cancel

Auto-Approve A Delegated Deployment In Power Platform Pipelines

Delegated deployments require an approval but we want the approval to be automatic. Set the flow trigger category to Power Platform Pipelines and use the action name OnApprovalStarted. Then add a condition where true=true so it will always evaluate to yes. Inside of the yes condition, perform an unbound action with the following parameters:

- Action Name – UpdateApprovalStatus
- Approval Properties – workflow()
- StageRunId – Action Inputs StageRunId
- Approval Status – 20 (approved)

Also, in the no condition perform the same unbound action with an Approval Status of 30 (rejected). Although it is not necessary in this scenario we have setup the no condition to understand how do it.

When an action is performed: OnApprovalStarted ...

* Catalog
Microsoft Dataverse Common

* Category
Power Platform Pipelines

* Table name
(none)

* Action name
OnApprovalStarted



Condition: Auto-Approve ...

☐ true is equal to ☐ true

+ Add

✓ If yes

Perform an unbound action: UpdateApprovalStatus Completed ...

* Action Name
UpdateApprovalStatus

ApprovalProperties
☐ workflow()

StageRunId
☐ ActionInputs St...

ApprovalComments

ApprovalStatus
20

Add an action

✗ If no

Perform an unbound action: UpdateApprovalStatus Rejected ...

* Action Name
UpdateApprovalStatus

ApprovalProperties
☐ workflow()

StageRunId
☐ ActionInputs St...

ApprovalComments

ApprovalStatus
30

Add an action

Enable Delegated Deployment For A Pipeline Deployment Stage

A delegated deployment can be setup from the deployment stage record in the deployment pipeline configuration app. Launch the app and go to the Pipelines page. Select the ALM Tutorial Pipeline and locate the deployment stages subgrid.

Power Apps

Deployment Pipeline Configuration

MD

Home

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Advanced Settings

ALM Tutorial Pipeline - Saved

Deployment Pipeline

General

Deployment stages

Run history

Related

Rows: 1

Deployment Stages (Deployment Pipeline)

☐

Name

☐

ALM Tutorial Deployment Stage (Test)

☐

ALM Tutorial Deployment Stage (Prod)

Deployment Pipeline

ALM Tutorial Pipeline

ALM Tutorial Pipeline

Rows: 2

Open the test deployment stage and check Is Delegated Deployment. Choose the Delegated Deployment Type Stage Owner and set the Owner to the Service Account. Save and close the record.

Power Apps

←

↗

Save

Save & Close

Share

ALM Tutorial Deployment Stage (Test) - Unsaved

Deployment Stage

GeneralRelated

Name

ALM Tutorial Pipeline

Owner

Matthew Devaney (Offline)

Description

Name

ALM Tutorial Deployment Stage (Test)

Owner

Service Account (Offline)

Description

Deployment Pipeline

ALM Tutorial Pipeline

Previous Deployment Stage

Target Deployment Environment ID

ALM Tutorial (Test)

Pre-Export Step Required

Pre-Deployment Step Required

Is Delegated Deployment

Delegated Deployment Type

Stage Owner

Do the same for the production deployment stage.

Power Apps

←

📁

Save

Save & Close

⋮

Share

ALM Tutorial Deployment Stage (Prod) - Saved

Deployment Stage

General Related

Name

ALM Tutorial Pipeline

Owner

MD Matthew Devaney

Description

Name

ALM Tutorial Deployment Stage (Prod)

Owner

SA Service Account

Description

Deployment Pipeline

ALM Tutorial Pipeline

Previous Deployment Stage

ALM Tutorial Deployment Stage (...)

Target Deployment Environment ID

ALM Tutorial (Prod)

Pre-Export Step Required

☐

Pre-Deployment Step Required

☐

Is Delegated Deployment

☒

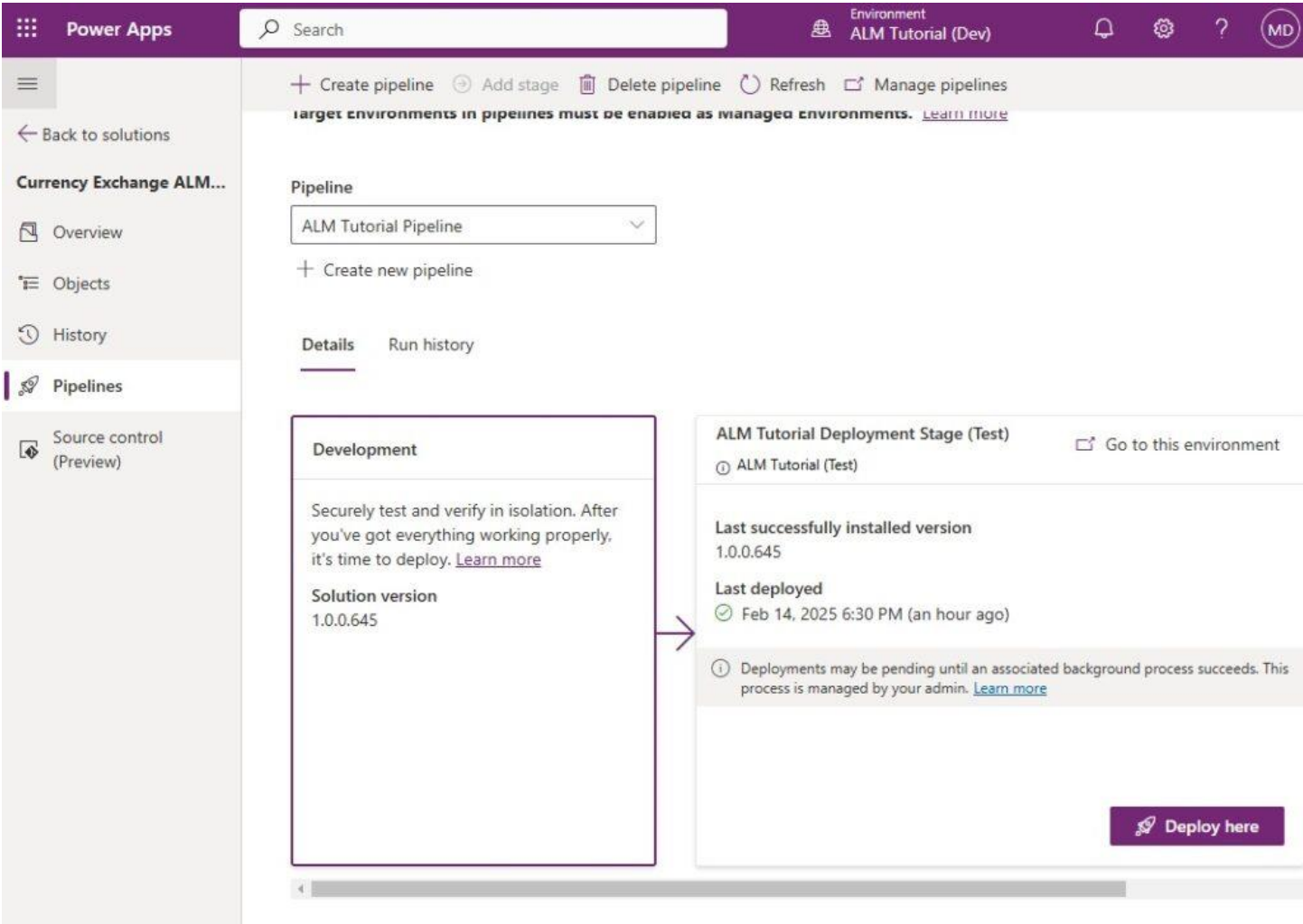
Delegated Deployment Type

Stage Owner

Deploy The Solution As The Service Account

The test and production deployment stages are now setup to perform delegated deployments. All future Power Platform Pipeline runs will be performed by the service account.

Go to the Pipelines menu for the sample solution and press the deploy here button to deploy the solution to the test environment.



Work through the deploying solution wizard and then press the deploy button on the final page.

Deploying Solution

Destination

Connections

Environment variables

Summary

Pipeline

ALM Tutorial Pipeline

Display name

Currency Exchange ALM Sample

Publisher

Matthew Devaney

Version *

1.0.0.646

Stage

ALM Tutorial Deployment Stage (Test)

Name

CurrencyExchangeALMSample

Prefix

md

Deployment notes

4. Entity - Currency Transaction: A table that records individual currency exchange transactions, capturing details necessary for tracking and reporting.

5. Entity - Exchange Rate: A table that stores exchange rate information between different currencies, essential for calculating transaction values.

6. Environment Variable Definition - Email Notifications Target: A configuration setting that specifies the target email address for sending notifications, ensuring that relevant stakeholders receive updates.

7. Modern Flow - Send Currency Rates Summary Email: An automated workflow that sends summary emails of currency rates, keeping users informed about current exchange rates.

Clear

AI-generated content can have mistakes. Make sure it's accurate and appropriate before using it. [Read terms](#)

Back

Deploy

Close

Perform A Delegated Deployment Using A Service Account

120

Review The Delegated Deployment Results

Wait for the Power Platform Pipeline to complete the solution deployment.

Search

Environment
ALM Tutorial (Dev)

+ Create pipeline

+ Add stage

+ Delete pipeline

+ Refresh

+ Manage pipelines

so the people using your app don't run into issues while you work on it. Once everything is ready, you'll move it to a more permanent place and share there. [Learn more](#)

Target Environments in pipelines must be enabled as Managed Environments. [Learn more](#)

Pipeline

ALM Tutorial Pipeline

+ Create new pipeline

Details

Run history

Development

Securely test and verify in isolation. After you've got everything working properly, it's time to deploy. [Learn more](#)

Solution version
1.0.0.645

ALM Tutorial Deployment Stage (Test)

ALM Tutorial (Test)

Go to this environment

Last successfully installed version
1.0.0.645

Last deployed
Feb 14, 2025 6:30 PM (an hour ago)

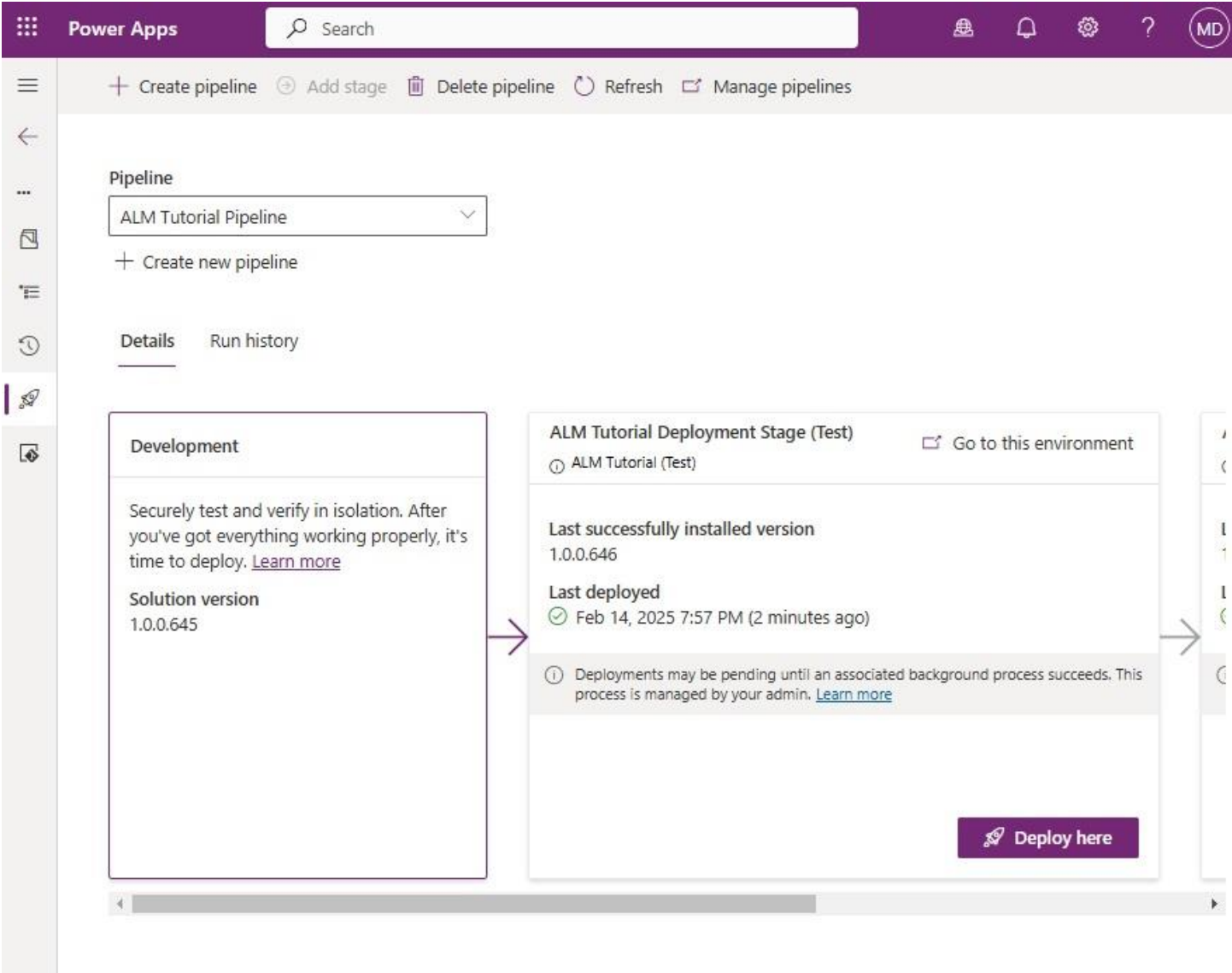
Currently exporting artifact version 1.0.0.646

Deploy here

Perform A Delegated Deployment Using A Service Account

121

Once completed, the deployment stage will show the new version number and last deployed date. Then go to the test environment to review the solution and ensure it deployed correctly.



Look at the latest run in the Approve Pipelines Deployment flow run history. It shows the auto-approval to the test deployment stage.

Power Apps

Search

←

Approve Delegated Deployment • Ran at 2/14/2025 7:56:55 PM

↺

Resubmit

✕

Cancel

✔

Your flow ran successfully.

When an action is performed: OnApprovalStarted

0s

✔

↓

Condition: Auto-Approve

1s

✔

INPUTS

Expression result

true

✔

If yes

Perform an unbound action: UpdateApprovalStatus Compelcted

1s

✔

✕

If no

Perform an unbound action: UpdateApprovalStatus Rejected

0s

✕

Perform A Delegated Deployment Using A Service Account

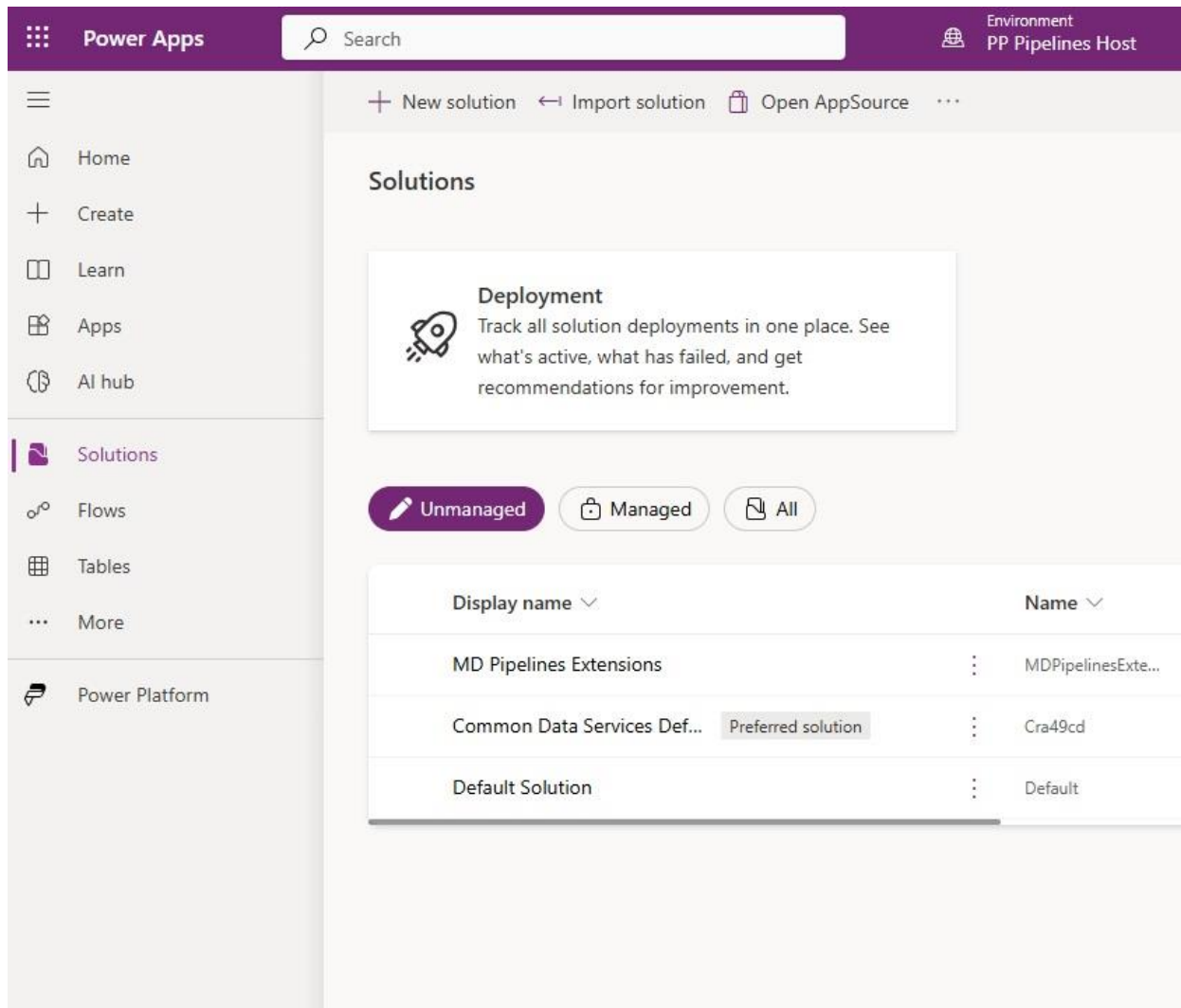
123

Configure Pre-Deployment Stage Approvals

Power Platform pipelines can be enhanced to include deployment stage approvals. Although there are no-built-in pipeline approvals it is possible to create one using a custom Dataverse table and a Power Automate flow. Stage approvals are important because they help the environment owner control what gets deployed into their environment. An approvals process should be a part of any Power Platform Pipelines process.

Create A Deployment Stage Approver Table

We want to allow a Power Platform pipelines administrator to add approvers to a deployment stage. To do this we will create a new Dataverse table. Go to the Pipelines Host environment and open the solutions menu.



The screenshot shows the Power Apps interface in the 'PP Pipelines Host' environment. The left sidebar contains navigation options: Home, Create, Learn, Apps, AI hub, Solutions (selected), Flows, Tables, More, and Power Platform. The main area displays the 'Solutions' section with a 'Deployment' card and a table of solutions.

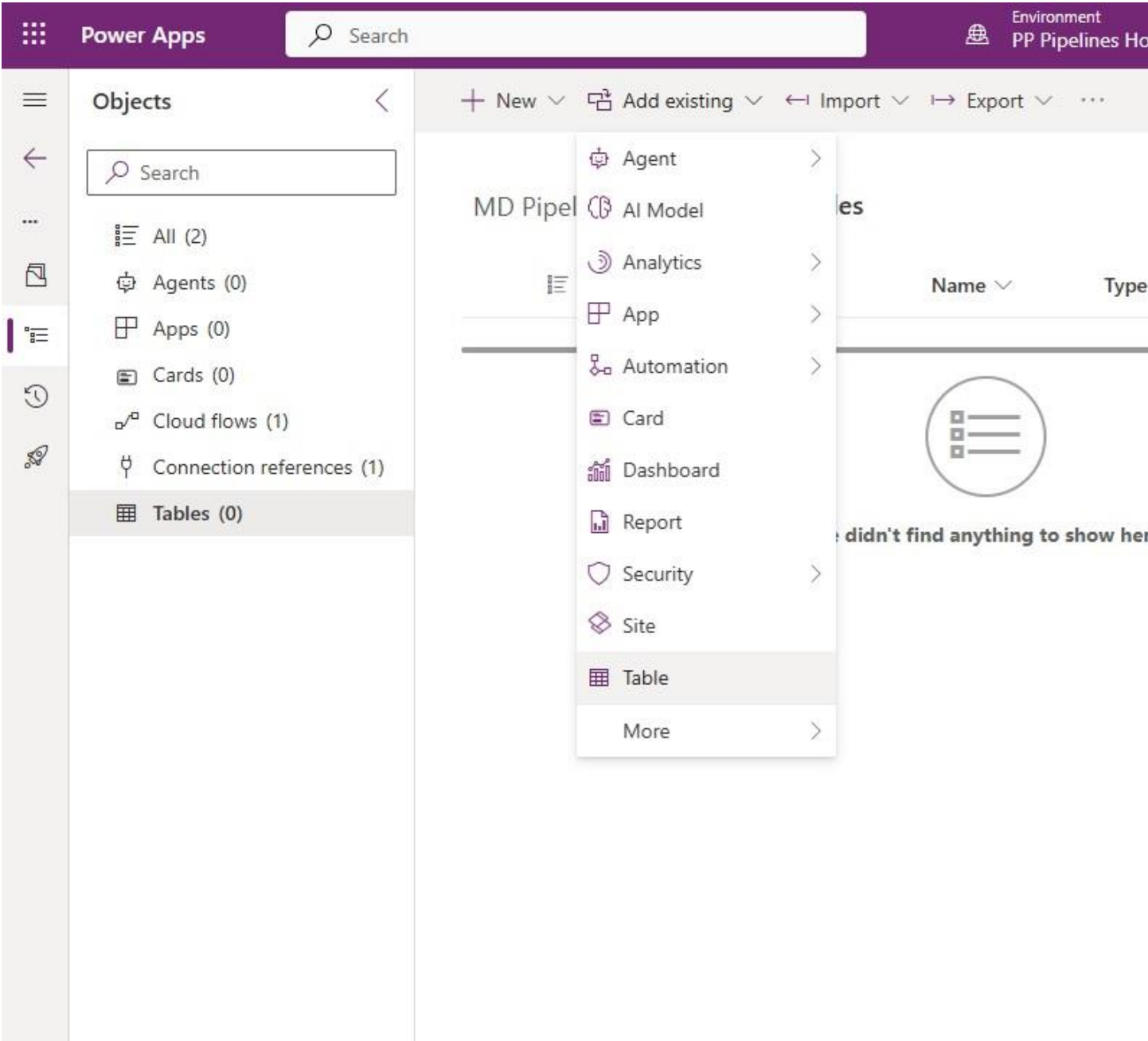
Solutions

Deployment
Track all solution deployments in one place. See what's active, what has failed, and get recommendations for improvement.

Unmanaged **Managed** **All**

Display name	Name
MD Pipelines Extensions	MDPipelinesExt...
Common Data Services Def...	Cra49cd
Default Solution	Default

Inside of a solution, add a new Dataverse table.



Name the table Deployment Stage Approver and press the Save button.

Power Apps

Search

Objects

Search

All (2)

Agents (0)

Apps (0)

Cards (0)

Cloud flows (1)

Connection references (1)

Tables (0)

New table

Use tables to hold and organize your data. Previousl
[Learn more](#)

Properties

Primary column

Display name *
Deployment Stage Approver

Plural name *
Deployment Stage Approvers

Description

☐ Enable attachments (including notes and files) ¹

Advanced options ▼

Save

Cancel

Power Apps will redirect us to the Deployment Stage Approver table landing page.

Power Apps

Search

Environment
PP Pipelines Host

⌵

⚙

?

M

Objects

Search

All (3)

Agents (0)

Apps (0)

Cards (0)

Cloud flows (1)

Connection references (1)

Tables (1)

Deployment Stage App...

Columns

Relationships

Keys

Forms

Views

Charts

Dashboards

Business rules

Commands

New

Add existing

Edit

Create an app

Using this table

MD Pipelines Extensions > Tables > Deployment Stage Approver

Table properties

Properties

Tools

Name	Primary column	Description
Deployment Stage Approver	Name	
Type	Last modified	
Standard	44 seconds ago	

Schema

Columns

Relationships

Keys

Data experiences

Forms

Views

Charts

Dashboards

Customizations

Business rules

Commands

Row summary

Deployment Stage Approver columns and data

Update forms and views

Edit

Created By

Created On

+13 more

Add Entra ID User & Deployment Stage Lookup Columns

A Power Platform pipelines administrator will configure approvals by adding a user and a deployment stage as a new record in the Deployment Stage Approvers table. Go to the columns page and select New column.

Power Apps

Search

Environment
PP Pipelines Host

Objects

Search

All (3)

Agents (0)

Apps (0)

Cards (0)

Cloud flows (1)

Connection references (1)

Tables (1)

Deployment Stage Appr...

Columns

Relationships

Keys

Forms

Views

Charts

Dashboards

Business rules

Commands

+ New column

Add existing column

Advanced

Search

MD Pipelines Extensions > Tables > Deployment Stage Approver > Column

Display name ↑		Name	Data type	Managed
Created By	:	CreatedBy	Lookup	No
Created By (Delegate)	:	CreatedOnBehalf...	Lookup	No
Created On	:	CreatedOn	Date and time	No
Deployment Stage Ap...	:	new_Deploymen...	Unique identifie	No
Import Sequence Nu...	:	ImportSequence...	Whole number	No
Modified By	:	ModifiedBy	Lookup	No
Modified By (Delegate)	:	ModifiedOnBeha...	Lookup	No
Modified On	:	ModifiedOn	Date and time	No
Name Primary name colum	:	new_Name	Single line of te:	No
Owner	:	OwnerId	Owner	No
Owning Business Unit	:	OwningBusiness...	Lookup	No
Owning Team	:	OwningTeam	Lookup	No
Owning User	:	OwningUser	Lookup	No

Name the column Entra ID user and select the data type Lookup. Make the field required. And choose the Microsoft Entra ID table. Then press the Save button.

+ New column

MD Pipe

New column

Previously called fields. [Learn more](#)

Display name *

Entra ID User

Description ⓘ

Data type * ⓘ

🔍 Lookup

Required ⓘ

Business required

☒ Searchable ⓘ

Related table *

Microsoft Entra ID

Advanced options

Save

Cancel

Create a second column named Deployment Stage as a Lookup data type. Make it a required field and set the related table to Deployment Stage. Press the Save button once finished.

New column

Previously called fields. [Learn more](#)

Display name *

Deployment Stage

Description ⓘ

Data type * ⓘ

Lookup

Required ⓘ

Business required

☒ Searchable ⓘ

Related table *

Deployment Stage

Advanced options

Save

Cancel

Setup The Deployment Stage Approver Main Form

We want the Power Platform pipelines approval to be configurable within the Deployment Pipelines Configuration app. Therefore we will need to setup a the table's main form. Switch to the Deployment Stage Approver table's Forms page and open the main form.

Power Apps

Search

Environment
PP Pipelines Host

Objects

Search

All (5)

Agents (0)

Apps (0)

Cards (0)

Cloud flows (1)

Connection references (1)

Tables (3)

New form

Add existing form

Search

Tables > Deployment Stage Approver > Forms

Name		Form type	Status
Information		Quick View	On
Information		Main	On
Information		Card	On

Add the Deployment Stage field and the Entra ID User field to the form. Save and publish the form then exit.

Power Apps | Form

Back

Delete

Form field

Component

Save a copy

Save and publish

Table columns

Search

New table column

Show only unused table columns

Created By

Created By (Delegate)

Created On

Modified By

Modified By (Delegate)

Modified On

Owning Business Unit

Status

Status Reason

New Deployment Stage Approver

Deployment Stage Approver

General

Related

Name

Owner

Deployment Stage

Entra ID User

Information

Responsive (1643 x 797)

Show hidden

Off

100 %

Configure The Test Deployment Stage Approver

We are now ready to configure the approver for a Power Platform pipelines deployment stage. Open the Deployment Pipeline Configuration app and browse to the Pipelines page. Select the test deployment stage record.

Power Apps

Deployment Pipeline Configuration

MD

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Run history

Solution artifacts

Settings

Security Teams

Advanced Settings

Save

Save & Close

New

Share

ALM Tutorial Pipeline - Saved

Deployment Pipeline

General

Deployment stages

Run history

Related

Active Deployment Stages

☐	Name	Deployment Pipeline
☒	ALM Tutorial Deployment Stage (Test)	ALM Tutorial Pipeline
☐	ALM Tutorial Deployment Stage (Prod)	ALM Tutorial Pipeline

Rows: 2

Selected: 1

Configure Pre-Deployment Stage Approvals

133

On the deployment stage form, check the pre-deployment step required checkbox.

Power Apps

Deployment Pipeline Configuration

MD

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ALM Tutorial Deployment Stage (Test) - Saved

Deployment Stage

General Related

Name

ALM Tutorial Deployment Stage (Test)

Owner

Service Account (Offline)

Description

Deployment Pipeline

ALM Tutorial Pipeline

Previous Deployment Stage

Target Deployment Environment ID

ALM Tutorial (Test)

Pre-Export Step Required

Pre-Deployment Step Required

Then go to the related tab and select Deployment Stage Approvers.

Power Apps

Deployment Pipeline Configuration

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Save

Save & Close

New

Deactivate

Delete

ALM Tutorial Deployment Stage (Test) - Saved

Deployment Stage

GeneralRelated

Related - Common

Deployment Stages

Deployment Stage Runs

Deployment Stage Approvers

Related - Process Sessions

Background Processes

Name *

ALM Tut

Owner *

MD Matt

Description

Name *

ALM Tutorial Deployment Stage (Test)

Owner *

SA Service Account (Offline)

Description

Select all

ALM Tutorial Deployment Stage (Test)_Curr
2/15/2025 1:55 AM
CurrencyExchangeALMSample

ALM Tutorial Deployment Stage (Test)_Curr
2/15/2025 12:27 AM
CurrencyExchangeALMSample

ALM Tutorial Deployment Stage (Test)_Curr
2/15/2025 12:25 AM

1 - 3 of 3

Press the New Deployment Stage Approvers button.

Power Apps

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Security Teams

Advanced Settings

Save

Save & Close

New

ALM Tutorial Deployment Stage (Test) - Saved

Deployment Stage

General

Deployment Stage Approvers

Related

Show Chart

New Deployment Stag...

Refresh

Flow

Deployment Stage Approver Associa...

Filter by key

Name

Created On

We didn't find anything to show here

Fill-in the Deployment Stage Approver form with the Test Deployment Stage and the Entra ID User who will be the approver.

Power Apps

Deployment Pipeline Configuration

MD

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Security Teams

Advanced Settings

Save

Save & Close

New

Flow

New Deployment Stage Approver - Unsaved

General

Name

ALM Tutorial Deployment Stage (Test) - Matthew Devaney

Owner

MD

Matthew Devaney (Offline) ×

Deployment Stage

ALM Tutorial Deployment Stage (Test) ×

Entra ID User

Matthew Devaney ×

Configure Pre-Deployment Stage Approvals

137

Assign The Production Deployment Stage Approver

The deployment to the production stage will also require an approver. Select the production deployment stage record.

Power Apps

Deployment Pipeline Configuration

MD

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Solution artifacts

Settings

Security Teams

Advanced Settings

ALM Tutorial Pipeline - Saved

Deployment Pipeline

GeneralDeployment stagesRun historyRelated

Active Deployment Stages

	Name	Deployment Pipeline
☐	ALM Tutorial Deployment Stage (Test)	ALM Tutorial Pipeline
☒	ALM Tutorial Deployment Stage (Prod)	ALM Tutorial Pipeline

Rows: 2 Selected: 1

Then on the main form check the pre-deployment stage required checkbox.

Power Apps

Deployment Pipeline Configuration

MD

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Environments

Pipelines

Deployments

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Solution artifacts

Settings

Security Teams

Advanced Settings

ALM Tutorial Deployment Stage (Prod) - Unsaved

Deployment Stage

GeneralRelated

Name

ALM Tutorial Deployment Stage (Prod)

Owner

SA

Service Account (Offline) ×

Description

Deployment Pipeline

*

ALM Tutorial Pipeline

Previous Deployment Stage

ALM Tutorial Deployment Stage (Test) ×

Target Deployment Environment ID

*

ALM Tutorial (Prod) ×

Pre-Export Step Required

?

☐

Pre-Deployment Step Required

?

☒

Next, open the related tab and choose Deployment Stage Approvers from the dropdown menu.

Power Apps

Deployment Pipeline Configuration

Search

Home

Recent

Pinned

Overview

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Pipeline Setup

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Pipelines

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Solution artifacts

Settings

Security Teams

Advanced Settings

Save

Save & Close

New

Deactivate

Delete

ALM Tutorial Deployment Stage (Prod) - Saved

Deployment Stage

GeneralRelated

Related - Common

Deployment Stages

Deployment Stage Runs

Deployment Stage Approvers

Related - Process Sessions

Background Processes

Name

ALM Tut

Owner

MD Matt

Description

Name

ALM Tutorial Deployment Stage (Prod)

Owner

SA Service Account (Offline)

Description

Deployment Pipeline

Select all

AT

ALM Tutorial Deployment Stage (Prod)_Curr

2/15/2025 12:30 AM

CurrencyExchangeALMSample

1 - 1 of 1

Press the New Deployment Stage Approver button.

Power Apps

Deployment Pipeline Configuration

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Solution artifacts

Settings

Security Teams

Advanced Settings

Save

Save & Close

New

Deactivate

Delete

ALM Tutorial Deployment Stage (Prod) - Saved

Deployment Stage

GeneralDeployment Stage ApproversRelated

Show Chart

New Deployment Stag...

Refresh

Flow

Run

Deployment Stage Approver Associated View

Name

Create

We didn't find anything to show here

Assign the approver to the Entra ID User field and select the production deployment stage.

Power Apps

Deployment Pipeline Configuration

Search

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Solution artifacts

Settings

Security Teams

Advanced Settings

New Deployment Stage Approver - Unsaved

General

Name

ALM Tutorial Deployment Stage (Prod) - Mary Baker

Owner

MDL

Matthew Devaney (Offline) X

Deployment Stage

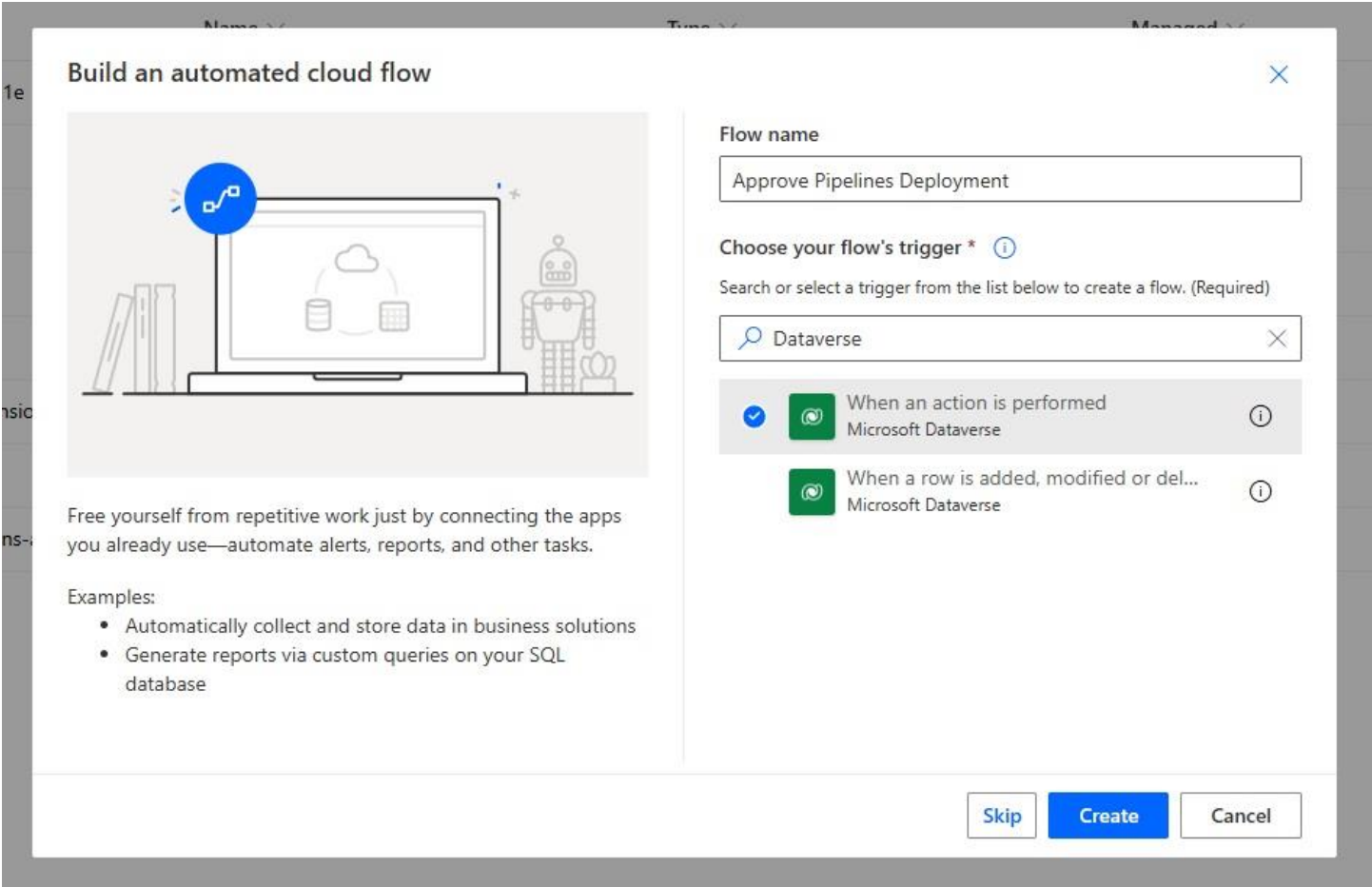
ALM Tutorial Deployment Stage (Prod) X

Entra ID User

Mary Baker X

Create A Power Platform Pipelines Approval Cloud Flow

We will use Power Automate approvals to provide an interface for stage owners to approve and reject deployments. In the solution, create a new automated cloud flow with the name Approve Pipelines Deployment. Select the Dataverse trigger when an action is performed.



Set the flow trigger category to Power Platform Pipelines and use the action name OnApprovalStarted

 When an action is performed: OnPreDeploymentStarted ...

* Catalog

Microsoft Dataverse Common

▼

* Category

Power Platform Pipelines

▼

* Table name

(none)

▼

* Action name

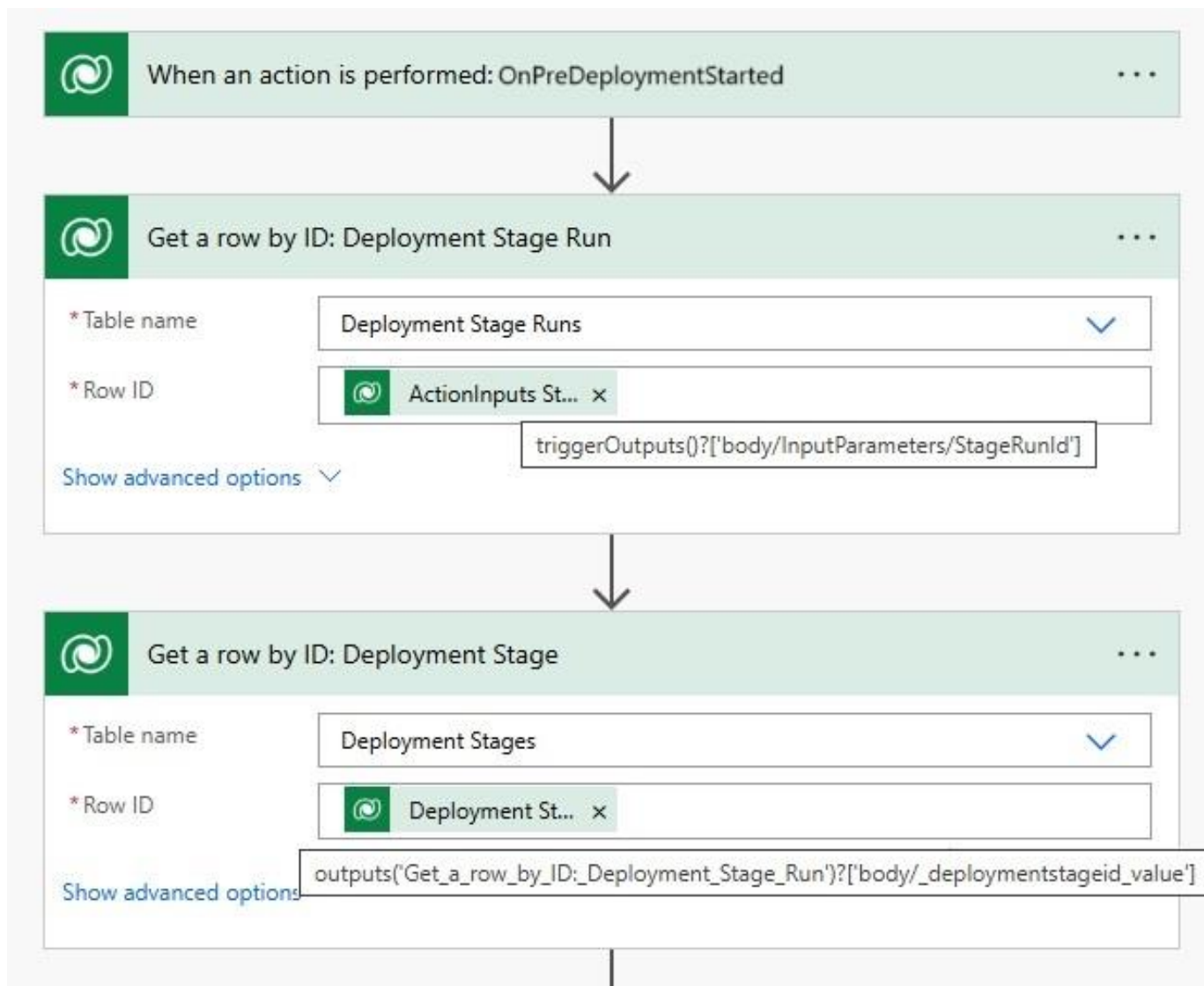
OnPreDeploymentStarted

▼

Get The Deployment Stage Approvers

After the cloud flow is triggered we want to get the list of deployment stage approvers so we can send them an approval request. Determining the approvers requires multiple flow actions.

Start by adding a Dataverse - Get A Row By ID Action and target the Deployment Stage Runs table. Use the StageRunId from the trigger as the Row ID. Then insert another Dataverse - Get A Row By ID action with the Deployment Stages table. Supply the deployment stage id from the previous action in the Row ID field.



There could be multiple deployment stage approvers so the next step requires a Dataverse - List Rows action.

Get a row by ID: Deployment Stage

List rows: Deployment Stage Approvers

*Table name: Deployment Stage Approvers

Select columns: Enter a comma-separated list of column unique names to limit which columns are returned

Filter rows: `_new_deploymentstage_value eq @{outputs('Get_a_row_by_ID:_Deployment_Stage')}`

Sort By: Columns to sort by in OData orderBy style (excluding lookups)

Fetch Xml Query: Enter a Fetch XML query for advanced customization

Row count: Enter the number of rows to be listed (default = 5000)

Show advanced options

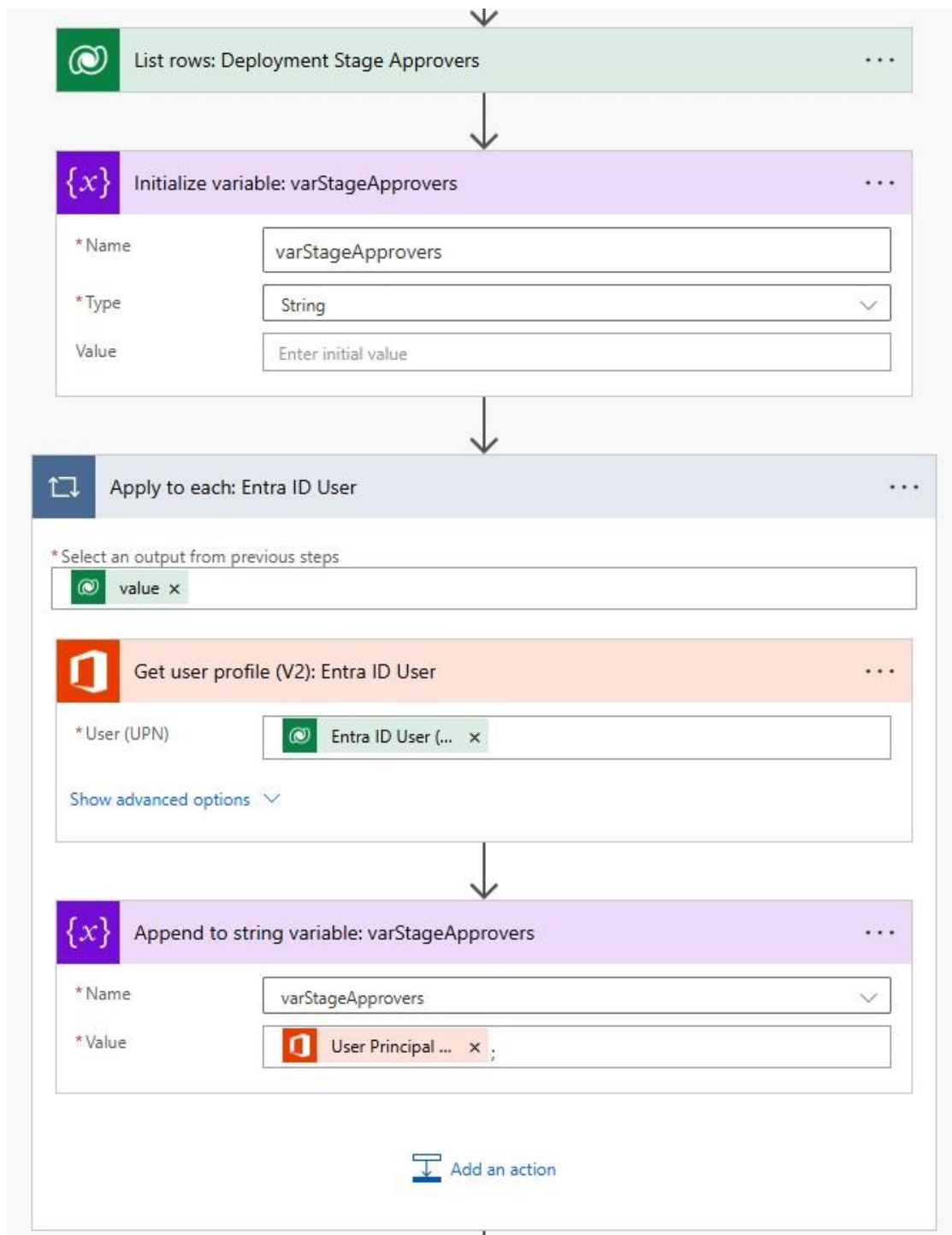
Write this code in the Filter Rows field to get all approvers with a matching deployment stage.

```
_new_deploymentstage_value eq @{outputs('Get_a_row_by_ID:_Deployment_Stage')}['body/deploymentstageid']
```

Obtain The Approvers User Principal Name

Now we have an array of deployment stage approvers. But what we really need is a semi-colon separated list of user principal names (emails) to use in an Approvals flow action.

Initialize a string variable named varStageApprovers. Then for each deployment stage approver, use the Office 365 Users - Get User Profile (V2) action and supply the Entra User ID to retrieve the user principal name. Append each user principal name to the string variable with a semi-colon at the end.



Send An Deployment Stage Approval To Teams

We have everything we need to send an approval request to the deployment stage approvers. Add an Approvals - Start and wait for an approval action and wait for an approval action with the following values:

- Approval type - Approve/Reject - First to respond
- Title: Deployment Approval Request
- Assigned to: varStageApprovers
- Details: Please approve deployment of the <Artifact Name> solution to the <Deployment Stage Name> stage.

The screenshot shows a workflow editor with two actions. The first action, 'Apply to each: Entra ID User', is highlighted with a blue bar and a refresh icon. An arrow points down to the second action, 'Start and wait for an approval', which is highlighted with a purple bar and a clock icon. The 'Start and wait for an approval' action has several fields: 'Approval type' is set to 'Approve/Reject - First to respond'; 'Title' is 'Deployment Approval Request'; 'Assigned to' is 'varStageApprovers'; 'Details' is 'Please approve deployment of the <Artifact Name> solution to the <Deployment Stage Name> stage.'; 'Item link' is 'Add a link to the item to approve'; and 'Item link description' is 'Describe the link to the item'. A 'Show advanced options' link is at the bottom.

Apply to each: Entra ID User

Start and wait for an approval

* Approval type: Approve/Reject - First to respond

* Title: Deployment Approval Request

* Assigned to: varStageApprovers

Details: Please approve deployment of the <Artifact Name> solution to the <Deployment Stage Name> stage.

Item link: Add a link to the item to approve

Item link description: Describe the link to the item

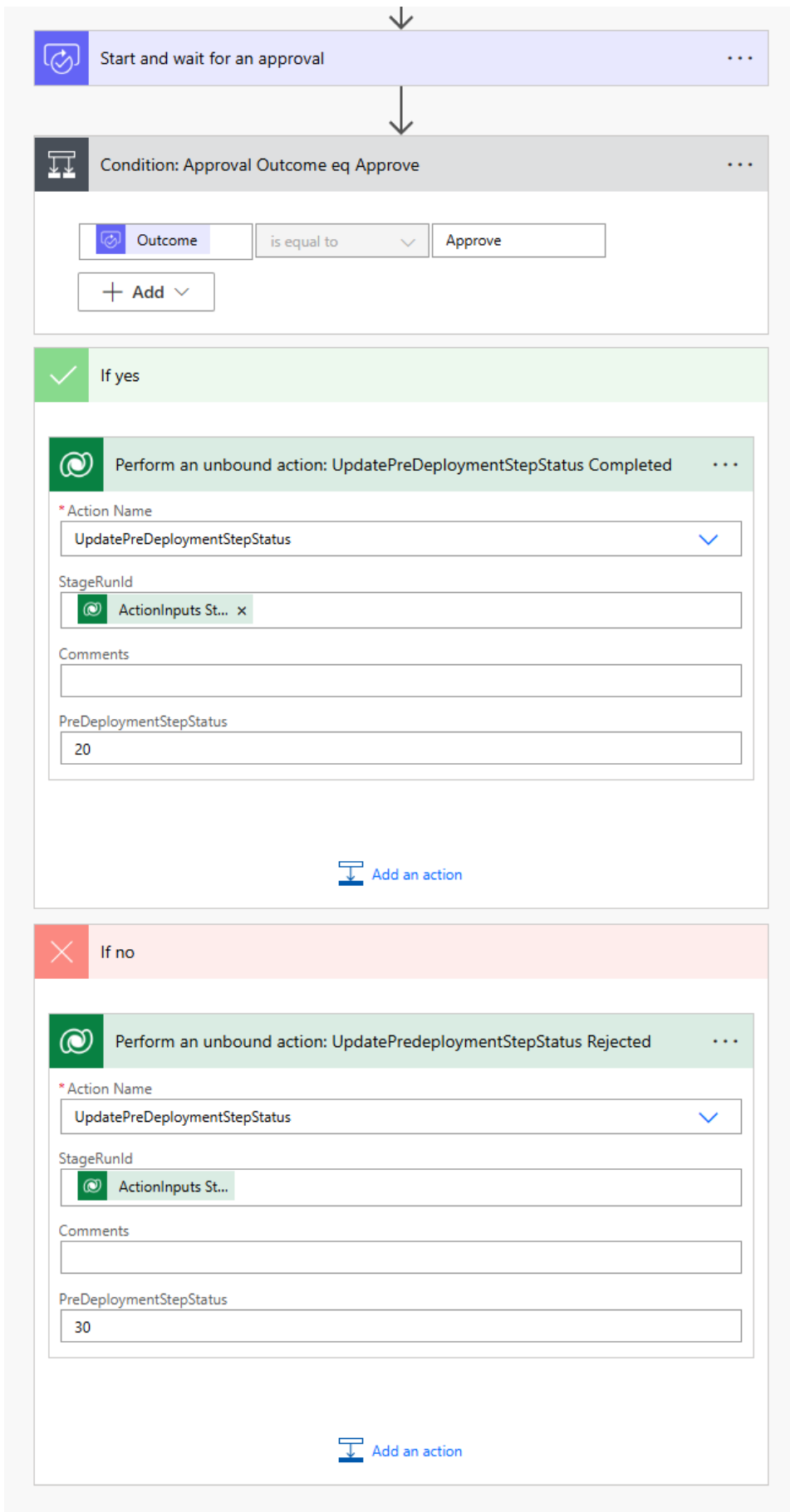
Show advanced options

Approve Or Reject The Request For Deployment

After the approvals action, insert a condition action to check the outcome of the approval. If yes, perform an unbound action to approve deployment. Fill-in the action with the following values:

- Action Name - UpdateApprovalStatus
- Approval Properties - workflow()
- StageRunId - Action Inputs StageRunId
- Approval Status - 20 (approved)

In the no condition perform the same unbound action with an Approval Status of 30 (rejected).



Deploy The Solution To The Test Deployment Stage

All that is left now is to run the pipeline make the sure the approvals we setup are working. Navigate to the sample solution in the development environment and open the pipelines menu. Press the deploy here button on the test stage.

Power Apps

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Environment
ALM Tutorial (Dev)

Create pipelineAdd stageDelete pipelineRefreshManage pipelines

Pipelines

As your app advances, you'll start sharing it to be used in your organization. As it gets widely adopted, it's important to practice healthy application lifecycle management by building securely in an isolated location so the people using your app don't run into issues while you work on it. Once everything is ready, you'll move it to a more permanent place and share there. [Learn more](#)

Target Environments in pipelines must be enabled as Managed Environments. [Learn more](#)

Pipeline

ALM Tutorial Pipeline

Create new pipeline

DetailsRun history

Development

Securely test and verify in isolation. After you've got everything working properly, it's time to deploy. [Learn more](#)

Solution version
1.0.0.644

ALM Tutorial Deployment Stage (Test)

ALM Tutorial (Test)

Go to this environment

Deploy here

Behind the scenes the Power Automate flow is triggered and sends an approval request to the deployment stage approver.



Approvals

Approval request details

Requested

Deployment Approval Request

Please approve deployment of the CurrencyExchangeALMSample solution to the ALM Tutorial Deployment Stage (Test) stage.

▼ Status: Requested

MD

Pending response
Matthew Devaney

MD

Requested by
Matthew Devaney

2/16/2025, 6:57:03 PM

Comments

Add your comments here

More actions ▼

Reject

Approve

Once the approver presses the Approve button the pipeline continues with deployment.

Power Apps

Search

Environment

ALM Tutorial (Dev)

Create pipeline

Add stage

Delete pipeline

Refresh

Manage pipelines

Pipelines

As your app advances, you'll start sharing it to be used in your organization. As it gets widely adopted, it's important to practice healthy application lifecycle management by building securely in an isolated location so the people using your app don't run into issues while you work on it. Once everything is ready, you'll move it to a more permanent place and share there. [Learn more](#)

Target Environments in pipelines must be enabled as Managed Environments.

[Learn more](#)

Pipeline

ALM Tutorial Pipeline

Create new pipeline

Details

Run history

Development

Securely test and verify in isolation. After you've got everything working properly, it's time to deploy. [Learn more](#)

Solution version

1.0.0.645

ALM Tutorial Deployment Stage (Test)

ALM Tutorial (Test)

Last successfully installed version

1.0.0.645

Last deployed

Feb 12, 2025 6:47 PM (an hour ago)

Go to this environment

Deploy here

Configure Pre-Deployment Stage Approvals

153

Setup Dataverse Git Integration For A Solution

A Power Platform solution can be connected to a Git repo for tracking and management of changes to code. This is known as source control. To do this, a Git repo must be setup inside an Azure DevOps project. Then you can choose whether to connect individual solutions or the entire environment to source control

Initialize A Git Repo In Azure DevOps

Go to Azure DevOps and open a project where you want the repo to be located. Then go to the Repos page and Initialize a new main branch with a readme.

The screenshot shows the Azure DevOps web interface. The top navigation bar includes 'Azure DevOps', 'Repos', 'Files', and 'ALM Tutorial Project'. The left sidebar lists various project features: Overview, Boards, Repos (selected), Files, Commits, Pushes, Branches, Tags, Pull requests, Advanced Security, Pipelines, Test Plans, and Artifacts. The main content area is titled 'ALM Tutorial Project is empty. Add some code!'. It contains four sections: 1. 'Clone to your computer' with buttons for 'HTTPS' and 'SSH', a text field showing the repository URL, a link to 'Learn more about HTTPS', and a 'Generate Git Credentials' button. 2. 'Push an existing repository from command line' with 'HTTPS' and 'SSH' buttons and a terminal command: `git remote add origin https://md-alm-tutorial@dev.azure.com/md-alm-tutorial/ALM%20Tutorial%20Project/_git/ALM%20Tutorial%20Project`. 3. 'Import a repository' with an 'Import' button. 4. 'Initialize main branch with a README or gitignore' with a checked 'Add a README' checkbox, a dropdown for 'Add a .gitignore: None', and an 'Initialize' button which is highlighted with a red rectangular box.

The main branch of the repo will appear with a readme file inside of it.

AP

+

ALM Tutorial Project

M README.md

main

Type to find a file or folder...

Files

Set up build

Clone

Contents

History

Name ↑	Last change	Commits
M README.md	Just now	8624d6b2 Added READ...

Introduction

TODO: Give a short introduction of your project. Let this section explain the objectives or the motivation behind this project.

Getting Started

TODO: Guide users through getting your code up and running on their own system. In this section you can talk about:

1. Installation process
2. Software dependencies
3. Latest releases
4. API references

Build and Test

TODO: Describe and show how to build your code and run the tests.

Connect The Environment To A Git Repo

We want to connect the development environment to our git repo. Go to the development environment and open the Solutions page. Select Connect to Git from the top-menu bar.

Power Apps

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ALM Tutorial (Dev)

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Power Platform

New solutionImport solutionOpen AppSource

Solutions

Deployment

Track all solution deployments in one place. See what's active, what has failed, and get recommendations for improvement.

Unmanaged

Managed

All

Display name	Name	Created
Currency Exchange ALM Sample	CurrencyExchang...	1 day
Common Data Services Def...	Cra49cd	1 week
Default Solution	Default	1 week

Publish all customizations

Set preferred solution

See history

Connect to Git

Switch to classic

Fill-in the Connect to Git menu with the following values the press the Next button.

- Connection Type – Solution
- Organization – the Azure DevOps organization where your repo is located
- Project – the Azure DevOps project where your repo is located
- Repository – the name of the repository
- Root Git Folder – “src” (which means source)

Power Apps

+ New solution

Solutions

Deploy
Track all
what's a
recomm

Unmanaged

Display name
Currency Ex
Common D
Default Sol

Connect to Git

You can connect this environment to a Git repository via Microsoft Azure DevOps.

i Solutions developed with source control integration can only be used within Managed Environments. [Learn more](#)

Connection type *

☐ Environment

☒ Solution

Connect different solutions in this environment to different branches and folders in source control. Solutions in this environment cannot be connected to different repositories.

Organization *

md-alm-tutorial

Select an organization to connect to Git.

Project *

ALM Tutorial Project

Select a project from your selected organization.

Repository *

ALM Tutorial Project

Select a Git repository from your selected project.

Root Git folder *

src

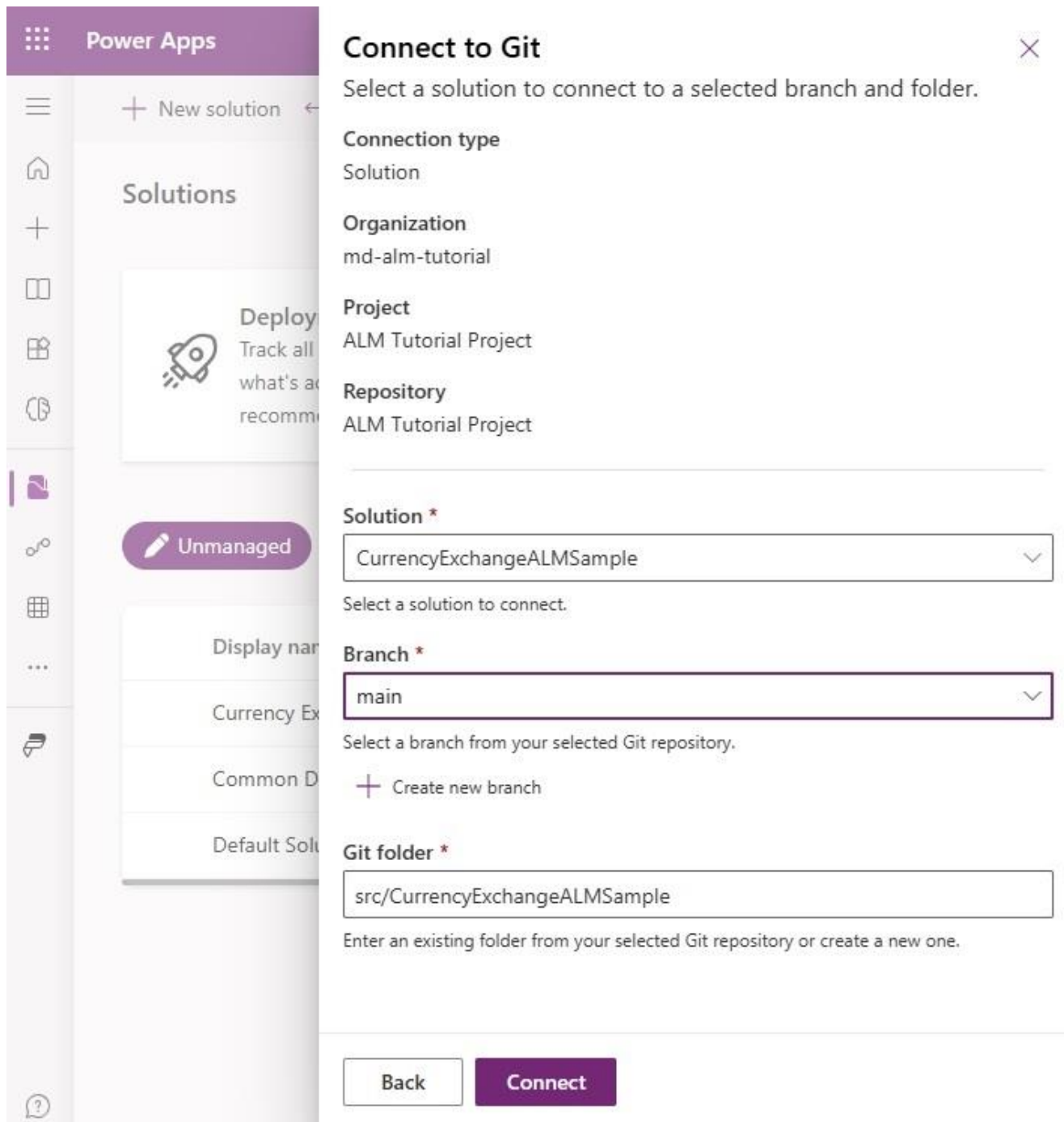
Enter an existing folder name from your selected Git repository or create a new one. This will be used

Next **Cancel**

Select A Solution To Connect To Git

On the next screen we must select a solution to connect to git and select a branch and folder. Use the following values and press the Connect button.

- Solution – YourSolutionName
- Branch – main
- Git folder – src/YourSolutionName



Power Apps

+ New solution

Solutions

Deploy
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Display nam
Currency Ex
Common D
Default Sol

Connect to Git

Select a solution to connect to a selected branch and folder.

Connection type
Solution

Organization
md-alm-tutorial

Project
ALM Tutorial Project

Repository
ALM Tutorial Project

Solution *
CurrencyExchangeALMSample

Select a solution to connect.

Branch *
main

Select a branch from your selected Git repository.

+ Create new branch

Git folder *
src/CurrencyExchangeALMSample

Enter an existing folder from your selected Git repository or create a new one.

Back Connect

After a few moments the solution will become connected to Git.

Power Apps

Search

MD

New solution

Import solution

Open AppSource

Search

CurrencyExchangeALMSample solution successfully connected to Git. [View source control](#)

Solutions

Deployment

Track all solution deployments in one place. See what's active, what has failed, and get recommendations for improvement.

Unmanaged

Managed

All

Display name	Name	Created
Currency Exchange ALM Sample	CurrencyExchang...	1 day ago
Common Data Services Def...	Cra49cd	1 week ago
Default Solution	Default	1 week ago

And the solution folder will appear in the repo. The folder will only include a readme file until we commit changes made to the solution.

AP

+

ALM Tutorial Project

src

CurrencyExchangeALMSample

Readme.md

README.md

main

Type to find a file or folder...

Files

Set up build

Clone

Contents

History

Name ↑	Last change	Commits
src	Just now	b2220416 Cr...
README.md	1h ago	8624d6b2 A...

Introduction

TODO: Give a short introduction of your project. Let this section explain the objectives or the motivation behind this project.

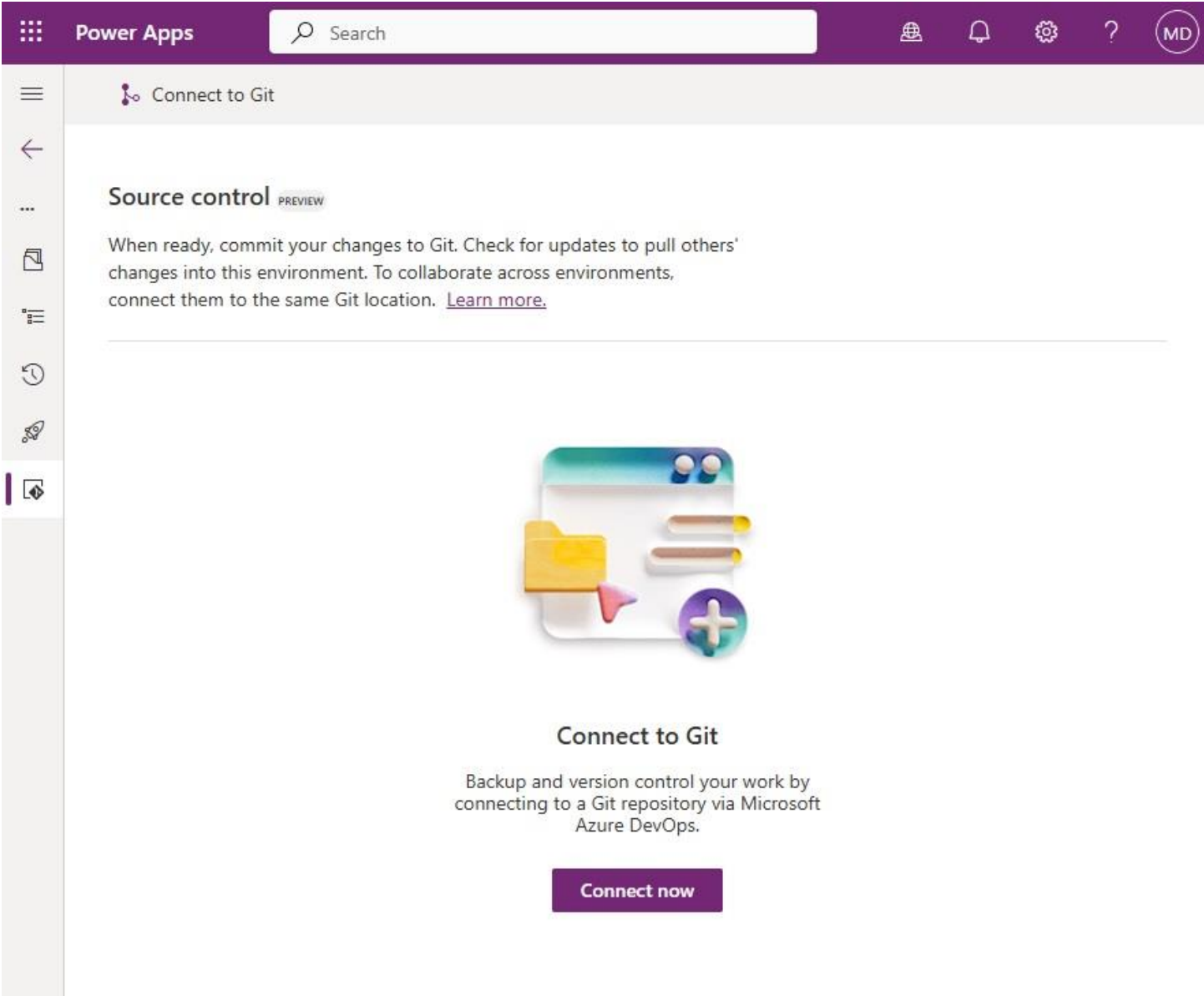
Getting Started

TODO: Guide users through getting your code up and running on their own system. In this section you can talk about:

1. Installation process
2. Software dependencies
3. Latest releases
4. API references

Connect Another Solution To Git

Our Dataverse Git integration was setup with the connection type Solution. This means we must connect individual solutions one-by-one. To connect another solution, open the solution and go to the source control page. Then press the Connect to Git button and go through the wizard steps once again.



Commit Code Changes To A Git Repository

The Dataverse Git Integration can be used to commit changes to a git repository. A git commit takes a snapshot of the project in its current state and saves it into the repo. It includes a message from the developer explaining the updates made in the commit. This makes it possible to revert back to that version if needed later on.

Commit Code Changes To A Git Repository

A git commit takes a snapshot of the project in its current state and saves it into source control. We want to use a git commit to perform the initial save of the solution to the repo. Open the solution we will save to the repo.

Power Apps

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Environment
ALM Tutorial (Dev)

Back to solutions

Currency Exchange ALM...

Overview

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Source control
(Preview)

Objects

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All (10)

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Cards (0)

Cloud flows (1)

Connection references (2)

Environment variables (1)

Site maps (1)

Tables (3)

New

Add existing

Publish all customizations

Currency Exchange ALM Sample > All

	Display name	Name
	Currency	md_currency
	Currency Exchange Ad...	md_Currenc
	Currency Exchange Ad...	md_Currenc
	Currency Transaction	md_currency
	Currency Transactions ...	md_currency
	Email Notifications Tar...	md_EmailNo
	Exchange Rate	md_exchang
	Microsoft Dataverse C...	md_sharedo
	Office 365 Outlook	md_Office36
	Send Currency Rates S...	Send Curren

Then open the source control menu for the solution. There will be a list of changes to be made to the current branch. When ready, press the commit button to create a save point.

Power Apps

Search

Environment
ALM Tutorial (Dev)

MD

Commit

Refresh

Check for updates

Git Connection

Back to solutions

Currency Exchange ALM...

Overview

Objects

History

Pipelines

Source control
(Preview)

Source control PREVIEW

When ready, commit your changes to Git. Check for updates to pull others' changes into this environment. To collaborate across environments, connect them to the same Git location. [Learn more.](#)

Current branch

main

Changes (134)

Updates (0)

Conflicts (0)

Items in this list are changes in your environment that need to be committed to this branch.

Display name	Name	Object	Git Operatio
CurrencyExchangeALMSample	CurrencyExchangeALMSample	Solution	Create
Microsoft Dataverse CurrencyE...	Microsoft Dataverse CurrencyExcha...	Connection Reference	Create
Office 365 Outlook	Office 365 Outlook	Connection Reference	Create
md_EmailNotificationsTarget	md_EmailNotificationsTarget	Environment Variable Defi...	Create
md_EmailNotificationsTarget	md_EmailNotificationsTarget	Environment Variable Value	Create
Exchange Rate Lookup View	Exchange Rate Lookup View	Saved Query	Create

Write A Commit Message For The Solution

All git commits must have a comment. They will help us to understand what changes were made for a specific commit when we need to know in the future. For the first git commit write the message initial commit. Then press the commit button.

A screenshot of a 'Commit' dialog box from a code editor. The dialog has a title bar with 'iron' and 'M T' on the left and a close button 'X' on the right. The main content area has a subtitle 'Review and write a comment about the item you are committing.' followed by a 'Comment *' label and a text input field containing 'Initial commit'. Below this is a 'Commit to branch' section with 'main' selected. At the bottom are two buttons: 'Commit' (highlighted in purple) and 'Cancel' (white with a grey border).

Commit X

Review and write a comment about the item you are committing.

Comment *

Initial commit

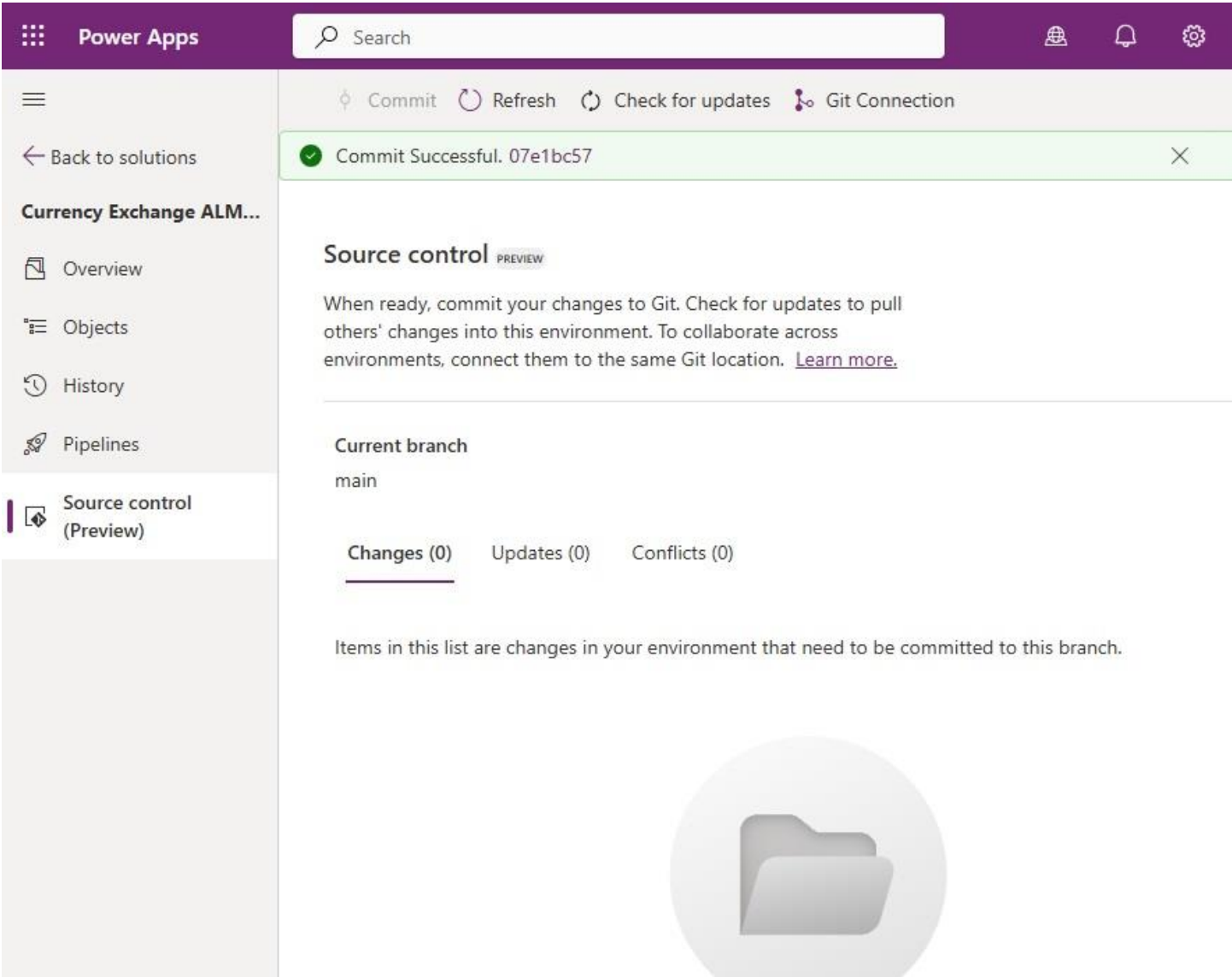
Commit to branch

main

Commit Cancel

View The Unpacked Solution In Source Control

Wait a few minutes for the git commit to complete and the Commit Successful banner to appear.



Once the git commit is completed we can see the unpacked solution in source control. Browse the unpacked solution to learn more about each components structure.

The screenshot displays the Azure DevOps web interface for the 'ALM Tutorial Project'. The left sidebar shows the file explorer with the following structure:

- src
 - CurrencyExchangeALMSample
 - appmodulesitemaps
 - canvasapps
 - connectionreferences
 - entities
 - entityrelationships
 - environmentvariabledefinition
 - modernflows
 - publishers
 - solutions
 - CurrencyExchangeALMS
 - missingdependencies
 - rootcomponents.yml
 - solution.yml
 - solutioncomponents.yml
- Readme.md

The main pane shows the 'solution.yml' file content, which is an XML schema definition for a solution manifest. The content is as follows:

```
1 ImportExportXml:
2   '@version': 9.2.24123.252
3   '@SolutionPackageVersion': 9.2
4   '@languagecode': 1033
5   '@generatedBy': CrmLive
6   '@xmlns:xsi': http://www.w3.org/2001/XMLSchema-instance
7 SolutionManifest:
8   UniqueName: CurrencyExchangeALMSample
9   LocalizedNames:
10     LocalizedName:
11       '@description': Currency Exchange ALM Sample
12       '@languagecode': 1033
13   Descriptions:
14     Version: 1.0.0.645
15     Managed: 0
16     Publisher:
17       UniqueName: MatthewDevaney
```